

Algebraic Topology 2

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Abstract

Algebraic Topology 2 concerns two new algebraic invariants for topological spaces. The homology groups of a space is more powerful than the fundamental group in identifying homeomorphic spaces. However it is also harder to compute, and requires more algebraic background to power the engine. The cohomology ring on the other hand lies in between the homotopy groups and the homology groups. It has a richer structure which enables it to distinguish more spaces, and it is easier to calculate than homology in some cases.

References:

- Notes on Algebraic Topology by Oscar Randal-Williams
- Notes on MA3H6 Algebraic Topology 2 by Martin Gallauer
- Algebraic Topology by Allen Hatcher

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1 The Categorical Viewpoint

Recall that the category of topological spaces $\mathbf{Top}$ is complete and cocomplete. This means that all kinds of limits and colimits exists in $\mathbf{Top}$. We have already seen the product space and disjoint union with their universal property as a limit / colimit. There are also more constructs that can be recognized / defined in terms of the universal property.

1.1 Categories of Spaces

Definition 1.1.1 (The Category of Pointed Topological Spaces) Define the category of pointed topological spaces $\mathbf{Top}_*$ to consist of the following data.

- The objects are a pair (X, x_0) where X is a topological space and $x_0 \in X$ is a chosen base point.
- For (X, x_0) and (Y, y_0) two pointed spaces, the morphisms

$$\mathrm{Hom}_{\mathbf{Top}_*}((X, x_0), (Y, y_0)) = \{f : X \rightarrow Y \mid f \text{ is continuous and } f(x_0) = y_0\}$$

are the continuous maps from X to Y such that base points are preserved.

- Composition is defined as the composition of functions such that base point is preserved.

Proposition 1.1.2 Let (X, x_0) and (Y, y_0) be pointed spaces. Then the product and coproduct of the two spaces in $\mathbf{Top}_*$ are

$$(X \times Y, (x_0, y_0)) \quad \text{and} \quad (X \vee Y, x_0 = y_0)$$

respectively.

Definition 1.1.3 (The Category of Pairs of Spaces) Define the category of pairs of topological spaces $\mathbf{Top}^2$ to consist of the following data.

- The objects are a pair (X, A) where X is a topological space $A \subseteq X$ is a subspace of X .
- For (X, A) and (Y, B) two pointed spaces, the morphisms

$$\mathrm{Hom}_{\mathbf{Top}^2}((X, A), (Y, B)) = \{f : X \rightarrow Y \mid f \text{ is continuous and } f(A) \subseteq B\}$$

are the continuous maps from X to Y such that subspaces are mapped to subspaces.

- Composition is defined as the composition of functions such that subspaces are mapped to subspaces.

Definition 1.1.4 (The Set of Homotopy Classes of Maps)

Let X, Y be spaces. Define the set of homotopy classes of maps from X to Y to be the quotient

$$[X, Y] = \frac{\{f : X \rightarrow Y \mid f \text{ is continuous}\}}{\sim}$$

where $f \sim g$ if f and g are homotopic.

Definition 1.1.5 (The Homotopy Category of Spaces)

Define the homotopy category of topological spaces $\mathbf{hTop}$ to consist of the following data.

- The objects are topological spaces.
- For X and Y two spaces, the morphisms

$$\mathrm{Hom}_{\mathbf{hTop}}(X, Y) = [X, Y]$$

are the homotopy classes of continuous maps from X to Y .

- Composition is defined by $[g] \circ [f] = [g \circ f]$ for $[f] \in \mathrm{Hom}_{\mathbf{hTop}}(X, Y)$ and $[g] \in \mathrm{Hom}_{\mathbf{hTop}}(Y, Z)$.

Definition 1.1.6 (The Set of Pointed Homotopy Classes of Maps)

Let $(X, x_0), (Y, y_0)$ be pointed spaces. Define the set of pointed homotopy classes of maps from (X, x_0) to (Y, y_0) to be the quotient

$$[(X, x_0), (Y, y_0)]_* = \frac{\{f : X \rightarrow Y \mid f \text{ is continuous and } f(x_0) = y_0\}}{\sim}$$

where $f \sim g$ if f and g are homotopic relative to x_0 .

Definition 1.1.7 (The Homotopy Category of Pointed Spaces)

Define the homotopy category of pointed spaces $\mathbf{hTop}_*$ to consist of the following data.

- The objects are pointed topological spaces.
- For two pointed spaces (X, x_0) and (Y, y_0) , the morphisms

$$\mathrm{Hom}_{\mathbf{hTop}_*}((X, x_0), (Y, y_0)) = [(X, x_0), (Y, y_0)]_*$$

are the pointed homotopy classes of continuous maps from (X, x_0) to (Y, y_0) .

- Composition is defined by $[g] \circ [f] = [g \circ f]$ for $[f] \in \mathrm{Hom}_{\mathbf{hTop}_*}((X, x_0), (Y, y_0))$ and $[g] \in \mathrm{Hom}_{\mathbf{hTop}_*}((Y, y_0), (Z, z_0))$.

1.2 Categorical Constructs in the Category of Spaces

We have seen pushouts and pullbacks in Category Theory 1. Let us first recall them.

Definition 1.2.1 (Adjunction Spaces) Let X, Y be spaces and $A \subseteq X$ a subspace. Let $f : A \rightarrow Y$ be a map. Define the adjunction space of X and Y to be the space

$$X \amalg_f Y = \frac{X \amalg Y}{a \sim f(a)}$$

together with the quotient topology.

The prototypical example of adjunction spaces come from CW complexes. It is easy to notice that the data of obtaining the $(n+1)$ -skeleton inductively is the same content of a pushout diagram:

$$\begin{array}{ccc} S_\alpha^{n-1} & \xrightarrow{\varphi_\alpha} & X^n \\ \downarrow & & \downarrow \\ D_\alpha^n & \xrightarrow{\Phi_\alpha} & X^{n+1} \end{array}$$

Indeed Φ_α is determined by φ_α as a property of the pushout. Collectively, we identify the attaching of all n -cells as the following pushout:

$$\begin{array}{ccc} \coprod_{\alpha \in I_n} S_\alpha^{n-1} & \xrightarrow{\coprod_{\alpha \in I_n} \varphi_\alpha} & X^n \\ \downarrow & & \downarrow \\ \coprod_{\alpha \in I_n} D_\alpha^n & \xrightarrow{\coprod_{\alpha \in I_n} \Phi_\alpha} & X^{n+1} \end{array}$$

The final CW complex X is then obtained by the union with the weak topology. But this is just the explicit description of the colimit of the n -skeletons.

Proposition 1.2.2 Let X, Y be spaces and $A \subseteq X$ a subspace of X . Let $f : A \rightarrow Y$ be a map. Then the adjunction space $X \amalg_f Y$ is a pushout of f and $i : A \rightarrow X$ in $\mathbf{Top}$.

Proposition 1.2.3 Let X, Y be spaces with chosen base point x_0 and y_0 respectively. Then the wedge product

$$X \vee Y = X \amalg_f Y$$

is an adjunction space with $Z = \{x_0\}$ and map $f : Z \rightarrow Y$ defined by $f(x_0) = y_0$.

Definition 1.2.4 (Mapping Cone) Let X, Y be spaces. Let $f : X \rightarrow Y$ be a map. Define the mapping cone of f to be

$$C_f = CX \amalg_f Y = \frac{CX \amalg Y}{\sim}$$

where the relation is generated by $(x, 0) \sim f(x)$ for $x \in X$, and CX is the unreduced cone of X .

1.3 Categories of CW Complexes

Definition 1.3.1 (The Category of CW Complexes)

Define the category of CW complexes $\mathbf{CW}$ to consist of the following data.

- The objects are CW complexes.
- For X and Y two CW complexes, the morphisms

$$\text{Hom}_{\mathbf{CW}}(X, Y) = \{f : X \rightarrow Y \mid f \text{ is a cellular map}\}$$

are the continuous maps from X to Y .

- Composition is defined as the composition of functions.

Definition 1.3.2 (The Category of Pointed CW Complexes)

Define the category of CW complexes $\mathbf{CW}_*$ to consist of the following data.

- The objects are pairs where (X, x_0) where X is a CW complex and $x_0 \in X^0$ lies in the zero skeleton.
- For (X, x_0) and (Y, y_0) two pointed CW complexes, the morphisms are given by

$$\text{Hom}_{\mathbf{CW}_*}(X, Y) = \{f : X \rightarrow Y \mid f \text{ is a cellular map and } f(x_0) = y_0\}$$

- Composition is defined as the composition of functions.

Definition 1.3.3 (The Category of Pairs of CW Complexes)

Define the category of pairs of CW complexes $\mathbf{CW}^2$ to consist of the following data.

- The objects are pairs (X, A) where X is a CW complex and A is a subcomplex of X .
- For (X, A) and (Y, B) two pairs of CW complexes, the morphisms are given by

$$\text{Hom}_{\mathbf{CW}^2}((X, A), (Y, B)) = \{f : X \rightarrow Y \mid f(A) \subseteq B, f \text{ and } f|_A \text{ are cellular maps}\}$$

- Composition is defined as the composition of functions.

2 Algebra of Chain Complexes

Homological algebra is the backbone for algebraic topology. In this chapter we will first develop all related notions of chain complexes and exact sequences before diving into the topology side of things.

2.1 Chain Complexes

We begin with a very important notion in homological algebra. A chain complex records a sequence of abelian groups together with group homomorphisms that connect them up. All of homology and cohomology starts with establish a chain complex out of a topological space.

Definition 2.1.1 (Chain Complex) Let R be a ring. A chain complex $(C_\bullet, \partial_\bullet)$ is a family of R -modules C_n for $n \in \mathbb{Z}$ and R -module homomorphisms $\partial_n : C_n \rightarrow C_{n-1}$ such that $\partial_n \circ \partial_{n+1} = 0$ for all n . Explicitly, we have a diagram as follows:

$$\cdots \longrightarrow C_{n+1} \xrightarrow{\partial_{n+1}} C_n \xrightarrow{\partial_n} C_{n-1} \longrightarrow \cdots$$

for which we require that

$$\text{im}(\partial_{n+1}) \subseteq \ker(\partial_n)$$

for each n .

We can relate short exact sequences with chain complexes. In particular, one can always extract short exact sequences from chain complexes.

Lemma 2.1.2 Let $(C_\bullet, \partial_\bullet)$ be a chain complex over $\mathbb{Z}$. Then for any n , the sequence

$$0 \longrightarrow Z_n(C_\bullet) \xrightarrow{\iota} C_n \xrightarrow{\partial_n} B_{n-1}(C_\bullet) \longrightarrow 0$$

is a short exact sequence. Moreover, if the terms of the chain complex consist of free abelian groups, then the sequence is split exact.

Proof The inclusion map is injective. ∂_n is also surjective on its image. Finally, $\ker(\partial_n)$ is by definition $Z_n(C_\bullet) \cong \text{im}(\iota)$. Now since subgroups of a free abelian group are free, we conclude that $B_{n-1} \leq C_n$ is free. Recall that if the third term of a short exact sequence is free, the sequence is split. Hence we conclude. $\blacksquare$

It is routine to also define maps between chain complexes. Indeed in Groups and Rings we introduced groups and group homomorphisms. In Topology we defined topological spaces and continuous maps. For chain complexes we have the notion of chain maps.

Definition 2.1.3 (Chain Map) Let R be a ring. Let $(C_\bullet, \partial_\bullet)$ and $(D_\bullet, \partial'_\bullet)$ be two chain complexes over R . A chain map $f_\bullet : C_\bullet \rightarrow D_\bullet$ is a family of R -module homomorphisms

$$f_n : C_n \rightarrow D_n$$

such that $\partial'_n \circ f_n = f_{n-1} \circ \partial_n$ for all n . Explicitly, we have the following commutative diagram:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & C_{n-1} \longrightarrow \cdots \\ & & \downarrow f_{n+1} & & \downarrow f_n & & \downarrow f_{n-1} \\ \cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial'_{n+1}} & D_n & \xrightarrow{\partial'_n} & D_{n-1} \longrightarrow \cdots \end{array}$$

It is customary to drop the $n \in \mathbb{N}$ in the notation as it is usually implicit. So for example the condition for chain map becomes $\partial' \circ f = f \circ \partial$.

Lemma 2.1.4 Let R be a ring. Let $C_\bullet, D_\bullet, E_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : D_\bullet \rightarrow E_\bullet$ be chain maps. Then the term-wise composition of f and g also gives a chain map $g \circ f : C_\bullet \rightarrow E_\bullet$.

Definition 2.1.5 (The Category of Chain Complexes) Let R be a ring. Define the category of chain complexes

$$\mathbf{Ch}(R)$$

over R to consist of the following data.

- The objects are the chain complexes over R
- The morphisms are the chain maps
- Composition is given by the term-wise composition of maps on all levels.

Definition 2.1.6 (Homology Group) Let R be a ring. Let $(C_\bullet, \partial_\bullet)$ be a chain complex over R .

- Define the the R -submodule of n -cycles of the chain complex to be

$$Z_n(C_\bullet) = \ker(\partial_n)$$

- Define the R -submodule of n -boundaries of the chain complex to be

$$B_n(C_\bullet) = \operatorname{im}(\partial_{n+1})$$

- Define the n th homology of the chain complex to be

$$H_n(C_\bullet) = \frac{Z_n(C_\bullet)}{B_n(C_\bullet)} = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})}$$

Definition 2.1.7 (Induced Homomorphisms) Let R be a ring. Let $C_\bullet$ and $D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ be a chain map. Define the induced map on homology

$$f_* : H_n(C_\bullet) \rightarrow H_n(D_\bullet)$$

by the formula $f_*([z]) = [f(z)]$.

Lemma 2.1.8 Let R be a ring. Let $C_\bullet$ and $C'_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow C'_\bullet$ be a chain map. Then the induced map on homology

$$f_* : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$$

is a well defined R -module homomorphism.

Proof For every map $f_n : C_n \rightarrow C'_n$, we can restrict the domain to cycles so that we obtain a map $f_n : Z_n(C_\bullet) \rightarrow C'_n$. Using the relation given between the boundary operator and the family of maps, we check that this map descends to a map in homology.

Firstly, $f_n(Z_n(C_\bullet)) \subseteq Z_n(C'_\bullet)$. Indeed let $x \in Z_n(C_\bullet)$. Then we have that

$$\partial'_n(f_n(x)) = f_{n-1}(\partial_n(x)) = f_{n-1}(0) = 0$$

which means that $f_n(x)$ lies in the kernel of ∂'_n . Now we have a map $f_n : Z_n(C_\bullet) \rightarrow Z_n(C'_\bullet)$. At the same time, f_n also restricts to a map $f_n : B_n(C_\bullet) \rightarrow B_n(C'_\bullet)$. Indeed if $b \in B_n(C_\bullet)$, then there exists some $c \in C_{n+1}$ such that $\partial_{n+1}(c) = b$. Applying f_n on both sides give

$$\begin{aligned} f_n(\partial_{n+1}(c)) &= f_n(b) \\ \partial'_{n+1}(f_{n+1}(c)) &= f_n(b) \end{aligned}$$

This means that $f_n(b)$ is the boundary of the element $f_{n+1}(c) \in C_{n+1}$, and so f_n restricts to a map of boundaries. Now $f_n : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$ is well defined. Indeed if $b_1, b_2 \in B_n(C_\bullet)$ lie in the same coset, then $b_1 - b_2 \in B_n(C_\bullet)$ so that $b_1 - b_2 \in B_n(C_\bullet)$. Then $f_n(b_1 - b_2) \in B_n(C'_\bullet)$ so that $f_n(b_1)$ and $f_n(b_2)$ lie in the same coset. Thus f_n is well defined. ■

We then have functoriality of the induced map.

Proposition 2.1.9 Let R be a ring. Let $C_\bullet, D_\bullet, E_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : D_\bullet \rightarrow E_\bullet$ be chain maps. Then the induced map on the homology groups satisfy the properties:

- $g_* \circ f_* = (g \circ f)_*$
- $\text{id}_* = \text{id}_{H_n}$

Proof Firstly, we have that

$$\partial \circ g_n \circ f_n = g_{n-1} \circ \partial \circ f_n = g_{n-1} \circ f_{n-1} \circ \partial$$

so that $g \circ f$ is indeed a chain map.

We have that $g_*(f_*([z])) = g_*([f(z)]) = [g(f(z))] = (g \circ f)_*([z])$. Also, we have that

$$\text{id}_*([z]) = [\text{id}(z)] = [z] = \text{id}_{H^n}([z])$$

and so we conclude. ■

Definition 2.1.10 (The Homology Functor) Let R be a ring. Define the homology functors

$$H_n(-) : \mathbf{Ch}(R) \rightarrow \mathbf{Ab}$$

for each n as follows.

- For each chain complex $C_\bullet$ over R , $H_n(C_\bullet)$ is the n th homology group of $C_\bullet$.
- For each chain map $f : C_\bullet \rightarrow C'_\bullet$,

$$H_n(f) : H_n(C_\bullet) \rightarrow H_n(C'_\bullet)$$

is the induced map on homology.

2.2 Chain Homotopies

In algebraic topology 1, we defined the notion of homotopies between two maps. In general, algebraic topologists are not only interested in maps between two objects such as groups or vector spaces, we are also interested in the maps between the maps. Therefore homotopies play such a crucial role in Algebraic Topology 1. In order to produce results similar to homotopy invariance of the fundamental groups, we will need the notion of homotopies between chain maps.

One can think of the chain complexes as our objects, chain maps as morphisms between chain complexes and morphisms between chain maps as chain homotopies. This is precisely the notion of the category of chain complexes.

Definition 2.2.1 (Chain Homotopy) Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : C_\bullet \rightarrow D_\bullet$ be chain maps. Then a chain homotopy from f to g is a collection of morphisms

$$\eta_n : C_n \rightarrow C'_{n+1} \quad \text{such that} \quad g_n - f_n = \partial'_{n+1}\eta_n + \eta_{n-1}\partial_n$$

for all $n \in \mathbb{Z}$. Explicitly, the following diagram is commutative:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & C_{n+1} & \xrightarrow{\partial_{n+1}} & C_n & \xrightarrow{\partial_n} & C_{n-1} \longrightarrow \cdots \\ & \searrow & \downarrow g-f & \swarrow \eta_n & \downarrow g-f & \swarrow \eta_{n-1} & \downarrow g-f \\ \cdots & \longrightarrow & D_{n+1} & \xrightarrow{\partial'_{n+1}} & D_n & \xrightarrow{\partial'_n} & D_{n-1} \longrightarrow \cdots \end{array}$$

In this case, f and g are said to be chain homotopic and we write $f \simeq g$.

Lemma 2.2.2 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Then the relation $\simeq$ on the chain maps from $C_\bullet$ to $D_\bullet$ is an equivalence relation.

Proof Let $f_\bullet, g_\bullet, h_\bullet : C_\bullet \rightarrow D_\bullet$ be chain maps. Then it is clear that $f_\bullet \simeq f_\bullet$ by the 0 map. Indeed we have that $\partial 0 + 0\partial = 0 = f - f$. Thus $\simeq$ is reflexive. if $f_\bullet$ and $g_\bullet$ are chain homotopic by η , then we have the identity

$$\partial\eta + \eta\partial = g - f$$

But then we have

$$\begin{aligned} f - g &= -\partial\eta - \eta\partial \\ &= \partial(-\eta) + (-\eta)\partial \end{aligned}$$

which means that $-\eta$ gives a chain homotopy from g to f .

Now if $g_\bullet$ and $h_\bullet$ are chain homotopic by ν , then we have that

$$\begin{aligned} h_\bullet - f_\bullet &= h_\bullet - g_\bullet + g_\bullet - f_\bullet \\ &= \partial\nu + \nu\partial + \partial\eta + \eta\partial \\ &= \partial(\nu + \eta) + (\nu + \eta)\partial \end{aligned}$$

so that $\nu + \eta$ defines a chain homotopy from f to h . ■

One consequence of chain homotopies is the following.

Lemma 2.2.3 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f : C_\bullet \rightarrow D_\bullet$ and $g : C_\bullet \rightarrow D_\bullet$ be chain maps. If f and g are chain homotopic, then f and g induce the same map

$$f_n = g_n : H_n(X) \rightarrow H_n(Y)$$

in homology.

Proof Let $c \in \ker(\partial_n)$ be an n -cycle. Using the equation for chain homotopy, we have that

$$\begin{aligned} b(c) - a(c) &= \partial'_{n+1}(\eta_n(c)) + \eta_{n-1}(\partial(c)) \\ &= \partial'_{n+1}(\eta(c)) \end{aligned}$$

is a boundary in $\text{im}(\partial'_{n+1}) \subseteq C'_n$. Thus $b_n(c)$ and $a_n(c)$ are of the same coset in $H_n(X)$. ■

Chain homotopies are also well defined in compositions.

Proposition 2.2.4 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Let $f_1, g_1 : C_\bullet \rightarrow D_\bullet$ and $f_2, g_2 : D_\bullet \rightarrow E_\bullet$ be chain maps. If $f_1 \simeq g_1$ and $f_2 \simeq g_2$, then $f_2 \circ f_1 \simeq g_2 \circ g_1$.

Proof The chain homotopies between f_1 and g_1 imposes the identity

$$\partial\eta + \eta\partial = g_1 - f_1$$

for $\eta : C_\bullet \rightarrow D_\bullet$ the given chain homotopy. Similarly, for $\nu : D_\bullet \rightarrow E_\bullet$ we have the identity

$$\partial\nu + \nu\partial = g_2 - f_2$$

Then we have that

$$\begin{aligned} g_2 \circ g_1 - f_2 \circ f_1 &= g_2 \circ g_1 - g_2 \circ f_1 + g_2 \circ f_1 - f_2 \circ f_1 \\ &= g_2(g_1 - f_1) + (g_2 - f_2) \circ f_1 \\ &= g_2(\partial\eta + \eta\partial) + (\partial\nu + \nu\partial) \circ f_1 \\ &= \partial g_2\eta + g_2\eta\partial + \partial\nu f_1 + \nu f_1\partial \\ &= \partial(g_2\eta + \nu f_1) + (g_2\eta + \nu f_1)\partial \end{aligned}$$

Thus $g_2\eta + \nu f_1 : C_n \rightarrow E_{n+1}$ would be a valid chain homotopy from $g_2 \circ g_1$ to $f_2 \circ f_1$. ■

While chain maps induce a map on the homology groups, we will see that chain homotopy equivalences induces isomorphisms on the homology groups, intuitively because they produce an inverse for each map on the level of homology.

Definition 2.2.5 (Chain Homotopy Equivalence) Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . We say that they are chain homotopy equivalence if there exists chain maps $f_\bullet : C_\bullet \rightarrow D_\bullet$ and $g_\bullet : D_\bullet \rightarrow C_\bullet$ such that there are chain homotopies

$$g_\bullet \circ f_\bullet \simeq \text{id}_{C_\bullet} \quad \text{and} \quad f_\bullet \circ g_\bullet \simeq \text{id}_{D_\bullet}.$$

In this case, we say that g is a chain homotopy inverse of f .

Lemma 2.2.6 Let R be a ring. Let $C_\bullet, D_\bullet$ be chain complexes over R . Suppose that $f : C_\bullet \rightarrow D_\bullet$ is a chain map that induces a chain homotopy equivalence between $C_\bullet$ and $D_\bullet$. Then f induces an isomorphism

$$H_n(C_\bullet) \cong H_n(D_\bullet)$$

in all degrees $n \in \mathbb{N}$.

Proof Suppose that g is the chain homotopy inverse of f . We know that $g_\bullet \circ f_\bullet \simeq \text{id}_{C_\bullet}$ which means that they induce the same map:

$$g_* \circ f_* = \text{id}_{H_n(C_\bullet)}$$

Similarly the chain homotopies $f_\bullet \circ g_\bullet \simeq \text{id}_{D_\bullet}$ induce the same map

$$f_* \circ g_* : \text{id}_{H_n(D_\bullet)}$$

as the identity. Then these two identities mean that f_* is both injective and surjective. ■

Similar to homotopies in Algebraic Topology, chain homotopy also defines an equivalence relation on all the chain maps.

Proposition 2.2.7 Chain homotopy equivalence defines an equivalence relation on all chain complexes.

Proof Clearly any chain complex is chain homotopy equivalent to itself by the identity map. If $C_\bullet$ and $D_\bullet$ are chain homotopy equivalent by the chain maps $a_\bullet : C_\bullet \rightarrow D_\bullet$ and $b_\bullet : D_\bullet \rightarrow C_\bullet$, then we have the identities $b_\bullet \circ a_\bullet = \text{id}_{C_\bullet}$ and $a_\bullet \circ b_\bullet = \text{id}_{D_\bullet}$. We can then read them in reverse so that $D_\bullet$ and $C_\bullet$ are chain homotopy equivalence by the maps $b_\bullet$ and $a_\bullet$.

Suppose further that $D_\bullet$ and $E_\bullet$ are chain homotopy equivalent via the maps $u_\bullet : D_\bullet \rightarrow E_\bullet$ and $v_\bullet : E_\bullet \rightarrow D_\bullet$. Then the maps $u_\bullet \circ a_\bullet$ and $b_\bullet \circ v_\bullet$ give a chain homotopy equivalence between $C_\bullet$ and $E_\bullet$. Indeed, upon composition, we have that they are chain homotopic to the identity maps. ■

Be careful that chain homotopy defines an equivalence relation on all chain maps between two fixed chain complexes, while chain homotopy equivalence defines an equivalence relation on all chain complexes. One is an equivalence relation on the objects and one is on the morphisms.

This is reminiscent to that of ordinary homotopies between continuous maps. Namely, homotopies defines an equivalence relation on all continuous maps between two fixed topological spaces, while homotopy equivalence defines an equivalence relation on all topological spaces.

2.3 Sequences of Chain Complexes

One can even define short exact sequences of chain complexes themselves.

Definition 2.3.1 (Short Exact Sequence of Chain Complexes) Let R be a ring. Let the following be a sequence of homomorphisms in $\mathbf{Ch}(R\mathbf{Mod})$.

$$0 \longrightarrow A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \longrightarrow 0$$

We say that it is short exact if for each $n \in \mathbb{N}$, the following sequence

$$0 \longrightarrow A_n \xrightarrow{f_n} B_n \xrightarrow{g_n} C_n \longrightarrow 0$$

is short exact.

Explicitly, we can see that a short exact sequence of chain complexes can be written in diagrams as:

$$\begin{array}{ccccccc} & & \vdots & & \vdots & & \vdots \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A_{n+1} & \xrightarrow{f_{n+1}} & B_{n+1} & \xrightarrow{g_{n+1}} & C_{n+1} \longrightarrow 0 \\ & & \partial \downarrow & & \partial \downarrow & & \partial \downarrow \\ 0 & \longrightarrow & A_n & \xrightarrow{f_n} & B_n & \xrightarrow{g_n} & C_n \longrightarrow 0 \\ & & \partial \downarrow & & \partial \downarrow & & \partial \downarrow \\ 0 & \longrightarrow & A_{n-1} & \xrightarrow{f_{n-1}} & B_{n-1} & \xrightarrow{g_{n-1}} & C_{n-1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \vdots & & \vdots & & \vdots \end{array}$$

Theorem 2.3.2 Let R be a ring. Let the following be a short exact sequence of chain complexes in $\mathbf{Ch}(R\mathbf{Mod})$.

$$0 \longrightarrow A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \longrightarrow 0$$

Then there exists a connecting homomorphism $d : H_n(C_\bullet) \rightarrow H_{n-1}(A_\bullet)$ such that the following sequence of homology groups

$$\cdots \longrightarrow H_{n+1}(C_\bullet) \xrightarrow{d} H_n(A_\bullet) \xrightarrow{(f_n)_*} H_n(B_\bullet) \xrightarrow{(g_n)_*} H_n(C_\bullet) \xrightarrow{d} H_{n-1}(A_\bullet) \longrightarrow \cdots$$

is long exact. Explicitly, d is defined as follows.

- Let $[c] \in H_n(C_\bullet)$. By surjectivity of g_n there exists $b \in B_n$ such that $g_n(b) = c$.
- We have $c \in Z_n(C_\bullet)$. Because $g_{n-1}(\partial_B(b)) = \partial_C(g_n(b)) = \partial_C(c) = 0$, we have that $\partial_B(b) \in \ker(g_{n-1}) = \text{im}(f_{n-1})$. So there exists $a \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b)$.

We then define

$$d([c]) = [a]$$

Proof We want to show that this is well defined. We have made two choices in the definition of d .

- Suppose that $b, b' \in B_n$ are such that $g_n(b) = c = g_n(b')$. Then $b - b' \in \ker(g_n) = \text{im}(f_n)$ means that there exists $a' \in A_n$ such that $f_n(a') = b - b'$. Using the same argument from the construction of the map d , we obtain $a, a' \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b)$ and $f_{n-1}(a') = \partial_B(b')$. This means that $f_{n-1}(a - a') = \partial_B(b - b') = \partial_B(f_n(a'))$. By commutativity of the diagram, we have that

$$\partial_B(f_n(a')) = f_{n-1}(\partial_A(a'))$$

so that we have $f_{n-1}(a - a') = f_{n-1}(\partial_A(a'))$. Since f_{n-1} is injective, we conclude that $a - a' = \partial_A(a')$. But this means that $[a] = [a']$. This implies that the choice of b did not matter in the definition of d .

- Suppose that $a, a' \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(b) = f_{n-1}(a')$. Then by injectivity of f_{n-1} we conclude that $a = a'$.

We now need to show that d is well defined under equivalence classes. Suppose that $c \in \text{im}(\partial_C)$. This implies that there exists $c' \in C_{n+1}$ such that $\partial_C(c') = c$. By surjectivity of g_{n+1} , there exists $b \in B_{n+1}$ such that $g_{n+1}(b) = c'$. By commutativity of the diagram, we have that

$$c = \partial_C(g_{n+1}(b)) = g_n(\partial_B(b))$$

Following the construction of d , we obtain $a \in A_{n-1}$ such that $f_{n-1}(a) = \partial_B(\partial_B(b)) = 0$. Hence $a = 0$. We conclude that d is a well defined function. We next show that d is an R -module homomorphism.

Suppose that $[c], [c'] \in H_n(C_\bullet)$. Following the construction of d we obtain $b, b' \in B_n$ and $a, a' \in A_{n-1}$ such that

$$g_n(b) = c \quad g_n(b') = c' \quad f_{n-1}(a) = \partial_B(b) \quad f_{n-1}(a') = \partial_B(b')$$

We notice that $g_n(b + b') = c + c'$ and $f_{n-1}(a + a') = \partial_B(b + b')$ so that

$$d([c + c']) = [a + a'] = [a] + [a'] = d([c]) + d([c'])$$

The prove is similar in showing that $d([rc]) = rd([c])$

We also need to show that the long sequence is exact. ■

Notice that the above theorem is essentially an application of the snake lemma. One can conclude immediately by the snake lemma.

The following theorem looks horrifying but it is not as terrible as it looks. In essence, we want to show a naturality condition. This means that given a morphism of short exact sequences of chain complexes, we want an induced map that satisfies some commutative square.

Theorem 2.3.3 Let R be a ring. Let the following be a short exact sequence of chain complexes in $\text{Ch}(R\text{Mod})$. Let $A_\bullet, B_\bullet, C_\bullet, A'_\bullet, B'_\bullet, C'_\bullet$ be six chain complexes and let the following

$$0 \longrightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{j} C_\bullet \longrightarrow 0$$

$$0 \longrightarrow A'_\bullet \xrightarrow{i'} B'_\bullet \xrightarrow{j'} C'_\bullet \longrightarrow 0$$

be two short exact sequence of chain complexes. Let the following diagram be a morphism of the two short exact sequence of chain complexes.

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 \cdots & \longrightarrow & A'_{n+1} & \xrightarrow{d} & A'_n & \xrightarrow{d} & A'_{n-1} \longrightarrow \cdots \\
 & \nearrow \alpha & \downarrow i' & & \downarrow i' & & \downarrow i' \\
 \cdots & \longrightarrow & A_{n+1} & \xrightarrow{d} & A_n & \xrightarrow{d} & A_{n-1} \longrightarrow \cdots \\
 & \searrow \beta & \downarrow i & & \downarrow i & & \downarrow i \\
 \cdots & \longrightarrow & B'_{n+1} & \xrightarrow{d} & B'_n & \xrightarrow{d} & B'_{n-1} \longrightarrow \cdots \\
 & \nearrow \beta & \downarrow j' & & \downarrow j' & & \downarrow j' \\
 \cdots & \longrightarrow & B_{n+1} & \xrightarrow{d} & B_n & \xrightarrow{d} & B_{n-1} \longrightarrow \cdots \\
 & \searrow \gamma & \downarrow j & & \downarrow j & & \downarrow j \\
 \cdots & \longrightarrow & C'_{n+1} & \xrightarrow{d} & C'_n & \xrightarrow{d} & C'_{n-1} \longrightarrow \cdots \\
 & \nearrow \gamma & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

Then the induced diagram

$$\begin{array}{ccccccc}
 \cdots & \longrightarrow & H_n(A) & \xrightarrow{i_*} & H_n(B) & \xrightarrow{j_*} & H_n(C) \xrightarrow{\partial} H_{n-1}(A) \longrightarrow \cdots \\
 & & \downarrow \alpha_* & & \downarrow \beta_* & & \downarrow \gamma_* \\
 \cdots & \longrightarrow & H_n(A') & \xrightarrow{i'_*} & H_n(B') & \xrightarrow{j'_*} & H_n(C') \xrightarrow{\partial} H_{n-1}(A') \longrightarrow \cdots
 \end{array}$$

is a commutative diagram.

Proof The first square is commutative since $\beta \circ i = i' \circ \alpha$ and i, i', α, β are all chain maps so that $\beta_* \circ i_* = i'_* \circ \alpha_*$ by proposition 1.1.5. Similarly, the second square is also commutative since $\gamma \circ j = j' \circ \beta$ implies $\gamma_* \circ j_* = j'_* \circ \beta_*$. Now recall that the connecting homomorphism is defined by $\partial([c]) = [a]$ where $j(b) = c$ and $i(a) = d(b)$. Since

$$\gamma(c) = \gamma(j(b)) = j'(\beta(b))$$

and

$$i'(\alpha(a)) = \beta(i(a)) = \beta(d(b)) = d(\beta(b))$$

we have that $\partial[\gamma(c)] = [\alpha(a)]$. Thus we have that

$$\partial(\gamma_*([c])) = \alpha_*([a]) = \alpha_*(\partial([c]))$$

and so we conclude. ■

This naturality condition is also satisfied by the exact sequence in the snake lemma.

3 Singular Homology

3.1 Singular n -Simplexes and Singular Homology

There are a few problems with simplicial homology. In particular, not every space has a *Delta*-complex structure. We would want to extend this definition to any space, with or without Δ -complex structures. Moreover, we have not yet shown that simplicial homology is independent of the choice of Δ -complex structures. Maps between Δ -complex structures may not necessarily define a map between its homology groups.

We therefore want a better version of homology that does all of this, and in particular, to allow any space to have a well defined homology groups.

Definition 3.1.1 (Singular n -Simplexes) A singular n -simplex in a topological space X is a continuous map $\sigma : \Delta^n \rightarrow X$ where Δ^n is an n -simplex.

We say that these are singular because we allow potential deformations of the faces.

It is clear that every Δ -complex consists of n -simplex so the inclusion map of the n -simplex to the Δ -complex defines a singular n -simplex. In fact by definition, none of these n -simplex are “singular” in the sense that some of their faces are degenerate.

Definition 3.1.2 (The Group of Singular n -Chains) Let X be a topological space. Define the group of singular n -chains on X to be the free abelian group

$$C_n(X) = \mathbb{Z}\langle \sigma : \Delta^n \rightarrow X \mid \sigma \text{ is a singular } n \text{ simplex} \rangle$$

on the set of singular n -chains.

This definition is reminiscent of that of Δ -sets and Δ -complexes. The important point is that to every set we can associate a free group on the set. As a set itself there is no algebraic invariants to study. But once we enrich it to include formal linear combinations, we obtain a group structure.

Definition 3.1.3 (The Boundary Operator) Let X be a topological space. Define the boundary operator $\partial_n : C_n(X) \rightarrow C_{n-1}(X)$ to be the homomorphism given by

$$\partial_n(\sigma) = \sum_{k=0}^n (-1)^k \sigma|_{\partial_k \Delta^n}$$

The same proof as in proposition 2.2.3 gives the following.

Proposition 3.1.4 Let X be a space. The family of abelian groups $C_n(X)$ of a simplicial complex and the boundary operator ∂ forms a chain complex.

Proof Let $\sigma \in C_{n+1}(X)$. Then we have that

$$(\partial_n \circ \partial_{n+1})(\sigma) = \partial_n \left(\sum_{j=0}^{n+1} (-1)^j \sigma|_{\partial_j \Delta^{n+1}} \right) = \sum_{i=0}^n \sum_{j=0}^{n+1} (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n}$$

Fix a pair $0 \leq i < j \leq n+1$. By definition 2.1.5, $A = (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n}$ and $B = (-1)^{i+j} (\sigma|_{\partial_i \Delta^{n+1}})|_{\partial_{j-1} \Delta^n}$ cancel out. Moreover every summand is of the form A or B and not

both so that the sum vanishes. In other words, we have that

$$\begin{aligned}
 & \sum_{i=0}^n \sum_{j=0}^{n+1} (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n} \\
 &= \sum_{0 \leq i < j \leq n+1} (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n} + \sum_{0 \leq j < i \leq n} (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n} \\
 &= \sum_{0 \leq i < j \leq n+1} (-1)^{i+j} (\sigma|_{\partial_i \Delta^{n+1}})|_{\partial_{j-1} \Delta^n} + \sum_{0 \leq j < i \leq n} (-1)^{i+j} (\sigma|_{\partial_j \Delta^{n+1}})|_{\partial_i \Delta^n} \\
 &= 0
 \end{aligned}$$

We conclude that $(C_\bullet, \partial_\bullet)$ forms a chain complex. ■

Definition 3.1.5 (The Singular Chain Complex) Let X be a space. Define the singular chain complex $(C_\bullet(X), \partial_\bullet)$ to be the chain complex given by the following data.

- The n th term is given by

$$C_n(X) = \mathbb{Z}\langle \sigma : \Delta^n \rightarrow X \mid \sigma \text{ is a singular } n \text{ simplex} \rangle$$

the group of singular n -chains of X .

- The boundary operator $\partial_n : C_n(X) \rightarrow C_{n-1}(X)$ is given by the formula

$$\partial_n(\sigma) = \sum_{k=0}^n (-1)^k \sigma|_{\partial_k \Delta^n}$$

Definition 3.1.6 (The Singular Homology Group) Let X be a topological space. Define the n th singular homology group of X to be

$$H_n(X) = H_n(C_\bullet(X), \partial_\bullet) = \frac{\ker(\partial_n : C_n(X) \rightarrow C_{n-1}(X))}{\text{im}(\partial_{n+1} : C_{n+1}(X) \rightarrow C_n(X))}$$

One can imagine that in order to deduce the homology groups of a space, the computations involved are highly non-trivial. For instance, one has first work out how the boundary maps are defined in order to deduce their kernel and images. But the set of singular n -simplexes of a space is a much larger set than that of Δ -complexes. When the dimension of the space X (whatever this means) is larger, one can also imagine that the singular homology groups $H_n(X)$ are non-trivial in most of $n \in \mathbb{N}$. We will see how to compute them in the next section.

Example 3.1.7 Let $X = *$ be a point. Then the homology of the one point space is

$$H_n(*) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof For each $n \geq 0$, there is a unique singular n -simplex, $c_n : \Delta^n \rightarrow *$ the constant map. Thus the singular chain complex becomes

$$\cdots \longrightarrow \mathbb{Z} \xrightarrow{\partial_2} \mathbb{Z} \xrightarrow{\partial_1} \mathbb{Z} \longrightarrow 0$$

Now notice that

$$\begin{aligned}
 \partial_n(c_n) &= \sum_{i=0}^n (-1)^i c_n|_{\partial_i \Delta^n} \\
 &= \sum_{i=0}^n (-1)^i c_{n-1} \\
 &= \begin{cases} c_{n-1} & \text{if } n > 0 \text{ is even} \\ 0 & \text{otherwise} \end{cases}
 \end{aligned}$$

This means that ∂_n is an isomorphism when n is even, ∂_n is the zero map when n is odd. When $n \neq 0$ is even, we have that

$$H_n(*) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \frac{\{0\}}{\{0\}} = 0$$

When n is odd, we have that

$$H_n(*) = \frac{\ker(\partial_n)}{\operatorname{im}(\partial_{n+1})} = \frac{\mathbb{Z}}{\mathbb{Z}} = 0$$

Finally when $n = 0$, we have that $H_0(*) = \mathbb{Z}$. ■

For now, we are concerned with some immediate consequences of this definition, such as functoriality.

Proposition 3.1.8 Let $f : X \rightarrow Y$ be a continuous map. Then f induces a chain map

$$f_* : H_n(X) \rightarrow H_n(Y)$$

defined by $[\sigma] \mapsto [f \circ \sigma]$ for each $\sigma : \Delta^n \rightarrow X$ a singular n -simplex. This map satisfies

- $\operatorname{id}_* = \operatorname{id}_{H_n(X)}$
- If $g : Y \rightarrow Z$ is another continuous map, then $(g \circ f)_* = g_* \circ f_*$

Proof Let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex in X . Then $f \circ \sigma : \Delta^n \rightarrow Y$ is a singular n -simplex in Y by continuity of f . We can linearly extend this to a group homomorphism $f_n : C_n(X) \rightarrow C_n(Y)$. The collection of these group homomorphisms lead to a chain map $f_* : C_\bullet(X) \rightarrow C_\bullet(Y)$: Indeed we have that

$$\begin{aligned} f_{n-1} \circ \partial_n(\sigma) &= f_{n-1} \left(\sum_{k=0}^n (-1)^k \sigma|_{\partial_k \Delta^n} \right) \\ &= \sum_{k=0}^n (-1)^k f_{n-1}(\sigma|_{\partial_k \Delta^n}) \\ &= \sum_{k=0}^n (-1)^k (\sigma|_{\partial_k(f_n(\Delta^n))}) \\ &= \partial_n(f_n(\sigma)) \end{aligned}$$

which shows that f is a chain map. By lemma 1.1.4, this yields a group homomorphism $f_n : H_n(X) \rightarrow H_n(Y)$ on each degree.

It is clear that the chain map $\operatorname{id}_* : C_\bullet(X) \rightarrow C_\bullet(Y)$ is the identity and so it also descends to the identity map on homology groups. The second property also follows immediately from the construction above. ■

Proposition 3.1.9 Let X be a topological space and that $\{X_\alpha | \alpha \in I\}$ is the path components of X . Then

$$H_n(X) = \bigoplus_{\alpha \in I} H_n(X_\alpha)$$

Proof Let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex. Then its image is path-connected and therefore lies entirely in one of the X_α . This means that

$$C_n(X) = \bigoplus_{\alpha \in I} C_n(X_\alpha)$$

Moreover, the boundary of σ is a linear combination of $(n-1)$ simplices which all lie in X_α . This means that the chain complex splits into

$$C_\bullet(X) = \bigoplus_{\alpha \in I} C_\bullet(X_\alpha)$$

This decomposition therefore passes down to cycles, boundaries and homology. ■

3.2 Relation to the Low Degree Homotopy Groups

We follow up with a geometric interpretation of H_0 and H_1 . This will make calculations slightly easier, especially since we are able to relate H_1 with the fundamental group, which we have seen various examples of in Algebraic Topology 1

Definition 3.2.1 (Homologous Elements) Let X be a topological space. Let $x, y \in C_n(X)$. We say that x and y are homologous if there exists $z \in C_{n+1}(X)$ such that $\partial_{n+1}(z) = x - y$.

Intuitively, two elements are homologous if they bound some sort of higher dimensional singular complex. In terms of Δ -complexes and CW complexes, two n -chains are homologous if they form the boundary of an $(n + 1)$ -chain.

Lemma 3.2.2 Let X be a path connected space. Then the 0th homology is the integers

$$H_0(X) \cong \mathbb{Z}$$

Proof Define a map $\deg : C_0(X) \rightarrow \mathbb{Z}$ by $\deg(x) = 1$ for every x in the generator of $C_0(X)$ and extend it linearity so that $\deg$ is a group homomorphism. Firstly $\deg$ is surjective since X is non-empty. Indeed there exists $x \in X$ with its image a generator in $\mathbb{Z}$.

Now we show $B_0(X) \subseteq \ker(\deg)$. Let $\gamma : \Delta^1 \rightarrow X$. Then

$$\begin{aligned} \deg(\partial_1 \gamma) &= \deg(\gamma(1) - \gamma(0)) \\ &= \deg(\gamma(1)) - \deg(\gamma(0)) \\ &= 1 - 1 \\ &= 0 \end{aligned}$$

Thus we are done. Now we show that $\ker(\deg) \subseteq B_0(X)$. Suppose that $L = \sum_{x \in X} \lambda_x \cdot x \in \ker(\deg)$ for $\lambda_x \in \mathbb{Z}$ and finitely many non-zero. Then we have the following:

$$\begin{aligned} L &= \sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x \cdot x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \cdot y \\ 0 &= \deg(L) = \deg \left(\sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x \cdot x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \cdot y \right) \\ &= \sum_{\substack{x \in X \\ \lambda_x \geq 0}} \lambda_x - \sum_{\substack{y \in X \\ \lambda_y < 0}} (-\lambda_y) \end{aligned}$$

This means that we can pair up the positives and the negatives so that

$$L = \sum (x_i - y_i)$$

for $x_i \in X$ with positive coefficient and $y_i \in X$ for negative coefficients.

Now observe the following: If $\gamma : [0, 1] = \Delta^1 \rightarrow X$ is a singular 1-simplex with $\gamma(0) = x$ and $\gamma(1) = y$ with $x, y \in C_0(X)$, then

$$\partial_1(\gamma) = \gamma|_{\partial_0 \Delta^1} - \gamma|_{\partial_1 \Delta^1} = \gamma(1) - \gamma(0) = y - x$$

Since X is path connected, for any x_i, y_i , there exists $\gamma_i : [0, 1] = \Delta^1 \rightarrow X$ such that $\gamma_i(0) = x_i$,

$\gamma_i(1) = y_i$. Then

$$\begin{aligned} L &= \sum (x_i - y_i) \\ &= \sum \partial_1 \gamma_i \\ &= \partial_1 \left(\sum_i \gamma_i \right) \end{aligned}$$

Thus $L \in B_0(X)$. Combining the fact that $\deg$ is surjective and $B_0(X) = \ker(\deg)$, we obtain

$$H_0(X) \cong \mathbb{Z}$$

by the first isomorphism theorem and thus we are done. ■

Using the lemma, we can now interpret $H_0(X)$ as the free group on the path components of X .

Corollary 3.2.3 Let X be a space. Then

$$H_0(X) \cong \mathbb{Z}\pi_0(X)$$

Proof For any space X , $\pi_0(X)$ is the path components of X . We know from proposition 3.2.2 that the zero homology of each path connected components is $\mathbb{Z}$. Proposition 3.1.7 shows that we can split the homology of X into direct sum of homology of path connected components. This leaves us with the claim. ■

The remainder of this section is dedicated to the first homology group $H_1(X)$ and its relation to the fundamental group $\pi_1(X, x)$. In fact we almost have a well defined map

$$h_1 : \pi_1(X, x) \rightarrow H_1(X)$$

given as follows. Start with a loop $\gamma : \Delta^1 \rightarrow X$ in $\pi_1(X, x)$. It follows that $\partial_1(\gamma) = \gamma(1) - \gamma(0) = x - x = 0$ so that γ is a cycle in $C_1(X)$. Since $\pi_1(X, x)$ is defined as an equivalence class of homotopic loops, we would like h_1 to send an equivalence class to $H_1(X)$ instead of just a loop itself. For this, we need to show that any two homotopic loops map to the same element in $H_1(X)$.

Lemma 3.2.4 Let γ_1, γ_2 be two paths in a space X that are homotopic relative to their end points. Then γ_1 and γ_2 are homologous elements of $C_1(X)$.

Proof Suppose that $H : I \times I \rightarrow X$ is the homotopy between γ_1 and γ_2 relative to end points x and y . Let c_x be the constant loop at x and vice versa for c_y . Define $\gamma(t) = H(t, t)$. Then $\gamma_1 \cdot c_y \cdot \bar{\gamma} = 0$ which means that it is the boundary of some 2-chain, say σ_1 . Similarly, $c_x \cdot \gamma_2 \cdot \bar{\gamma} = 0$ and so it is the boundary of a 2-chain, say σ_2 . We then have that

$$\begin{aligned} \partial_2(\sigma_2 - \sigma_1) &= \gamma_2 - \gamma + c_x - c_y + \gamma - \gamma_1 \\ &= (\gamma_2 - \gamma_1) + (c_x - c_y) \end{aligned}$$

Our goal is to show that $c_x - c_y \in B_1(X)$ so that γ_1 and γ_2 are homologous via the 2-chain $\sigma_2 - \sigma_1$.

Let $\sigma : \Delta^2 \rightarrow X$ be the constant map with value x . Then

$$\partial_2(\sigma) = c_x - c_x + c_x = c_x$$

shows that $c_x \in B_1(X)$. This is the same for c_y and so every constant path on X lie in $B_1(X)$. ■

Thus now we have a map of sets

$$h_1 : \pi_1(X, x) \rightarrow H_1(X)$$

we still need to show that it is a group homomorphism. The following lemma will help in the proof.

Lemma 3.2.5 Let γ_1, γ_2 be two paths in X with $\gamma_1(1) = \gamma_2(0)$. Then $\gamma_1 \cdot \gamma_2$ is homologous to $\gamma_1 + \gamma_2$. Moreover, $\bar{\gamma}$ is homologous to $-\gamma$.

Proof It is clear that $\gamma_1, \gamma_2, \gamma_1 \cdot \gamma_2$ form the boundary of a 2-simplex, say $\sigma = [v_0, v_1, v_2]$ since $\gamma_1 \cdot \gamma_2 \cdot \overline{\gamma_1 \cdot \gamma_2} = 0$. Now project v_1 orthogonally down to the face $[v_0, v_2]$ to get a new two simplex

$$\sigma : [v_0, v_1, v_2] \rightarrow [v_0, v_2] \rightarrow X$$

Then we have $\partial_2(\sigma) = \gamma_1 + \gamma_2 - \gamma_1 \cdot \gamma_2$ which shows that $\gamma_1 + \gamma_2$ and $\gamma_1 \cdot \gamma_2$ are homologous.

Now we have that $\gamma + \bar{\gamma}$ is homologous to $\gamma \cdot \bar{\gamma}$, and this is homologous to the trivial loop. This means that $\bar{\gamma}$ is homologous to $-\gamma$. ■

This gives the following proposition.

Proposition 3.2.6 Let X be a topological space. The map

$$h_1 : \pi_1(X, x) \rightarrow H_1(X)$$

defined by $h_1([\gamma]) = [\gamma]$ for $[\gamma] \in \pi_1(X, x)$ is a group homomorphism.

Proof We have shown from lemma 3.2.4 that every constant path on X lie in $B_1(X)$. Then $h_1([c_x]) = (0 + B_1(X)) \in H_1(X)$ implies that h_1 maps units to units. Now let γ_1, γ_2 be two loops based at x and $\gamma_1 \cdot \gamma_2$ be their concatenation. Our goal is to show that $h_1([\gamma_1] \cdot [\gamma_2]) = h_1([\gamma_1]) + h_1([\gamma_2])$. This amounts to showing that $\gamma_1 \cdot \gamma_2$ is homologous to $\gamma_1 + \gamma_2$. Then by the above lemma, we are done. ■

In general, $h_1 : \pi_1(X, x) \rightarrow H_1(X)$ is a map to an abelian group $H_1(X)$. However as we have seen in Algebraic Topology 1, not every space has an abelian fundamental group. However, if we forcefully abelianize the fundamental group, we in fact obtain an isomorphism.

Theorem 3.2.7 Let X be a non-empty path connected topological space. Then there is an isomorphism

$$\pi_1(X, x)^{\text{ab}} \cong H_1(X)$$

Proof Since X is path connected, for every $y \in X$, we can once and for all, choose a path η_y from x to y . Given any path $\gamma : \Delta^1 \rightarrow X$, we associate a loop based at x , as the following concatenation:

$$g(\gamma) = \eta_{\gamma(0)} \cdot \gamma \cdot \eta_{\gamma(1)}^{-1}$$

We can extend this map linearly to obtain a homomorphism

$$g : Z_1(X) \subseteq C_1(X) \rightarrow \pi_1(X, x)^{\text{ab}}$$

Now we want $g(b) = 0$ for any boundary $b \in B_1(X)$. Now notice that for a singular 2-simplex with boundary $\gamma_1 : I \rightarrow [v_0, v_1], \gamma_2 : I \rightarrow [v_1, v_2], \gamma_3 : I \rightarrow [v_0, v_2]$, we have that $\gamma_1 \cdot \gamma_2$ is homotopic relative to their end points. It follows that for $\partial_2(\sigma) \in B_1(X)$, we have

$$\begin{aligned} g(\partial_2(\sigma)) &= g(\gamma_1 + \gamma_2 - \gamma_3) \\ &= g(\gamma_1) + g(\gamma_2) - g(\gamma_3) \\ &= [\eta_{v_0} \cdot \gamma_1 \cdot \eta_{v_1}^{-1}] + [\eta_{v_1} \cdot \gamma_2 \cdot \eta_{v_2}^{-1}] + [\eta_{v_2} \cdot \bar{\gamma}_3 \cdot \eta_{v_0}^{-1}] \\ &= [\eta_{v_0} \cdot \gamma_1 \cdot \gamma_2 \cdot \bar{\gamma}_3 \cdot \eta_{v_0}^{-1}] \\ &= [\eta_{v_0} \eta_{v_0}^{-1}] \\ &= 0 \end{aligned}$$

This means that $\bar{g} : H_1(X) \rightarrow \pi_1(X, x)^{\text{ab}}$ is well defined.

It remains to show that the two composites are the identity. Let $[\gamma] \in \pi_1(X, x)^{\text{ab}}$. Then we have

$$\begin{aligned}\bar{g}(\bar{h}_1([\gamma])) &= [\gamma] \\ &= [\eta_x \cdot \gamma \cdot \eta_x^{-1}] \\ &= [\eta_x] + [\gamma] + [\eta_x^{-1}] \\ &= [\gamma]\end{aligned}$$

Thus $\bar{g} \circ \bar{h}_1 = \text{id}$. Now let $L = \sum \lambda_\gamma \gamma \in Z_1(X)$ by a 1-cycle where $\lambda_\gamma \in \mathbb{Z}$. By replacing $-\gamma$ by $\bar{\gamma}$ if necessary (Lemma 3.2.5), we may assume that $\lambda_\gamma \in \mathbb{N} \setminus \{0\}$. Relabelling gives

$$L = \sum_{i=1}^n \gamma_i$$

where γ_i can possibly repeat. If γ_1 is not a loop, then there must exist $i > 1$ such that $\gamma_1(1) = \gamma_i(0)$. Replacing $\gamma_1 + \gamma_i$ by the concatenation $\gamma_1 \cdot \gamma_i$ (By lemma 3.2.5) and doing induction reduces the claim for $L = \gamma$ a single loop, say based at y . In this case we have that

$$\begin{aligned}\bar{h}_1(\bar{g}([\gamma])) &= [\eta_y \cdot \gamma \cdot \eta_y^{-1}] \\ &= [\eta_y] + [\gamma] - [\eta_y] \\ &= [\gamma]\end{aligned}$$

and this completes the proof. ■

This gives a rather nice interpretation of the first homology group: It is the abelianization of the fundamental group. Intuitively, since $H_1(X)$ is abelian, it makes detecting differences in this invariant easier, when compared to the non-abelian $\pi_1(X, x)$, which also depends on base point.

We have an immediate application based on calculations of the fundamental group.

Corollary 3.2.8 If X is simply connected then $H_1(X) = 0$.

Proof In this case the fundamental group is trivial and by the above theorem, we have that $H_1(X) = 0$. ■

3.3 Reduced Homology

Using corollary 3.2.3, we see that path-connected spaces will have non-trivial 0th homology group. This motivates the minor modification into reduced homology since the intuition should show that homology groups of single points should be equal to 0.

We shall see in later chapters that this also simplifies the statements of homology in many cases.

Definition 3.3.1 (The Augmented Chain Complex) Let X be a space. Define the augmented chain complex $(\tilde{C}_\bullet(X), \partial_\bullet)$ of X to be the chain complex given by

$$\cdots \xrightarrow{\partial_{n+1}} C_n(X) \xrightarrow{\partial_n} \cdots \xrightarrow{\partial_2} C_1(X) \xrightarrow{\partial_1} C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

where $\varepsilon : C_0 \rightarrow \mathbb{Z}$ is the map given by

$$\varepsilon \left(\sum_{i \in I} n_i \sigma_i \right) = \sum_{i \in I} n_i$$

Definition 3.3.2 (Reduced Homology) Let X be a space. Define the n th reduced homology group of X to be

$$\tilde{H}_n(X) = H_n(\tilde{C}_\bullet(X), \partial_\bullet)$$

It is clear that for $n \geq 1$, singular homology and reduced homology gives the same homology groups, but not for $n = 0$.

Lemma 3.3.3 Let X be a topological space. Then for $n \geq 1$, the homology and the reduced homology are equal. This means that

$$\tilde{H}_n(X) \cong H_n(X)$$

for all $n \geq 1$.

Proof The chain complex is entirely the same for $n \geq 1$ so their homology groups will also be the same. ■

When $n = 0$, we must have that

$$\tilde{H}_0(X) \cong \frac{\ker(\varepsilon)}{\text{im}(\partial_1)}$$

There is in fact an alternate definition of the reduced homology groups, given as follows.

Proposition 3.3.4 Let X be a space. Let $\pi : X \rightarrow *$ be the unique map to the one point space. Then there is a split exact sequence

$$0 \longrightarrow \tilde{H}_0(X) \xrightarrow{f} H_0(X) \xrightarrow{\pi_*} \mathbb{Z} \longrightarrow 0$$

where f is the inclusion map.

Proof We first show that f is well defined. Notice that $\ker(\varepsilon) \leq C_0(X) \rightarrow \frac{C_0(X)}{\text{im}(\partial_1)} = H_0(X)$ so that there is a canonical map $\ker(\varepsilon) \rightarrow H_0(X)$ given by $x \mapsto [x]$. Then kernel of this map is precisely $\ker(\varepsilon) \cap \text{im}(\partial_1) = \text{im}(\partial_1)$. Since $\text{im}(\partial_1) \leq \ker(\varepsilon)$ our map descends to a well defined map $\tilde{H}_0(X) = \frac{\ker(\varepsilon)}{\text{im}(\partial_1)}$ which is precisely the inclusion map. The inclusion map is injective and π is surjective since it is a retract. We are left to show that $\text{im}(f) = \ker(\pi_*)$.

Now π_* sends all generators of $H_0(X)$ to the unique element of the one point set, hence π_* is the map given by $[\sum_{k=0}^n m_k x_k] \mapsto \sum_{k=1}^n m_k$. Since ε has the exact same effect as π_* we conclude that $\ker(\varepsilon) = \ker(\pi_*)$.

Finally since the third term of the short exact sequence is free, we conclude that the short exact sequence splits. ■

Note that the upcoming two theorems: homotopy invariance and Mayer Vietoris also holds for reduced homology.

3.4 Homotopy Invariance

The goal of this subsection is to establish homotopy invariance for singular homology. This is not at all obvious compared to that of fundamental groups. We will split the proof into the part involving topology, and the part involving algebra. In fact, we have already completed the proof on the algebra side in chapter 1. This is the notion of chain homotopy. In category terms they represent homotopy between morphisms of objects. Therefore the notion of chain maps is important in proving any theorem involving homotopies.

For the topological side, we define the prism operator on Δ -sets first.

Definition 3.4.1 (Prism Operator) Let $\Delta^n = [v_0, \dots, v_n]$ be an n -simplex. Define the prism operator by

$$P(\Delta^n) = \sum_{i=0}^n (-1)^i [v_{00}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \in C_{n+1}(\Delta^n \times [0, 1])$$

where v_{ij} denotes the i th vertex of the j th Δ^n simplex for $0 \leq i \leq n$ and $0 \leq j \leq 1$.

As we have already seen that chain homotopies produce equal maps, our goal is now to establish a chain homotopy from two homotopic maps $f, g : X \rightarrow Y$ of spaces. Given a homotopy $H : X \times I \rightarrow Y$ of f and g and a simplex $\sigma : \Delta^n \rightarrow X$ in X , we can compose them to obtain a map:

$$H \circ (\sigma \times \text{id}) : \Delta^n \times I \rightarrow Y$$

so that we now have a homotopy between two singular n -simplexes, namely $f \circ \sigma : \Delta^n \rightarrow Y$ and $g \circ \sigma : \Delta^n \rightarrow Y$. Recalling from the definition of chain homotopies, we need to produce an $(n+1)$ -simplex from this datum. This is done by considering the prism in between $f \circ \sigma$ and $g \circ \sigma$. In particular, $\sigma \times I$ is a prism, and we could divide up the prism into simplexes. This is precisely the content of the Prism operator.

The sign changes in the prism operator is defined so that the boundary of the prism produces the following equality, which is reminiscent with that of chain homotopies.

Lemma 3.4.2 For every $n \geq 0$, we have in $C_n(\Delta^n \times [0, 1])$ that

$$\partial P(\Delta^n) = [v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}] - P(\partial \Delta^n)$$

Proof We have that

$$\begin{aligned} \partial_{n+1} P(\Delta^n) &= \sum_{j \leq i} (-1)^{i+j} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \\ &\quad + \sum_{j \geq i} (-1)^{i+j+1} [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}] \end{aligned}$$

Notice that for $i = j$, we get

$$\sum_{i=0}^n [v_{00}, \dots, v_{(i-1)0}, v_{i1}, \dots, v_{n1}] - \sum_{i=0}^n [v_{00}, \dots, v_{i0}, v_{(i+1)1}, \dots, v_{n1}]$$

All but two of which cancel out, leaving us with

$$[v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}]$$

For $i \neq j$, apply the prism operator P to each face $[v_0, \dots, \hat{v}_j, \dots, v_n]$ of Δ^n to get

$$\begin{aligned} P([v_0, \dots, \hat{v}_j, \dots, v_n]) &= \sum_{i < j} (-1)^i [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}] \\ &\quad + \sum_{j < i} (-1)^{i+1} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \end{aligned}$$

Taking the alternating sum over all j , we get

$$\begin{aligned} P(\partial_n \Delta^n) &= \sum_{i < j} (-1)^{i+j} [v_{00}, \dots, v_{i0}, v_{i1}, \dots, \hat{v}_{j1}, \dots, v_{n1}] \\ &\quad + \sum_{j < i} (-1)^{i+j+1} [v_{00}, \dots, \hat{v}_{j0}, \dots, v_{i0}, v_{i1}, \dots, v_{n1}] \end{aligned}$$

This is precisely the negative terms of $\partial_{n+1} P(\Delta^n)$ yet to be accounted for (terms for which $i \neq j$) Combining the results concludes the proof. ■

We can now prove homotopy invariance of the singular homology groups.

Theorem 3.4.3 (Homotopy Invariance) Suppose $f, g : X \rightarrow Y$ are homotopic continuous maps. Then they induce the same homomorphism

$$f_n = g_n : H_n(X) \rightarrow H_n(Y)$$

In particular, if X and Y are homotopy equivalent then $H_n(X) \cong H_n(Y)$.

Proof Suppose that $H : X \times I \rightarrow Y$ is a homotopy from f to g . Let $\sigma : \Delta^n \rightarrow X$ be a singular n -simplex in X . Consider $H \circ (\sigma \times \text{id}) : \Delta^n \times [0, 1] \rightarrow Y$. This map induces a map on $(n+1)$ -chains:

$$(H \circ (\sigma \times \text{id}))_* : C_{n+1}(\Delta^n \times [0, 1]) \rightarrow C_{n+1}(Y)$$

Then define a chain homotopy $\eta_n : C_n(X) \rightarrow C_{n+1}(Y)$ by

$$\eta_n(\sigma) = (H \circ (\sigma \times \text{id}))_*(P(\Delta^n))$$

Indeed the calculation

$$\begin{aligned} \partial(\eta_n(\sigma)) &= \partial(H \circ (\sigma \times \text{id}))_*(P(\Delta^n)) && \text{(Definition of } \eta_n) \\ &= H \circ (\sigma \times \text{id})_*(P(\Delta^n)) && \text{(Chain map)} \\ &= H \circ (\sigma \times \text{id})_*([v_{01}, \dots, v_{n1}] - [v_{00}, \dots, v_{n0}] - P(\partial\Delta^n)) && \text{(By the above lemma)} \\ &= g_*(\sigma) - f_*(\sigma) - \eta_{n-1}(\partial(\sigma)) \end{aligned}$$

shows that η_n satisfies the chain homotopy equation. Since chain homotopy induces the same map in homology, we are done. ■

The following result is immediate from homotopy invariance. In fact, it is the important result that we will use all the time. For example, by deformation retracting a space into a known subspace, we can deduce at once its singular homology groups.

Corollary 3.4.4 Let X and Y be homotopy equivalent, then $H_n(X) \cong H_n(Y)$ are isomorphic.

Proof Suppose that the homotopy equivalence is given by $f : X \rightarrow Y$ and $g : Y \rightarrow X$. That is, we have $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. Then by proposition 3.1.6 and homotopy invariance, we have that

$$f_* \circ g_* = (f \circ g)_* = (\text{id}_Y)_* = \text{id}$$

and similarly $g_* \circ f_* = \text{id}$. Thus g_* and f_* are inverses of each other and so $H_n(X) \cong H_n(Y)$. ■

Homotopy invariance is also true for reduced homology.

Example 3.4.5 Let X be a contractible space. Then

$$H_n(X) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof Follows from the homology of one point space and homotopy invariance. ■

3.5 Mayer-Vietoris Sequence

Seifert-van Kampen theorem allows the fundamental group of a space to be computed by considering appropriate subspaces. There is a similar method for homology provided by the Mayer-Vietoris sequence.

The Mayer-Vietoris sequence is another powerful for breakdown spaces into subspaces in order to compute homology groups. We will make use of the notion of Barycentric subdivisions to proof the theorem.

We can transfer the definition of barycentric subdivision simply by pushing forward.

Definition 3.5.1 (Barycentric Subdivision of Singular Simplices) Let X be a space and $\sigma : \Delta^n \rightarrow X$ a singular n -simplex. Let $\sigma_* : C_n(\sigma) \rightarrow C_n(X)$ be the induced map given on the basis by $[v_0, \dots, v_n] \mapsto \sigma|_{[v_0, \dots, v_n]}$. Define the barycentric subdivision of σ to be the n -chain

$$S(\sigma) = \sigma_*(S^\Delta(\Delta^n))$$

Extend it to a map $S : C_n(X) \rightarrow C_n(X)$ defined on the basis.

Proposition 3.5.2 Let $n \in \mathbb{N}$. Define $T(\Delta^n) \in C_{n+1}(\Delta^n \times I)$ as follows.

- $T^\Delta(\Delta^0) = \mathcal{B}(\Delta^0) = [B(\Delta^0), v] \in C_1(\Delta^0 \times I)$
- For $n > 0$, define $T^\Delta(\Delta^n) = \mathcal{B}(\Delta^n \times \{0\}) - \mathcal{B}(T(\partial(\Delta^n \times \{0\}))) \in C_{n+1}(\Delta^n \times I)$

Then we have

$$\partial(T^\Delta(\Delta^n)) + T(\partial(\Delta^n \times \{0\})) = \Delta^n \times \{0\} - S(\Delta^n \times \{1\})$$

Proof It is trivially true for $n = 0$. We then induct on n . Suppose that the previous cases are true. We have that

$$\begin{aligned} \partial(T(\Delta^n)) &= \partial(\mathcal{B}\Delta_0^n) - \partial(\mathcal{B}(T(\partial\Delta_0^n))) \\ &= \Delta_0^n - \mathcal{B}(\partial\Delta_0^n) - T(\partial\Delta_0^n) + \mathcal{B}(\partial T(\partial\Delta_0^n)) && \text{(Lemma 4.2.4)} \\ &= \Delta_0^n - T(\partial\Delta_0^n) + \mathcal{B}(\partial T(\partial\Delta_0^n) - \partial\Delta_0^n) \\ &= \Delta_0^n - T(\partial\Delta_0^n) - \mathcal{B}(S(\partial\Delta_1^n) + T(\partial^2\Delta_0^n)) && \text{(By induction)} \\ &= \Delta_0^n - T(\partial\Delta_0^n) - S(\Delta_1^n) \end{aligned}$$

■

Proposition 3.5.3 Let X be a space. Then the barycentric subdivision map $S : C_n(X) \rightarrow C_n(X)$ is a chain map that is chain homotopic to the identity map via the map $T : C_n(X) \rightarrow C_{n+1}(X)$ defined as follows.

Consider the map $\phi_\sigma : \Delta^n \times I \xrightarrow{\text{proj}} \Delta^n \xrightarrow{\sigma} X$ given by projection. Write $\sigma_*^\sigma : C_{n+1}(\Delta^n \times I) \rightarrow C_{n+1}(X)$ the induced map defined by $\sigma_*(\tau) = \phi_\sigma \circ \tau$. Then define

$$T(\sigma) = \sigma_*(T^\Delta(\Delta^n))$$

Proof We have that

$$\begin{aligned} \partial S(\sigma) &= \partial(\sigma_*(S\Delta^n)) \\ &= \sigma_*(\partial(S\Delta^n)) && (\sigma_* \text{ is a chain map}) \\ &= \sigma_*(S(\partial\Delta^n)) && \text{(Lemma 4.2.4)} \\ &= \sum_{i=0}^n (-1)^i \sigma_*(S(\partial_i \Delta^n)) \\ &= \sum_{i=0}^n (-1)^i S(\partial_i \sigma) && \text{(Definition of } S) \\ &= S(\partial\sigma) \end{aligned}$$

which shows that S is chain map.

Since T^Δ satisfies the chain homotopy relation and T and σ_* is linear, we have that T also satisfies the chain homotopy relation so that it is a chain homotopy from S to id . ■

We now come to the final ingredient of the proof. The subgroup of n -chains consists of linear combinations of n -chains contained in U_1 and U_2 .

Definition 3.5.4 (Subgroup of n -Chains from Subspaces) Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Define

$$C_n(U_1 + U_2) = \left\{ \sum_{i \in I} \sigma_i + \sum_{j \in J} \tau_j \mid \sigma_i \in C_n(U_1) \text{ and } \tau_i \in C_n(U_2) \right\}$$

the subgroup of $C_n(X)$ of n -chains that can be written as the sum of n -chains in U_1 and n -chains in U_2 .

We can form a short exact sequence using the subgroup of n -chains. This allows to use the fact that short exact sequences produces a long exact sequence in homology groups.

Proposition 3.5.5 Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Let $j_1 : U_1 \rightarrow X$, $j_2 : U_2 \rightarrow X$ and $i_1 : U_1 \cap U_2 \rightarrow U_1$, $i_2 : U_1 \cap U_2 \rightarrow X$ be inclusions. Then the sequence of chain complexes

$$0 \longrightarrow C_\bullet(U_1 \cap U_2) \xrightarrow{(i_1)_* - (i_2)_*} C_\bullet(U_1) \oplus C_\bullet(U_2) \xrightarrow{(j_1)_* + (j_2)_*} C_\bullet(U_1 + U_2) \longrightarrow 0$$

is exact.

Proof We have to show exactness at the three spots in each degree n .

- Since the inclusion $(i_1)_* : C_n(U_1 \cap U_2) \rightarrow C_n(U_1)$ is already injective, so is $(i_1)_* - (i_2)_*$.
- We have that the composite map is equal to

$$((j_1)_* + (j_2)_*) \circ ((i_1)_* - (i_2)_*) = (j_1)_* \circ (i_1)_* - (j_2)_* \circ (i_2)_* = (j_1 \circ i_1)_* - (j_2 \circ i_2)_* = 0$$

since $j_1 \circ i_1$ and $j_2 \circ i_2$ are both inclusions from $U_1 \cap U_2$ to X . This shows that $\text{im}((i_1)_* - (i_2)_*) \subseteq \ker((j_1)_* + (j_2)_*)$.

Conversely, if $(c_1, c_2) \in C_n(U_1) \oplus C_n(U_2)$, such that $(j_1)_*(c_1) + (j_2)_*(c_2) = 0$, then we have that $(j_1)_*(c_1) = (j_2)_*(-c_2)$ so that c_1 and c_2 must both be in $U_1 \cap U_2$. This means that there exists some $c \in U_1 \cap U_2$ such that $k_*(c) = (j_1)_*(c_1) = (j_2)_*(-c_2)$ for some $k_* : U_1 \cap U_2 \rightarrow X$. By injectivity of $(i_1)_*$ and $(i_2)_*$, we deduce that $(i_1)_*(c) = c_1$ and $(i_2)_*(c) = c_2$.

- The subgroup $C_n(U_1 + U_2)$ is defined precisely as the image of $(j_1)_* + (j_2)_*$, so this map is surjective.

And so the sequence of chain complexes is exact. ■

Note that the choice of the minus sign can be place arbitrarily. The important point is that the place of the minus is so that we can prove $\text{im}((i_1)_* - (i_2)_*) = \ker((j_1)_* + (j_2)_*)$.

Proposition 3.5.6 Let X be a space and U_1, U_2 be open such that $X = U_1 \cup U_2$. Then the inclusion

$$C_\bullet(U_1 + U_2) \hookrightarrow C_\bullet(X)$$

induces isomorphisms in homology.

Proof Let $\sigma \in C_n(X)$ be a singular n -simplex of X . Consider $\sigma^{-1}(U_1)$ and $\sigma^{-1}(U_2)$ as an open cover of Δ^n . Let δ be the Lebesgue number of the open cover. Suppose that $v_k \in S^k(\Delta^n)$ lie in the k -times barycentric subdivision of σ . By lemma 3.5.5, we have that

$$\text{diam}(v_k) \leq \left(\frac{n}{n+1} \right)^k \text{diam}(\Delta^n)$$

As $k \rightarrow \infty$, $\text{diam}(v_k) \rightarrow 0$. So there exists $k \in \mathbb{N}$ such that $\text{diam}(v_k) \leq \delta$. By definition of the Lebesgue cover this means that v_k lie completely in U_1 or U_2 . Hence there exists a smallest number k such that $S^k(\sigma) \in C_n(U_1 + U_2)$.

Let $z \in Z_n(X)$ be an n -cycle in X . Thus z is a linear combination of finitely many n -simplexes $\sigma_k : \Delta^n \rightarrow X$. Choose $k \in \mathbb{N}$ such that $S^k(\sigma_i) \in C_n(U_1 + U_2)$. Since $S \simeq \text{id}$ by the proposition 4.3.3 and composition of chain homotopic maps are chain homotopic, we have that $S^k \simeq \text{id}$ via some chain homotopy η . Thus we have that

$$z - S^k(z) = \partial\eta(z) + \eta\partial(z) = \partial\eta(z)$$

since $\partial z = 0$. In particular, this shows that z and $S^k(z)$ are homologous in $H_n(X)$ so that $[z] = \iota([S^k(z)])$.

For injectivity, let $w \in Z_n(C_\bullet(U_1 + U_2))$ such that $\iota([w]) = 0$. This means that $w = \partial(z)$ for some $z \in C_{n+1}(X)$. Similar to the above, there exists $k \in \mathbb{N}$ such that $S^k(z) \in C_{n+1}(U_1 + U_2)$ and η such that $z - S^k(z) = \partial\eta(z) + \eta\partial(z)$. But then $\partial S^k(z) = \partial(z) - \partial^2\eta(z) - \partial\eta\partial(z) = w - \partial\eta(w)$ so that $[w] = 0$ as required. ■

We can now combine all the results to prove the Mayer-Vietoris sequence.

Theorem 3.5.7 (Mayer-Vietoris Sequence) Let $X = A \cup B$ be the union of two open subspaces with $j_1 : U_1 \rightarrow X$ and $j_2 : U_2 \rightarrow X$ the inclusion maps. Let $i_1 : A \cap B \rightarrow A$ and $i_2 : A \cap B \rightarrow B$ also be the inclusion maps. Then there exists connecting homomorphisms $\partial : H_n(X) \rightarrow H_{n-1}(U_1 \cap U_2)$ such that

$$\cdots \longrightarrow H_{n+1}(X) \xrightarrow{\partial} H_n(U_1 \cap U_2) \xrightarrow{(i_1)_* - (i_2)_*} H_n(U_1) \oplus H_n(U_2) \xrightarrow{(j_1)_* + (j_2)_*} H_n(X) \xrightarrow{\partial} H_{n-1}(U_1 \cap U_2) \longrightarrow \cdots$$

is a long exact sequence.

Proof The short exact sequence in proposition 4.3.6 induces a long exact sequence by theorem 1.4.2. By the above proposition, we can replace $H_n(C_\bullet(U_1 + U_2))$ by $H_n(X)$ and so we are done. ■

Note that the Mayer-Vietoris sequence also holds for reduced homology.

Our first application of the Mayer-Vietoris sequence comes with the computation of the homology groups of the k -spheres.

Example 3.5.8 Let $k \in \mathbb{N}$. Then the homology of the k -sphere S^k is

$$H_n(S^k) \cong \begin{cases} \mathbb{Z} & \text{if } n = k, 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof We first consider the case of S^1 . Since S^1 is path connected, $H_0(S^1) \cong \mathbb{Z}$. Moreover, $\pi_1(S^1, 1) \cong \mathbb{Z}$ is already abelian and so $H_1(X) \cong \mathbb{Z}$. Let U_1 be the upper half of S^1 and U_2 the lower half. It is clear that U_1 and U_2 are contractible, and that $U_1 \cap U_2 \simeq * \amalg *$. By Mayer Vietoris sequence and homotopy invariance, we have

$$\cdots \longrightarrow H_{n+1}(S^1) \longrightarrow H_n(* \amalg *) \longrightarrow H_n(*) \oplus H_n(*) \longrightarrow H_n(S^1) \longrightarrow \cdots$$

Combining the homology of one point space and the fact that homology can be decomposed into the homology of its path connected components, we have an exact sequence

$$0 \longrightarrow H_n(S^1) \longrightarrow 0$$

for all $n \geq 1$. This shows that

$$H_n(S^1) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and so we have computed the homology groups of S^1 .

We now induct on k where S^k means the k -sphere in $\mathbb{R}^{k+1}$. Suppose that the homology of the sphere S^{k-1} is given by the formula. Write S^k as the union of the open upper hemisphere U_1 and the open lower hemisphere U_2 each containing the equator. Then $U_1 \cap U_2 \simeq S^{k-1}$ and U_1, U_2 are

contractible. By Mayer Vietoris sequence and homotopy invariance, we have

$$\cdots \longrightarrow H_{n+1}(S^k) \longrightarrow H_n(S^{k-1}) \longrightarrow H_n(*) \oplus H_n(*) \longrightarrow H_n(S^k) \longrightarrow \cdots$$

S^k is path connected and so $H_0(S^k) = \mathbb{Z}$. We know that $\pi_1(S^k, x) = 0$ and thus $H_1(S^k) = 0$. Now consider the case of $n > 1$. Combining the homology of one point space and induction hypothesis, we have an exact sequence

$$0 \longrightarrow H_n(S^k) \longrightarrow H_{n-1}(S^{k-1}) \longrightarrow 0$$

Again using induction hypothesis, we see that

$$H_n(S^k) \cong H_{n-1}(S^{k-1}) \cong \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

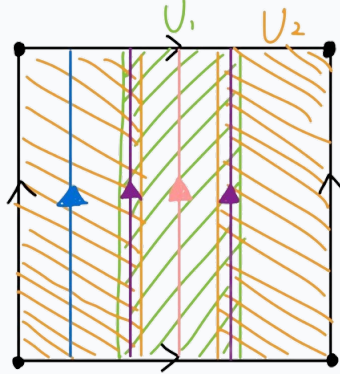
and so we conclude. ■

Recall that the torus and the Klein bottle can be expressed by a quotient of $I \times I$. Together with Mayer-Vietoris sequence and homotopy invariance we can compute the singular homology groups of the two.

Example 3.5.9 Let $\mathbb{T} = S^1 \times S^1$ denote the torus. Then the homology of the torus $\mathbb{T}$ is

$$H_k(\mathbb{T}) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

Proof Cover the torus by two open sets.



The two open sets U_1, U_2 each deformation retract to a circle and $U_1 \cap U_2$ deformation retracts to a disjoint union of two circles. Using homotopy invariance, we obtain

$$H_k(U_1), H_k(U_2) \cong \begin{cases} \mathbb{Z} & \text{if } k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

and since the disjoint union of two circles consists of two path connected components, together with proposition 3.1.7, we have that

$$H_k(U_1 \cap U_2) \cong \begin{cases} \mathbb{Z} \oplus \mathbb{Z} & \text{if } k = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

By the Mayer-Vietoris sequence, the non-trivial terms of the sequence are precisely given by

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \tilde{H}_1(U_1 \cap U_2) \xrightarrow{i} \tilde{H}_1(U_1) \oplus \tilde{H}_1(U_2) \xrightarrow{j} \tilde{H}_1(T) \xrightarrow{\partial} \tilde{H}_0(U_1 \cap U_2) \longrightarrow 0$$

where ∂ are the connecting homomorphisms, i and j are induced by the inclusion maps $i = (\iota_1)_* - (\iota_2)_*$ and $j = (j_1)_* + (j_2)_*$. Also because U_1, U_2 and T are all path connected the end terms are 0. Rewriting them into known homology groups, we obtain

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \mathbb{Z}^2 \xrightarrow{i} \mathbb{Z}^2 \xrightarrow{j} \tilde{H}_1(T) \xrightarrow{\partial} \mathbb{Z} \longrightarrow 0$$

Consider the sequence

$$0 \longrightarrow \tilde{H}_2(T) \xrightarrow{\partial} \mathbb{Z}^2 \xrightarrow{i} \text{im}(i) \longrightarrow 0$$

This sequence is exact by definition. Moreover, since $\text{im}(i)$ is a subgroup of the abelian group $\mathbb{Z}^2$, $\text{im}(i)$ is finite free and so is isomorphic to $\mathbb{Z}^n$ for some $0 \leq n \leq 2$. Then by proposition 1.2.6, the sequence is split exact and by proposition 1.2.5, we obtain that

$$\mathbb{Z}^2 \cong \tilde{H}_2(T) \oplus \text{im}(i)$$

Since $\text{im}(i)$ is a subgroup of $\mathbb{Z}^2$ we can divide both sides by $\text{im}(i)$ to obtain

$$\tilde{H}_2(T) \cong \frac{\mathbb{Z}^2}{\text{im}(i)} \cong \ker(i)$$

Also, by considering the split exact sequence

$$0 \longrightarrow \ker(\partial) \xrightarrow{\iota} \tilde{H}_1(T) \xrightarrow{\partial} \mathbb{Z} \longrightarrow 0$$

we obtain an isomorphism $\tilde{H}_1(T) \cong \mathbb{Z} \oplus \ker(\partial)$. Now $\ker(\partial) = \text{im}(j)$ since the sequence is exact. Also, we have that

$$\frac{\mathbb{Z}^2}{\ker(j)} \cong \text{im}(j)$$

by the first isomorphism theorem. Using the fact that the sequence being exact implies $\text{im}(i) = \ker(j)$, we have that

$$\tilde{H}_1(T) \cong \mathbb{Z} \oplus \text{im}(j) \cong \mathbb{Z} \oplus \frac{\mathbb{Z}^2}{\ker(j)} \cong \frac{\mathbb{Z}^2}{\text{im}(i)} \cong \mathbb{Z} \oplus \text{coker}(i)$$

We now have to compute what i does to the generators of $\tilde{H}_1(U_1 \cap U_2) \cong \mathbb{Z}$ marked in purple. Recall that i is defined as $(\iota_1)_* - (\iota_2)_*$. i then sends the generators to the positive generator (pink) in U_1 . It sends the generators also to the positive generator (blue) in U_2 . This means that our map i can be written as

$$i = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

with basis vectors the two generators in $\tilde{H}_1(U_1 \cap U_2)$ and the two in $\tilde{H}_1(U_1) \oplus \tilde{H}_2(U_2)$. The smith normal form of this map reduces it to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Its kernel is one copy of $\mathbb{Z}$ and its cokernel is also one copy of $\mathbb{Z}$. Thus we conclude that

$$H_k(\mathbb{T}) = \begin{cases} \mathbb{Z} & \text{if } k = 0, 2 \\ \mathbb{Z}^2 & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

and so we are done. ■

Note that the choice of the orientation of the generators is arbitrary, one can choose the opposite orientation and deduce the same homology groups.

Example 3.5.10 Let K denote the Klein bottle. Then the homology of the Klein bottle K is

$$H_k(K) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \\ \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}$$

3.6 Three Applications of Singular Homology

In Algebraic Topology 1, we have seen Brouwer fixed point theorem for D the unit disc in $\mathbb{R}^2$. Using homology groups, we can finally state and prove the result for n -dimensional discs.

Lemma 3.6.1 For any $k \in \mathbb{N}$, S^{k-1} is not a retract of D^k .

Proof Assume to the contrary that $i : S^{k-1} \hookrightarrow D^k$ admits a retraction $r : D^k \rightarrow S^{k-1}$. Then $\text{id}_{S^k} = r \circ i$ so that

$$\tilde{H}_{k-1}(S^{k-1}) \xrightarrow{i_*} \tilde{H}_{k-1}(D^k) \xrightarrow{r_*} \tilde{H}_{k-1}(S^{k-1})$$

is the identity map. Substituting the homology of S^{k-1} and D^k , we have that

$$\mathbb{Z} \xrightarrow{i_*} 0 \xrightarrow{r_*} \mathbb{Z}$$

is the identity map which is impossible. ■

Theorem 3.6.2 (Brouwer Fixed-point Theorem) Every continuous map $f : D^k \rightarrow D^k$ has a fixed point.

Proof Suppose not, then the ray starting at $f(x)$ in the direction of x meets S^{k-1} in exactly one point $g(x) \neq f(x)$. Then $g : D^k \rightarrow S^{k-1}$ defines a retraction. By the above corollary, this is a contradiction. ■

Using topological properties preserved by homeomorphisms, we are only able to prove that $\mathbb{R}^1$ and $\mathbb{R}^2$ are not homeomorphic using the argument on the number of connected components. Using homology, we can prove the full form of invariance of domain.

Corollary 3.6.3 (Invariance of Domain) If $n \neq m$, then $\mathbb{R}^n$ is not isomorphic to $\mathbb{R}^m$.

Proof Assume that $\mathbb{R}^n \cong \mathbb{R}^m$. Then $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^m \setminus \{0\}$ is also a homeomorphism. Since $\mathbb{R}^n \setminus \{0\} \simeq S^{n-1}$ and $\mathbb{R}^m \setminus \{0\} \simeq S^{m-1}$, we have that f_* induces an isomorphism

$$\mathbb{Z} \cong \tilde{H}_{n-1}(S^{n-1}) \cong \tilde{H}_{n-1}(S^{m-1})$$

by homotopy invariance and the computation of the homology groups of the n -sphere. This can only be true when $n = m$ by our computations. ■

The Jordan curve theorem is a highly non-trivial result that looks deceptively easy to prove. Using homology we can give an algebraic proof.

Definition 3.6.4 (Jordan Curve) A Jordan Curve is a simple closed curve in $\mathbb{R}^2$.

Lemma 3.6.5 Let $\gamma : I \rightarrow S^2$ be an injective continuous map with image $C = \gamma(I)$. Then

$$H_n(S^2 \setminus C) = \begin{cases} \mathbb{Z} & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof Let $J \subseteq I$ be an interval. Define $C_J = \gamma(J)$. Let $U = S^2 \setminus C_{[0,1/2]}$ and $V = S^2 \setminus C_{[1/2,1]}$. Note that $U \cap V = S^2 \setminus C$ and

$$U \cup V = S^2 \setminus C_{[1/2,1/2]} \cong \mathbb{R}^2 \simeq *$$

By Mayer-Vietoris, we obtain isomorphisms

$$H_n(S^2 \setminus C) \cong H_n(S^2 \setminus C_{[0,1/2]}) \oplus H_n(S^2 \setminus C_{[1/2,1]})$$

for $n \geq 1$ together with a short exact sequence

$$0 \longrightarrow H_0(S^2 \setminus C) \longrightarrow H_0(S^2 \setminus C_{[0,1/2]}) \oplus H_0(S^2 \setminus C_{[1/2,1]}) \longrightarrow \mathbb{Z} \longrightarrow 0$$

Assume for a contradiction that for some $n \geq 1$ there exists $\sigma \in H_n(S^2 \setminus C)$ which is non-zero. By the isomorphism above, it remains non-zero in one of the two groups of the direct sum. By repeating this argument cutting the interval where its homology group contains σ in half, we obtain a nested sequence of intervals

$$I \supset J_1 \supset J_2 \supset \cdots$$

with non zero intersection $p \in \bigcap_{i=1}^{\infty} J_i$ such that $\sigma \neq 0$ in all of $H_n(S^2 \setminus C_{J_i})$. However we have that $S^2 \setminus C_{[p,p]} \cong \mathbb{R}^2$ is contractible hence σ vanishes in $H_n(S^2 \setminus \{\gamma(p)\})$. Let $\tau \in C_{n+1}(S^2 \setminus C_{[p,p]})$ such that $\partial\tau = \sigma$. Write τ as a finite linear combination of singular simplexes. Each of the singular simplexes has compact image in $S^2 \setminus C_{[p,p]}$ since Δ^{n+1} is compact. The union of these images is covered by open subsets $(S^2 \setminus C_{J_l})_l$ so by compactness, there exists l such that $\tau \in C_{n+1}(S^2 \setminus C_{J_l})$. But then $\sigma = 0$ in $H_n(S^2 \setminus C_{J_l})$, which is a contradiction. Thus $H_n(S^2 \setminus C) = 0$ for all $n \geq 1$.

For $n = 0$, assume that $x, y \in S^2 \setminus C$ are distinct in different path components. Then similar to the above we obtain a nested sequence of intervals J_l such that x and y are in different path components of $S^2 \setminus C_{J_l}$ for all l . Since $S^2 \setminus \{\gamma(p)\}$ is contractible, it must contain a path connecting x and y . By compactness, this path misses C_{J_l} for large l . Thus x and y lie in the same path component of $S^2 \setminus C_{J_l}$ for large l , which is a contradiction. We deduce that $H_0(S^2 \setminus C) = \mathbb{Z}$. ■

Theorem 3.6.6 (Jordan Curve Theorem) Let $\gamma : [0, 1] \rightarrow \mathbb{R}^2$ be a Jordan curve with image C . Then $\mathbb{R}^2 \setminus C$ has two connected components.

Proof Choose S_+^1 and S_-^1 the upper and lower semicircles of S^1 so that the intersection is S^0 . Define the following: $X = S^2 \setminus \gamma(S^0)$, $U_+ = S^2 \setminus \gamma(S_+^1)$, $U_- = S^2 \setminus \gamma(S_-^1)$ and $U_+ \cap U_- = S^2 \setminus C$. Since $S_{\pm}^1 \cong I$, the homology groups of $U_{\pm}$ is given by the above lemma. Also $X \simeq S^1$. By Mayer-Vietoris applied to U, V, X , we obtain an exact sequence

$$0 \longrightarrow H_{n+1}(S^2 \setminus \gamma(S^0)) \longrightarrow H_n(S^2 \setminus C) \longrightarrow H_n(U_+) \oplus H_n(U_-)$$

for $n > 0$ in which the first and third term vanish so that $H_n(S^2 \setminus C) = 0$. For $n = 0$, we have an exact sequence

$$0 \longrightarrow H_1(S^2 \setminus \gamma(S^0)) \longrightarrow \tilde{H}_0(S^2 \setminus C) \longrightarrow \tilde{H}_0(U_+) \oplus \tilde{H}_0(U_-)$$

where the last term vanishes. Thus we have an isomorphism $\tilde{H}_0(S^2 \setminus C) \cong \mathbb{Z}$. It follows that $H_0(S^2 \setminus C) \cong \mathbb{Z}^2$. So now we have

$$H_n(S^2 \setminus C) = \begin{cases} \mathbb{Z}^2 & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Now consider $X = S^2 \setminus C$ and $U = \mathbb{R}^2 \setminus C$ identified in $\mathbb{R}^2 \cup \{\infty\} \cong S^2$. Let V be an open disk around ∞ that does not meet C . Using Mayer-Vietoris sequence, the only interesting terms are this sequence

$$0 \longrightarrow H_n(\mathbb{R}^2 \setminus C) \longrightarrow 0$$

for $n \geq 2$, which gives $H_n(\mathbb{R}^2 \setminus C) = 0$ and the lower degree terms give an exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow H_1(\mathbb{R}^2 \setminus C) \longrightarrow 0 \longrightarrow 0 \longrightarrow \tilde{H}_0(\mathbb{R}^2 \setminus C) \longrightarrow \mathbb{Z} \longrightarrow 0$$

which is due to the fact that $U \cap V \simeq S^1$ and V is contractible. Then it is clear that $H_1(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}$ and $\tilde{H}_0(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}$ so that $H_0(\mathbb{R}^2 \setminus C) \cong \mathbb{Z}^2$ and so we obtain

$$H_n(\mathbb{R}^2 \setminus C) = \begin{cases} \mathbb{Z}^2 & \text{if } n = 0 \\ \mathbb{Z} & \text{if } n = 1 \\ 0 & \text{if } n > 1 \end{cases}$$

Since $\mathbb{R}^2 \setminus C$ is locally path connected, by corollary 3.2.3 we thus have that $\mathbb{R}^2 \setminus C$ has two connected components. ■

4 Relative Homology

4.1 Relative Homology Groups

Given a subspace A of a space X , we know that the inclusion map $A \hookrightarrow X$ induces an inclusion $C_n(A) \hookrightarrow C_n(X)$. Unfortunately, this does not induce an injection $H_n(A) \rightarrow H_n(X)$. Relative homology gives a precise measure of the failure of injectivity and surjectivity of the map in homology.

Definition 4.1.1 (Relative Homology Group) Let X be a topological space and $A \subseteq X$ a subspace. Define the relative homology group $H_n(X, A)$ to be the homology group of the chain complex

$$\cdots \longrightarrow C_n(X, A) \xrightarrow{\partial} C_{n-1}(X, A) \longrightarrow \cdots$$

where $C_n(X, A)$ denotes the quotient group $C_n(X)/C_n(A)$. In other words,

$$H_n(X, A) = \frac{\ker(\partial : C_n(X, A) \rightarrow C_{n-1}(X, A))}{\operatorname{im}(\partial : C_{n+1}(X, A) \rightarrow C_n(X, A))} = H_n(C_\bullet(X, A))$$

Elements of $Z_n(X, A)$ are called relative n -cycles, while elements of $B_n(X, A)$ are called relative n -boundaries.

Geometrically, relative n -cycles are n -cycles in $C_n(X)$ such that $\partial z \in C_{n-1}(A)$ which means that the boundary of z is contained in the subspace A . Intuitively, we are treating cycles in the subspace A as 0 and so the homology groups measure the homology of X without A .

TBA: Functoriality

Relative homology also satisfies the homotopy invariance property and has the Mayer-Vietoris sequence.

Theorem 4.1.2 (Homotopy Invariance) Let $(X, A), (Y, B)$ be pairs of spaces. Let $f, g : (X, A) \rightarrow (Y, B)$ be homotopic maps. Then the induced map on relative homology are the same:

$$f_* = g_* : H_n(X, A) \rightarrow H_n(Y, B)$$

Corollary 4.1.3 Let $f : (X, A) \rightarrow (Y, B)$ be a map such that both $f : X \rightarrow Y$ and $f|_A : A \rightarrow B$ are homotopy equivalences. Then

$$f_* : H_n(X, A) \rightarrow H_n(Y, B)$$

induces an isomorphism for all $n \in \mathbb{N}$.

Definition 4.1.4 (Triplet of Spaces) Let X be a space. A triple (X, A, B) is said to be a triplet of spaces if $B \subseteq A \subseteq X$.

Definition 4.1.5 (Maps of Triplets) Let (X, A, B) and (Y, C, D) be triplets of spaces. Let $f : X \rightarrow Y$ be a map. We say that f is a map of the triplets if $f(A) \subseteq C$ and $f(B) \subseteq D$.

Proposition 4.1.6 Let (X, A, B) be a triplet of spaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Then

$$0 \longrightarrow C_\bullet(A, B) \xrightarrow{i} C_\bullet(X, B) \xrightarrow{j} C_\bullet(X, A) \longrightarrow 0$$

is a short exact sequence of chain complexes.

TBA: Naturality

Proposition 4.1.7 (Long Exact Sequence of a Triple) Let (X, A, B) be a triplet of spaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Then there is a long exact sequence

$$\cdots \longrightarrow H_n(A, B) \xrightarrow{i_*} H_n(X, B) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \longrightarrow \cdots$$

in relative homology induced by the inclusion maps and the connecting homomorphism ∂ from the snake lemma.

Theorem 4.1.8 (Relative Mayer-Vietoris Sequence) Let (X, Y) be a pair of space such that $A, B \subset X$ cover X and $C, D \subset Y$ cover Y with $C \subset A$ and $D \subset B$. Then there is an exact sequence

$$\cdots \longrightarrow H_n(A \cap B, C \cap D) \xrightarrow{\Phi} H_n(A, C) \oplus H_n(B, D) \xrightarrow{\Psi} H_n(X, Y) \longrightarrow \cdots$$

in relative homology.

4.2 Relation to Singular Homology

Theorem 4.2.1 (Exact Sequence of Pairs of Spaces) Let (X, A) be a pair of spaces. Then there is an exact sequence

$$\cdots \longrightarrow H_n(A) \xrightarrow{\iota_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \longrightarrow \cdots$$

where

- $\iota : A \rightarrow X$ is the inclusion map
- $j : X \rightarrow X \setminus A$ is the quotient map
- The connecting homomorphism $\partial : H_n(X, A) \rightarrow H_{n-1}(A)$ is defined by $[z] \mapsto [dz]$ for $z \in C_n(X)$ a relative cycle.

Moreover, the exact sequence is natural with respect to maps $f : (X, A) \rightarrow (Y, B)$.

Proof Notice that

$$0 \longrightarrow C_\bullet(A) \xrightarrow{\iota} C_\bullet(X) \xrightarrow{j} C_\bullet(X, A) \longrightarrow 0$$

is a short exact sequence by construction. Thus it induces a long exact sequence in homology groups.

Consider $[z] = z + C_n(A) \in C_n(X, A)$ a cycle in $C_n(X, A)$. The surjective map j is the projection map so $j(z) = [z]$. Now $d(z) \in \ker(j)$ because

$$j(d(z)) = d(j(z)) = d([z]) = 0$$

by assumption. So $d(z) \in \ker(j) = \text{im}(i)$. Since i is the inclusion, $[d(z)] \in C_{n-1}(A)$ is precisely the element that ∂ maps $[z]$ to. ■

We obtain an immediate result of the singular homology of the space X and its subspace A .

Lemma 4.2.2 Let (X, A) be a pair of spaces. The inclusion map $\iota : A \rightarrow X$ induces an isomorphism

$$H_n(A) \cong H_n(X)$$

for all $n \in \mathbb{N}$ if and only if $H_n(X, A) = 0$ for all n .

Proof If $H_n(X, A) = 0$ for all $n \in \mathbb{N}$, then the long exact sequence above shows that there are isomorphisms $H_n(A) \cong H_n(X)$. If $H_n(A) \cong H_n(X)$ in the long exact sequence above, then the map $H_n(X) \rightarrow H_n(X, A)$ is the zero map and the map $H_n(X, A) \rightarrow H_{n-1}(A)$ is the zero map. Thus for $n \geq 0$ there is an exact sequence

$$0 \longrightarrow H_n(X, A) \longrightarrow 0$$

showing that $H_n(X, A) = 0$. ■

Using the lemma, we have a low degree interpretation for relative homology.

Proposition 4.2.3 Let (X, A) be a pair of space. Then the following are true.

- $H_0(X, A) = 0$ if and only if every path component of X contains at least one path component of A .
- $H_1(X, A) = 0$ if and only if $H_1(A) \rightarrow H_1(X)$ is surjective and each path component of X contains at most one path-component of A .

Lemma 4.2.4 Let X be space and $x \in X$ be a point. Then there is an isomorphism

$$H_n(X, \{x\}) \cong \tilde{H}_n(X)$$

of homology groups for all $n \in \mathbb{N}$.

Proof We have a long exact sequence as in theorem 4.1.3. In particular, since $H_n(\{x\}) = 0$ for all $n \geq 1$, we have an isomorphism

$$H_n(X, \{x\}) \cong H_n(X)$$

which descends to an isomorphism in reduced homology. We are now left with the exact sequence:

$$0 \longrightarrow H_1(X) \xrightarrow{j_*} H_1(X, x) \xrightarrow{\partial} H_0(x) \xrightarrow{\iota_*} H_0(X) \xrightarrow{j_*} H_0(X, x) \longrightarrow 0$$

Now since $\iota : \{x\} \rightarrow X$ is the inclusion map, the projection map $p : X \rightarrow \{x\}$ is such that $p \circ \iota = \text{id}$. By proposition 3.1.6 we have $p_* \circ \iota_* = \text{id}_*$ and thus ι_* is injective and has kernel 0. Exactness then implies that $\text{im}(\partial) = \ker(\iota_*) = 0$ and thus ∂ is the zero map. This means that $\tilde{H}_1(X) \cong H_1(X) \cong H_1(X, x)$.

We are now left with a short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\iota_*} H_0(X) \xrightarrow{j_*} H_0(X, x) \longrightarrow 0$$

Since $H_0(*) \cong \mathbb{Z}$. The map $p_* : H_0(X) \rightarrow H_0(x)$ has the property that $p_* \circ \iota_* = \text{id}_*$. This means that the short exact sequence is split exact and we have that

$$H_0(X) \cong \mathbb{Z} \oplus H_0(X, x)$$

Consider the map

$$H_0(X) \xrightarrow[p_* \times j_*]{\cong} H_0(x) \oplus H_0(X, x) \xrightarrow{\pi} H_0(x)$$

where $\pi : H_0(x) \oplus H_0(X, x) \rightarrow H_0(x)$ is defined by $\pi(a, c) = a$. This map sends $b \in H_0(X)$ to $p_*(b)$. Then we have

$$\begin{aligned} \tilde{H}_0(X) &\cong \ker(H_0(X) \rightarrow H_0(x)) \\ &\cong \ker(H_0(X) \oplus H_0(X, x) \xrightarrow{p_*} H_0(x)) \\ &= H_0(X, x) \end{aligned}$$

Therefore we conclude. ■

4.3 The Excision Theorem

The excision theorem is a powerful theorem for computing relative homology groups. This statement is derived directly from Mayer-Vietoris.

Theorem 4.3.1 (The Excision Theorem) Let X be a space and Z, A be subspaces of X such that $\bar{Z} \subseteq A^\circ$. Then the inclusion map $(X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces an isomorphism

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$$

for all n . Moreover, the excision isomorphism is natural in maps $f : (X, A, Z) \rightarrow (Y, B, C)$.

Proof Let $B = X \setminus Z$. Then notice that $A \cap B = A \setminus Z$, $A^\circ \cup B^\circ = A^\circ \cup (X \setminus \bar{Z}) = X$. Moreover, we have that

$$\begin{aligned} C_n(X \setminus Z, A \setminus Z) &= C_n(B, A \cap B) \\ &= \frac{C_n(B)}{C_n(A \cap B)} && \text{(By definition)} \\ &\cong \frac{C_n(A + B)}{C_n(A)} && \text{(Second Isomorphism Theorem)} \end{aligned}$$

This implies that

$$0 \longrightarrow C_\bullet(A) \longrightarrow C_\bullet(A + B) \longrightarrow C_\bullet(X \setminus Z, A \setminus Z) \longrightarrow 0$$

is a short exact sequence of chain complexes. Moreover, by considering inclusion maps, we have the following commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_\bullet(A) & \longrightarrow & C_\bullet(A + B) & \longrightarrow & C_\bullet(X \setminus Z, A \setminus Z) \longrightarrow 0 \\ & & \downarrow = & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C_\bullet(A) & \longrightarrow & C_\bullet(X) & \longrightarrow & C_\bullet(X, A) \longrightarrow 0 \end{array}$$

Considering that the rows are short exact sequences of chain complexes, they induce long exact sequences in homology by naturality in theorem 1.4.3:

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & H_n(A) & \longrightarrow & H_n(A + B) & \longrightarrow & H_n(X \setminus Z, A \setminus Z) & \longrightarrow & H_{n-1}(A) & \longrightarrow & H_{n-1}(A + B) \longrightarrow \cdots \\ & & \downarrow = & & \downarrow & & \downarrow & & \downarrow = & & \downarrow \\ \cdots & \longrightarrow & H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) & \longrightarrow & H_{n-1}(A) & \longrightarrow & H_{n-1}(X) \longrightarrow \cdots \end{array}$$

where the vertical arrows are naturally induced by inclusion. By proposition 8.2.3, the second the fifth arrows are isomorphisms. Thus by the five lemma, we have that

$$H_n(X \setminus Z, A \setminus Z) \cong H_n(X, A)$$

and so we are done. ■

Lemma 4.3.2 The Excision Theorem is equivalent to the following version: If $A, B \subset X$ are subspaces such that $X = A^\circ \cup B^\circ$, then the inclusion $(B, A \cap B) \rightarrow (X, A)$ induces isomorphisms

$$H_n(B, A \cap B) \xrightarrow{\cong} H_n(X, A)$$

for all n .

When X and A in relative homology is nice enough, we can interpret the result from relative homology as the homology groups of the quotient space. The following notation is due to Hatcher.

Definition 4.3.3 (Good Pairs) Let X be a space and A a closed subspace of X . We say that the pair (X, A) is a good pair if there exists an open set V such that $A \subset V$ and V deformation retracts to A .

It is clear by definition that any CW-complex X and subcomplex A forms a good pair.

Lemma 4.3.4 Both CW-complexes and Δ -complexes are good pairs.

Proposition 4.3.5 Let X be a space and A a closed subspace of X such that (X, A) is a good pair. Then the quotient map $X \rightarrow X/A$ induces a natural isomorphism

$$H_n(X, A) \cong \tilde{H}_n(X/A)$$

in homology groups together with the excision isomorphism.

Proof Let V be a open neighbourhood of A satisfying the good pair condition. Then there is an obvious inclusion map $(X, A) \rightarrow (X, V)$. Applying the long exact sequence of relative homology for both (X, A) and (X, V) , we obtain the following diagram by naturality in theorem 1.4.3:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & H_n(A) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, A) & \longrightarrow & H_{n-1}(A) & \longrightarrow & H_{n-1}(X) & \longrightarrow & \cdots \\ & & \downarrow \cong & & \downarrow \cong & & \downarrow & & \downarrow \cong & & \downarrow \cong & & \\ \cdots & \longrightarrow & H_n(V) & \longrightarrow & H_n(X) & \longrightarrow & H_n(X, V) & \longrightarrow & H_{n-1}(V) & \longrightarrow & H_{n-1}(X) & \longrightarrow & \cdots \end{array}$$

where the isomorphisms come from the fact that V deformation retracts onto A since (X, A) is a good pair. By the five lemma, we obtain an isomorphism

$$H_n(X, A) \cong H_n(X, V)$$

By excision, we obtain another isomorphism

$$H_n(X, A) \cong H_n(X, V) \cong H_n(X \setminus A, V \setminus A)$$

Repeating the argument with $(X/A, A/A)$, we obtain an isomorphism

$$H_n(X/A, A/A) \cong H_n(X \setminus A, V \setminus A) \cong H_n\left(\left(\frac{X}{A}\right) \setminus \left(\frac{A}{A}\right), \left(\frac{V}{A}\right) \setminus \left(\frac{A}{A}\right)\right)$$

Combining the two gives the desired result. ■

Most pairs of spaces are good pairs. We provide some examples that are not.

The Hawaiian earrings with its wedge point is not a good pair since any open neighbourhood of the wedge point contains an infinite number of circles and so cannot be contractible.

The interval together with $A = \{\frac{1}{n} \mid n \in \mathbb{N} \setminus \{0\}\}$ is not a good pair. In fact, one can show that the pair $(X/A, A/A)$ is homeomorphic to the Hawaiian earring with the wedge point.

Proposition 4.3.6 Let $k \geq 0$, we have that

$$H_n(\Delta^k, \partial\Delta^k) = \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{otherwise} \end{cases}$$

Proof Since Δ^k is also a CW-complex, $(\Delta^k, \partial\Delta^k)$ is a good pair thus we can apply proposition 6.2.4 to obtain

$$\tilde{H}_n(\Delta^k, \partial\Delta^k) \cong \tilde{H}_n(S^k)$$

and so the desired results follows. ■

Lemma 4.3.7 Assume that each (X_i, x_i) is a good pair. Then we have an isomorphism

$$\tilde{H}_n\left(\bigvee_{i \in I} (X_i, x_i)\right) = \bigoplus_{i \in I} \tilde{H}_n(X_i)$$

in reduced homology.

4.4 Local Homology Groups

Local homology groups forgets every homological information about the space X , under than a small neighbourhood around some subset $A \subseteq X$.

Definition 4.4.1 (The Local Homology Groups) Let X be a space. Let $A \subseteq X$ be a subspace. Define the local homology group of X in A to be the homology groups

$$H_n(X | A) = H_n(X, X \setminus A)$$

for each $n \in \mathbb{N}$.

Elements of the local homology group are represented by cycles in X whose boundary lie outside of x . It is called local because it only captures local topological data of X surrounding x .

Proposition 4.4.2 For all $n, k \in \mathbb{N}$, there is an isomorphism

$$H_n(\mathbb{R}^k | \{*\}) \cong \begin{cases} \mathbb{Z} & \text{if } n = k \\ 0 & \text{if } n \neq k \end{cases}$$

Proof Now if $k = 0$, the results are clear. If $k \geq 1$, then the long exact sequence of the pair $(\mathbb{R}^k, \mathbb{R}^k \setminus *)$ together with the fact that $\mathbb{R}^k \setminus * \simeq S^{k-1}$ and $\mathbb{R}^k \simeq *$ gives

$$H_n(\mathbb{R}^k | *) = 0$$

for $n > k$ and $n < k$. When $n = k$, we have an exact sequence

$$0 \longrightarrow H_k(\mathbb{R}^k | *) \longrightarrow H_{k-1}(S^{k-1}) \longrightarrow H_{k-1}(\mathbb{R}^k)$$

when $k > 1$ since $H_{k-1}(\mathbb{R}^k) = 0$. Thus $H_k(\mathbb{R}^k | *) \cong \mathbb{Z}$. If $k = 1$, then the last map $H_0(S^0) \rightarrow H_0(\mathbb{R})$ is given by the matrix $\begin{pmatrix} 1 & 1 \end{pmatrix} : \mathbb{Z}^2 \rightarrow \mathbb{Z}$ thus also giving isomorphism. ■

The excision theorem gives the following stronger version of invariance of domain.

Corollary 4.4.3 Let $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ be non empty open subsets. If $U \cong V$ then $m = n$.

Proof Let $x \in U$. By excision, we obtain an isomorphism

$$H_k(U | \{x\}) \cong H_k(\mathbb{R}^m | \{x\}) \cong \begin{cases} \mathbb{Z} & \text{if } m = k \\ 0 & \text{if } m \neq k \end{cases}$$

and similarly for $H_k(V | \{\phi(x)\})$ if $\phi : U \rightarrow V$ gives the homeomorphism. Then it is clear that if the homology groups are equal, then $m = n$. ■

This shows that no connected manifold can have atlases mapping to different dimensions of $\mathbb{R}$ so that the notion of dimension is well defined.

5 Degree Theory and Cellular Homology

5.1 Degree of Continuous Maps

The degree is a main component of the explicit construction of maps in cellular homology. Therefore we develop some of the basic theory here. Degree theory is also an important subject in its own right.

Definition 5.1.1 (Degree of a Continuous Map) Let $f : S^k \rightarrow S^k$ be a continuous map. Let

$$f_* : \tilde{H}_k(S^k) \cong \mathbb{Z} \rightarrow \tilde{H}_k(S^k) \cong \mathbb{Z}$$

be induced by f and let $f_*(1) = n$. Define the degree of f to be $\deg(f) = n$.

Lemma 5.1.2 Let $f, g : S^k \rightarrow S^k$ be continuous maps. Then the following are true regarding the degree.

- $\deg(\text{id}) = 1$
- $\deg(g \circ f) = \deg(g) \cdot \deg(f)$
- If $f \simeq g$ then $\deg(g) = \deg(f)$
- If f is a homotopy equivalence then $\deg(f) = \pm 1$
- If f is not surjective then $\deg(f) = 0$.

Proof

- The identity map on the k -sphere induces the identity map on the homology groups.
- Direct consequence of functoriality of the induced map.
- Direct consequence of homotopy invariance.
- If f is a homotopy equivalence that $f \simeq \text{id}$. By the third and first point we are done.
- Let $x \in S^k$ not be in the image of f . Then we obtain a factorization of f .

$$\begin{array}{ccc} S^k & \xrightarrow{f} & S^k \\ & \searrow & \nearrow \\ & S^k \setminus \{x\} & \end{array}$$

Passing to reduced homology, we see that f_* factors through 0:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{f_*} & \mathbb{Z} \\ & \searrow & \nearrow \\ & 0 & \end{array}$$

so that f_* is the 0 map.

and so we conclude. ■

It is easy to compute all the possible degrees when $k = 0$ or 1. When $k = 0$, there are two points to map to and so there are two obvious maps: the identity and the continuous map taking one point to the other. Any other map must be homotopic to one of these two thus the only possible degree maps of S^0 is just the identity with degree 0 and the non-trivial map with degree 1.

When $k = 1$, the set of maps $f : S^1 \rightarrow S^1$ defined by $z \mapsto z^n$ has degree n . Indeed from Algebraic Topology 1 we have seen that every $\omega_n : I \rightarrow S^1$ defined by $t \mapsto e^{2\pi i n t}$ induces a map of sending the generator in $H_1(S^1) \cong \pi_1(S^1, 1)$ to n times the generator. Explicitly, if $\sigma : I \rightarrow S^1$ is the generator in $H_1(S^1)$ defined by $\sigma(t) = e^{2\pi i t}$, then $f \circ \sigma$ is homologous to $n\sigma$ again via $\pi_1(S^1, 1) \cong H_1(S^1)$.

Proposition 5.1.3 For all $k \geq 1$ and for all $n \in \mathbb{Z}$, there exists a continuous map $f : S^k \rightarrow S^k$ of degree n .

Proof Define $SX = \frac{X \times [-1, 1]}{X \times \{-1\}, X \times \{1\}}$ (the suspension of X). The two open sets

$$C_+X = \frac{X \times (\epsilon, 1]}{X \times \{1\}} \quad \text{and} \quad C_-X = \frac{X \times [-1, \epsilon)}{X \times \{-1\}}$$

are contractible. By Mayer-Vietoris sequence, we obtain an isomorphism

$$H_{k+1}(SX) \xrightarrow{\partial} H_k(X)$$

It is clear that every map $f : X \rightarrow Y$ induces a map $Sf : SX \rightarrow SY$ by $Sf(x, t) = (f(x), t)$. Now notice that the following diagram

$$\begin{array}{ccc} H_{k+1}(SX) & \xrightarrow{Sf_*} & H_{k+1}(SY) \\ \partial \downarrow & & \downarrow \partial \\ H_k(X) & \xrightarrow{f_*} & H_k(Y) \end{array}$$

commutes by naturality in theorem 1.4.3. Applying this with $X = Y = S^{k-1}$ and since $S(S^{k-1}) \cong S^k$, we deduce that $\deg(Sf) = \deg(f)$ so if we use induction, we are done. But the base case is already treated by the above paragraph, and so we conclude. ■

Using the fundamental class of a sphere, we can compute the degree of a reflection.

Lemma 5.1.4 Let $S^k \subseteq \mathbb{R}^{k+1}$ be the unit circle. Let $f : S^k \rightarrow S^k$ be the reflection of a hyperplane through 0 in $\mathbb{R}^{k+1}$. Then $\deg(f) = -1$.

Proof The top S^k_+ and bottom S^k_- half of the sphere cut by the hyperplane are homeomorphic to k -simplices. Choose k -simplices $\sigma_+ : \Delta^k \xrightarrow{\cong} S^k_+$ and $\sigma_- : \Delta^k \xrightarrow{\cong} S^k_-$ such that $\partial\Delta^k$ is mapped homeomorphically to the equator $S^k_+ \cap S^k_-$ in both cases, and that the composition

$$\partial\Delta^k \xrightarrow{\sigma_+} S^k_+ \cap S^k_- \xrightarrow{(\sigma_-)^{-1}} \partial\Delta^k$$

is the identity.

It is clear that it is a cycle since it is a boundary in the chain complex $C_\bullet(\Delta^{k+1})$. I claim that $[\sigma_+ - \sigma_-]$ is a generator of $H_k(S^k)$.

We proceed by induction. When $k = 0$, the statement is clear. So suppose that $k > 0$. Let U_1, U_2 be open subspaces of Δ^{k+1} as follows. U_1 is an open neighbourhood of the last face of $\partial_{k+1}\Delta^{k+1}$ which deformation retracts onto $\partial\Delta^{k+1}$. U_2 is an open neighbourhood of the remaining faces $U_2 = \bigcup_{i=0}^k \partial_i\Delta^{k+1}$ which deformation retracts onto $\bigcup_{i=0}^k \partial_i\Delta^{k+1}$. Moreover, choose them in such a way that $U_1 \cap U_2$ deformation retract onto $\partial\Delta^{k+1} = \partial[v_0, \dots, v_k]$ and $U_1 \cup U_2$ deformation retracts onto $\partial[v_0, \dots, v_{k+1}]$. By induction hypothesis, we know that $\tilde{H}_{k-1}(U_1 \cap U_2) \cong \mathbb{Z}$ is generated by

$$\partial([v_0, \dots, v_k]) = \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

From Mayer-Vietoris sequence, since $U_1 \cup U_2$ deformation retracts onto $\partial\Delta^{k+1}$, the connecting homomorphism

$$\tilde{H}_k(U_1 \cup U_2) \rightarrow \tilde{H}_{k-1}(U_1 \cap U_2)$$

since U_1 and U_2 are contractible so we only need to show that $\partial\sigma$ is sent to the generator or its negative.

For this we will explicitly compute the connecting homomorphism. It is clear that

$$\left((-1)^{k+1} [v_0, \dots, v_k], \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_{k+1}] \right) \in C_k(U_1) \oplus C_k(U_2)$$

is such that it is a lift of the cycle $\partial\sigma$. Its image under the connecting homomorphism is the unique $(k-1)$ -cycle τ in $U_1 \cap U_2$ which satisfies

$$((i_1)_*(\tau), -(i_2)_*(\tau)) = \left((-1)^{k+1} \partial([v_0, \dots, v_k]), \sum_{i=0}^k (-1)^i \partial([v_0, \dots, \hat{v}_i, \dots, v_{k+1}]) \right)$$

It is clear that $\tau = (-1)^{k+1} \partial([v_0, \dots, v_k])$ is a generator in $\tilde{H}_k(U_1 \cap U_2)$.

We have seen that $s = [\sigma_+ - \sigma_-]$ generates $H_k(S^k)$. Thus

$$f_*(s) = [f \circ \sigma_+] - [f \circ \sigma_-] = [\sigma_-] - [\sigma_+] = -f_*(s)$$

and thus the degree of f is -1 since it send a the generator to the negative of itself. $\blacksquare$

Proposition 5.1.5 Let $T : \mathbb{R}^{k+1} \rightarrow \mathbb{R}^{k+1}$ be a linear orthogonal transformation. The restriction of T to the homeomorphism

$$f : S^k \rightarrow S^k$$

has degree $\deg(f) = \det(T)$.

Proof Choose an orthonormal basis for $\mathbb{R}^{k+1}$ such that T can be represented by a block sum matrix where each block is either a 2×2 rotation or a 1×1 matrix with entry ± 1 . This is possible from the notes Geometry. Any rotation is homotopic to the identity by undoing the rotation by rewinding. Thus T is homotopic to a diagonal matrix with m entries having -1 and $k+1-m$ entries having 1 . But then T is just the composition of m reflections and $\det(T) = (-1)^m$. By the above lemma, we conclude. $\blacksquare$

As a result, we can compute the degree of the following two maps.

Corollary 5.1.6 Let $f : S^k \rightarrow S^k$ be the antipodal map. Then $\deg(f) = (-1)^{k+1}$.

Proof The antipodal map is a composition of $n+1$ reflections. By the above proposition we conclude. $\blacksquare$

Corollary 5.1.7 Let $f : S^k \rightarrow S^k$ have no fixed points. Then $\deg(f) = (-1)^{k+1}$.

Proof The line through $f(x)$ and $-x$ passes through the origin if and only if $f(x) = x$. Since f has no fixed points, the line never passes through the origin so that the map $g : S^k \times I \rightarrow S^k$ defined by

$$g(x, t) = \frac{tf(x) + (1-t)(-x)}{\|tf(x) + (1-t)(-x)\|}$$

defines a homotopy from f to the antipodal map. By homotopy invariance and the above corollary, we conclude that $\deg(f) = (-1)^{k+1}$. $\blacksquare$

We can now prove a famous result from manifold theory. Recall that a tangent vector field $v : M \rightarrow TM$ on a smooth manifold is a vector field such that $v(x)$ and x is orthogonal.

Theorem 5.1.8 (Hairy Ball Theorem) Every tangent vector field on an even-dimensional sphere vanishes at some point.

Proof Assume the contrary. This means that there exists a vector field $v : S^k \rightarrow \mathbb{R}^{k+1}$ that is non-zero everywhere, where k is even. Consider the map $g : S^k \times I \rightarrow \mathbb{R}^{k+1}$ defined by

$$g(x, t) = \cos(\pi t)x + \sin(\pi t)v(x)$$

Since x and $v(x)$ are orthogonal and $v(x) \neq 0$, the two vectors x and $v(x)$ are linearly independent. It follows that the map lands in $\mathbb{R}^{k+1} \setminus \{0\}$ and we can divide by the norm to obtain a homotopy $S^k \times I \rightarrow S^k$. This homotopy is in fact a homotopy from the identity to the antipodal map. They have degree 1 and -1 respectively. But by homotopy invariance of the degree, it is a contradiction. ■

5.2 Local Degree

It is not at all obvious that given a map of spheres, how one would compute the degree of the map. We can do this by computing locally what the degree of the map looks like. These are called the local degree of f .

Definition 5.2.1 (Local Degree) Let $f : S^k \rightarrow S^k$ be a continuous map. Assume that $f^{-1}(y) = \{x_1, \dots, x_n\}$. Let U_i be an open neighbourhood of x_i such that $U_i \cap U_j = \emptyset$ for each $i \neq j$. Define the local degree of f to be the degree of the map

$$f|_{x_i} : H_k(U_i | x_i) \rightarrow H_k(S^k | y)$$

denoted by $\deg(f|_{x_i})$.

Notice that this indeed makes sense. Indeed by excision, we have an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(U_i, U_i \setminus \{x_i\})$$

by excising the piece $S^k \setminus U$. By using excision again, we obtain an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(\mathbb{R}^k, \mathbb{R}^k \setminus *)$$

Indeed S^k is a smooth manifold and so there exists some $V \subseteq S^k$ which maps homeomorphically to an open subset of $\mathbb{R}^k$. Excision then gives

$$H_k(S^k \setminus (S^k \setminus V), (S^k \setminus \{x\}) \setminus (S^k \setminus V)) \cong H_k(S^k, S^k \setminus \{x\})$$

This translates to $H_k(V, V \setminus \{x\}) \cong H_n(\mathbb{R}^k, \mathbb{R}^k \setminus *)$ so that we now have

$$H_k(S^k, S^k \setminus \{x_i\}) \cong \mathbb{Z}$$

by proposition 6.2.2. However, there is an obvious isomorphism which makes the definition clearer.

Lemma 5.2.2 The local degree of a map $f : S^k \rightarrow S^k$ is well defined.

Proof By excision, we have an isomorphism

$$H_k(S^k, S^k \setminus \{x_i\}) \cong H_k(U_i, U_i \setminus \{x_i\})$$

by excising the piece $S^k \setminus U$.

By using the long exact sequence for relative homology, we have that

$$\dots \longrightarrow H_k(S^k \setminus \{x_i\}) \longrightarrow H_k(S^k) \longrightarrow H_k(S^k, S^k \setminus \{x_i\}) \longrightarrow H_{k-1}(S^k \setminus \{x_i\}) \longrightarrow \dots$$

But the first and latter terms are 0 since $S^k \setminus \{x_i\} \simeq \mathbb{R}^k$ so that we have an isomorphism

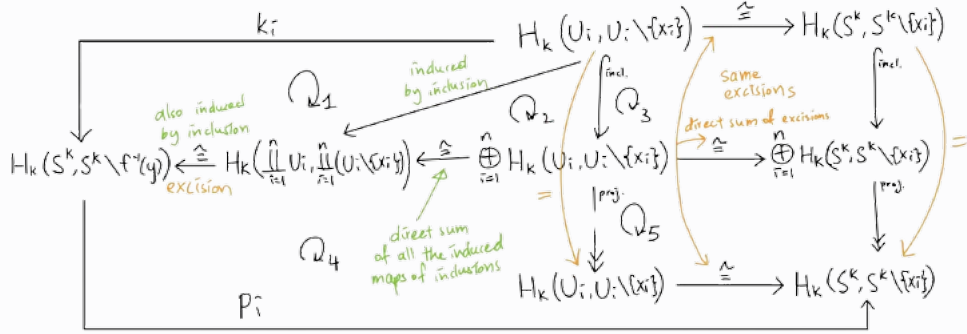
$$H_k(S^k) \cong H_k(S^k, S^k \setminus \{x_i\})$$

This is true for both the domain and the codomain of $f|_{x_i}$. ■

Proposition 5.2.3 Let $f : S^k \rightarrow S^k$ be a map. Let $f^{-1}(y) = \{x_1, \dots, x_n\}$. Then

$$\deg(f) = \sum_{i=1}^n \deg(f|_{x_i})$$

Proof Let U_i be an open neighbourhood of x_i for which $U_i \cap U_j = \emptyset$ for $i \neq j$. Let $f(U_i) \subseteq V$ for all i so that V is a neighbourhood of y . Consider the following diagram.

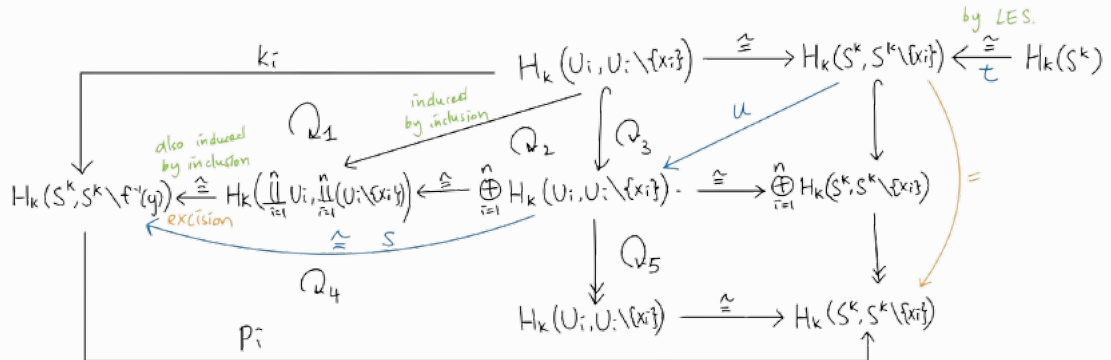


$k_i : H_k(U_i, U_i \setminus \{x_i\}) \rightarrow H_k(S^k, S^k \setminus f^{-1}(y))$ is by definition of the composition of the obvious maps and so the upper left part of the diagram is commutativity. The upper middle part of the diagram commutes since both are induced by inclusions and a direct sum of the induced maps of inclusions. The upper right and lower right part are commutative since they are induced by the same excisions and the direct sum of the very same excisions. Finally, $p_i : H_k(S^k, S^k \setminus f^{-1}(y)) \rightarrow H_k(S^k, S^k \setminus \{x_i\})$ is defined to be the composition of the obvious maps and so the bottom left diagram is commutative by definition.

All of the above shows that the upper left of the following diagram commutes:

$$\begin{array}{ccccc}
 & & H_k(U_i, U_i \setminus \{x_i\}) & \xrightarrow{f_* = \times \deg(f|_{x_i})} & H_k(V, V \setminus \{y\}) \\
 & \nwarrow \cong & \downarrow k_i & & \downarrow \cong \\
 H_k(S^k, S^k \setminus \{x_i\}) & \xleftarrow{p_i} & H_k(S^k, S^k \setminus f^{-1}(y)) & \xrightarrow{f_*} & H_k(S^k, S^k \setminus \{y\}) \\
 & \nwarrow \cong & \uparrow j_i & & \uparrow \cong \\
 & & H_k(S^k) & \xrightarrow{f_* = \times \deg(f)} & H_k(S^k)
 \end{array}$$

The bottom left of this diagram is commutative because:



and $s \circ u \circ t$ is the map j . The isomorphism on the top right is obtained by excision. k_i is induced by the inclusion $(S^k, S^k \setminus \{x_i\}) \hookrightarrow (S^k, S^k \setminus f^{-1}(y))$ and then by the very same excision

$$H_k(U_i, U_i \setminus \{x_i\}) \cong H_k(S^k, S^k \setminus \{x_i\})$$

so that the top right square commutes. The isomorphism on the bottom right is given by the long exact sequence for relative homology. It is commutative by the naturality of long exact sequences for relative homology.

By treating the middle term as

$$H_k(S^k, S^k \setminus f^{-1}(y))$$

one can see that k_i is just the inclusion into the direct summands. Similarly, using the first diagram of the proof we see that some part of p_i is a projection of the direct summands. Commutativity of the third diagram shows that $p_i(j(1)) = 1$. Since p_i is the projection from the direct summand, we must have that $p_i(0, \dots, 0, 1, 0, \dots, 0) = 1$ which shows that $j(1) = (1, \dots, 1)$. At the same time, k_i is the inclusion into the direct summands thus $(1, \dots, 1) = \sum_{i=1}^n k_i(1)$. Thus we conclude that $j(1) = \sum_{i=1}^n k_i(1)$.

Now recall that $\deg(f) = f_*(1)$. By the lower part of the second diagram, we have that

$$f_*(1) = f_*(j(1)) = f_*\left(\sum_{i=1}^n k_i(1)\right) = \sum_{i=1}^n f_*(k_i(1))$$

By the top part of the second diagram, we conclude that $f_*(k_i(1)) = \deg(f|_{x_i})$ for each i . Thus we have shown that

$$\deg(f) = \sum_{i=1}^n \deg(f|_{x_i})$$

and so we are done. ■

5.3 Cellular Homology

Singular homology is not often easy to calculate. A more combinatorial result comes from cellular homology, which actually gives the same result as singular homology.

Recall that CW complexes and subcomplexes always forms good pairs.

Lemma 5.3.1 Let X be a CW-complex with n -cells $\{D_\alpha^n\}$ for $n \geq 0$. Then

$$H_k(X^n, X^{n-1}) \cong \begin{cases} \bigoplus_\alpha \mathbb{Z} \cdot \{D_\alpha^n\} & \text{if } k = n \\ 0 & \text{otherwise} \end{cases}$$

Proof We have that

$$H_k(X^n, X^{n-1}) \cong \tilde{H}_k(X^n/X^{n-1}) \cong \tilde{H}_k\left(\bigvee_\alpha D_\alpha^n/\partial D_\alpha^n\right) \cong \bigoplus_\alpha \tilde{H}_k(D_\alpha^n/\partial D_\alpha^n)$$

and so we conclude. ■

In fact, one can explicitly describe the isomorphism as follows. For D_α^n in the n -skeleton in X , it is isomorphic to Δ^n an n -simplex. Thus we have a continuous map

$$\Delta^n \cong D_\alpha^n \xrightarrow{\Phi_\alpha} X$$

which is in fact a relative cycle for the pair (X^n, X^{n-1}) . Its relative homology class then generates the copy of $\mathbb{Z}$ corresponding to D_α^n .

Lemma 5.3.2 Let X be a CW-complex. Then the following are true regarding the singular homology of X .

- If X is of dimension k then $H_n(X) = 0$ for all $n > k$.
- The map $H_n(X^m) \rightarrow H_n(X)$ induced by the inclusion $X^m \rightarrow X$ is an isomorphism if $n < m$ and surjective if $m = n$.

Proof We proceed by induction. When $k = 0$, it is obvious. Assume the statement is true for $k - 1$. Consider the long exact sequence for relative homology:

$$\cdots \longrightarrow H_{n+1}(X^k, X^{k-1}) \longrightarrow H_n(X^{k-1}) \longrightarrow H_n(X^k) \longrightarrow H_n(X^k, X^{k-1}) \longrightarrow \cdots$$

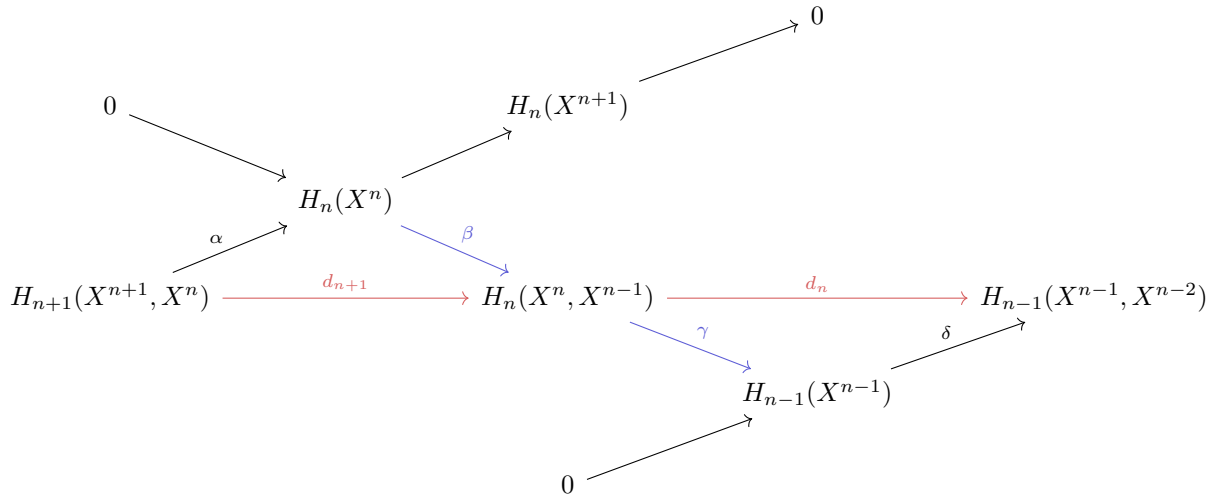
By induction, the second term vanishes for $n > k$. By the previous lemma, the last term vanishes for $n > k$. Thus we obtain the required isomorphisms.

Similarly, by the above long exact sequence, when $n < k - 1$, both outer terms vanish and the middle arrow becomes an isomorphism. For $n < k$, the last term vanishes and the middle arrow is a surjection. Thus for $n < m$, we have isomorphisms:

$$H_n(X^m) \longrightarrow H_n(X^{m+1}) \longrightarrow H_n(X^{m+2}) \longrightarrow \cdots$$

By properties of the geometric realization, we must have that every compact subspace of X must lie inside some X^t for $t \in \mathbb{N}$. For any $\tau \in C_{n+1}(X)$, we thus have $\tau \in C_{n+1}(X^t)$ for some $t > n$. Hence τ maps to 0 in $H_n(X^t)$ if τ is an n -cycle. Thus all these groups must be equal to $H_n(X)$. On the other hand, if $n = m$, then the first arrow $H_n(X^m) \rightarrow H_n(X^{m+1})$ is only surjective while the remaining ones are isomorphisms. Thus we are done. ■

Using the above lemma, we now have a spliced diagram :



where the diagonals come from the long exact sequence in the proof of the first part of lemma 7.4.3. The d_n is then defined through the composition of the diagonal arrows.

Lemma 5.3.3 The composition $d_n \circ d_{n+1}$ is zero.

Proof The composition $d_n \circ d_{n+1}$ factors through the two blue arrows in the diagram. They are part of a long exact sequence and so their composition is 0. ■

Definition 5.3.4 (Cellular Chain Complex) Let X be a CW-complex. Define the cellular chain complex $C_\bullet^{\text{CW}}(X)$ where

$$C_n^{\text{CW}}(X) = H_n(X^n, X^{n-1})$$

together with differentials d_n as defined above. Define the cellular homology groups of this chain complex to be

$$H_n^{\text{CW}}(X) = H_n(C_\bullet^{\text{CW}}(X))$$

Lemma 5.3.5 For any CW-complex X , there are canonical isomorphisms

$$H_n^{\text{CW}}(X) \cong H_n(X)$$

Proof Since δ is injective, we have that

$$\ker(d_n) = \ker(\gamma) = \text{im}(\beta)$$

Also by lemma 7.3.2, we have that $H_n(X^{n+1}) \cong H_n(X)$. This means that

$$H_n(X) \cong \text{coker}(\alpha) \cong \frac{\text{im}(\beta)}{\text{im}(\beta \circ \alpha)} = \frac{\ker(d_n)}{\text{im}(d_{n+1})}$$

where the second isomorphism comes from the fact that β is injective. ■

Recall that a CW-complex X is defined recursively through the use of attaching maps $\{\phi_\alpha : S_\alpha^{n-1} \rightarrow X^{n-1} | \alpha \in I_n\}$ for each n , such that

$$X^n = \left(X^{n-1} \amalg \coprod_{\alpha \in I_n} D_\alpha^n \right) / \sim$$

where the equivalence relation is $x \sim \phi_\alpha(x)$ for all $x \in \partial D_\alpha^n$ (This is an amalgamated product over $\phi_\alpha(x)$). Notice that we have

$$\frac{X^n}{X^{n-1}} \cong \bigvee_{\alpha \in I_n} \frac{D_\alpha^n}{\partial D_\alpha^n} = \bigvee_{\alpha \in I_n} S_\alpha^n$$

implies that there are canonical quotient maps obtain by collapsing all sphere other than one into a single point by an equivalence relation:

$$\pi_\alpha : X^n \twoheadrightarrow \frac{X^n}{X^{n-1}} \cong \bigvee_{\alpha \in I_n} S_\alpha^n \twoheadrightarrow S_\alpha^n$$

To compute the cellular homology of a CW complex, we already know that the chain complex has terms given by lemma 7.3.2. It remains to compute what the differentials look like. We will use this map in the following formula.

Theorem 5.3.6 (Cellular Boundary Formula) Let X be a CW-complex with attaching maps denoted ϕ_α , and $C_\bullet^{\text{CW}}(X)$ the cellular chain complex of X with generators $\{[D_\alpha^n] | \alpha \in I_n\}$ for each degree n . The boundary operator of the chain complex is given by the following formula:

- In degree $n = 1$, we have

$$d_1([D_\alpha^1]) = [\phi_\alpha(1)] - [\phi_\alpha(0)]$$

- In degrees $n > 1$, we have

$$d_n([D_\alpha^n]) = \sum_{\beta \in I_{n-1}} d_{\alpha\beta} \cdot [D_\beta^{n-1}]$$

where

$$d_{\alpha\beta} = \deg \left(\Delta_{\alpha\beta} : S_\alpha^{n-1} \xrightarrow{\phi_\alpha} X^{n-1} \xrightarrow{\pi_\beta} S_\beta^{n-1} \right)$$

On computation, one crucial fact to notice is that the map π_β is a projection to the quotient topology. This means that after applying π_β , the map

$$\Delta_{\alpha\beta} = \pi_\beta \circ \phi_\alpha : S_\alpha^{n-1} \rightarrow S_\beta^{n-1}$$

forgets about every boundary of the attached discs to X^n other than the boundary of the disc D_β^n . Also notice that $\alpha \in I_n$ loops over all attaching maps and disks on the n th dimension, while $\beta \in I_{n-1}$ loops over that of the $n - 1$ dimension. Notice that in the formula, β is used to indicate the S_β^{n-1} which are of dimension n . This makes sense because in π_β , there is a projection $X^{n-1} \rightarrow X^{n-1}/X^{n-2}$.

$$\frac{X^{n-1}}{X^{n-2}} = \bigvee_{\beta \in I_{n-1}} \frac{D_\beta^{n-1}}{\partial D_\beta^{n-1}}$$

is then a wedge sum of all disks in dimension $n - 1$ modulo boundary.

We can now compute the singular homology of more complicated spaces such as the projective space.

Example 5.3.7 Let $n \geq 1$. Then the real projective space $\mathbb{RP}^n$ has the following homology groups:

$$H_k(\mathbb{RP}^n) = \begin{cases} \mathbb{Z} & \text{if } k = 0 \text{ and } k = n \text{ is odd} \\ \mathbb{Z}/2\mathbb{Z} & \text{if } 0 < k < n \text{ and } k \text{ is odd} \\ 0 & \text{otherwise} \end{cases}$$

Proof Recall that the CW complex structure of $\mathbb{RP}^n$ consists of one k -cell for each $0 \leq k \leq n$. We have to determine the boundary operator. This is done by induction.

The base case is obtained easily since $S^1 \cong \mathbb{RP}^1$.

Suppose that the attaching maps are defined for all lower values of k . Recall that the attaching map of each k -cell is defined by the two-to-one map $f : S^{k-1} \rightarrow \mathbb{RP}^{k-1}$. This forms the first part of the map $\Delta_{\alpha\beta}$ of the formula for the cellular boundary maps. To proceed, we quotient out the X^{k-2} skeleton to obtain

$$S^{k-1} \rightarrow \mathbb{RP}^{k-1} \rightarrow \frac{\mathbb{RP}^{k-1}}{\mathbb{RP}^{k-2}} \cong S^{k-1}$$

We compute the degree of this map using local degrees. Since it is a two-to-one covering map, we concern ourselves with the north / south pole of S^{k-1} . Choose your favourite homeomorphism $D^{k-1}/\partial D^{k-1} \cong S^{k-1}$. Then the map $f : S^{k-1} \rightarrow \mathbb{RP}^{k-1}$ is a local homeomorphism since it is a covering space. Thus the degree of the map is ± 1 , fixed by the choice of the homeomorphism. This is similar for the south pole. In fact, the map is identical to that of the local homeomorphism at the north pole and so the local homeomorphism is just the antipodal map composed with the local homeomorphism at the north pole.

Denote N and S the north and south pole respectively. Since $\deg(f|_N) = \pm 1$ is fixed and the antipodal map has degree $(-1)^k$, we obtain

$$\deg(f) = \deg(f|_N) + (-1)^n \deg(f|_N) = \begin{cases} \pm 2 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

Thus the cellular chain complex together with induction, is of the form

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_k} \mathbb{Z} \xrightarrow{d_{k-1}} \cdots \xrightarrow{\pm 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{\pm 2} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \longrightarrow 0$$

where $d_k = \pm 2$ if k is even and $d_k = 0$ if k is odd. It follows that

$$\deg(f) = \deg(f|_N) + (-1)^n \deg(f|_N) = \begin{cases} \pm 2 & \text{if } n \text{ is even} \\ 0 & \text{if } n \text{ is odd} \end{cases}$$

■

6 Variants of Homology

6.1 The Tensor Product Functor

Let R be a ring. Let M, N be R -modules. Recall that we defined the tensor product of M and N via the universal property in Rings and Modules: The tensor product is an R -module $M \otimes_R N$ such that for any R -bilinear map $\varphi : M \times N \rightarrow L$ where L is an R -module, there exists a unique map $\rho : M \otimes_R N \rightarrow L$ such that

$$\varphi = \rho \circ u$$

where $u : M \times N \rightarrow M \otimes_R N$ is the map $(m, n) \mapsto (m \otimes n)$.

We specialize the case to where $R = \mathbb{Z}$. In this case, M and N are just abelian groups.

Definition 6.1.1 (The Tensor Product Functor) Let G be an abelian group. Define the tensor product functor over G to be the functor

$$- \otimes G : \mathbf{Ab} \rightarrow \mathbf{Ab}$$

to consist of the following data.

- For each abelian group $H \in \mathbf{Ab}$, the functor assigns the abelian group $H \otimes G$.
- For each group homomorphism $f : A \rightarrow B$, define the induced homomorphism $f \otimes G : A \otimes G \rightarrow B \otimes G$ by

$$a \otimes g \mapsto f(a) \otimes g$$

Proposition 6.1.2 Let A, B, C, G be abelian groups. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be group homomorphisms. If $A \rightarrow B \rightarrow C \rightarrow 0$ is an exact sequence, then the sequence

$$A \otimes G \xrightarrow{f \otimes G} B \otimes G \xrightarrow{g \otimes G} C \otimes G \longrightarrow 0$$

is exact.

6.2 Homology with Coefficients

We would now like to generalize singular homology so that the coefficients of the chain complex does not take values in $\mathbb{Z}$, but instead in an arbitrary group G . For this, we first modify the group of n -chains.

Definition 6.2.1 (Singular n -Chains with Coefficients in a Group) Let X be a space. Let G be an abelian group. Define the group of singular n -chains with coefficients in G to be

$$C_n(X; G) = C_n(X) \otimes G$$

Let $A \subseteq X$ be a subspace. Define the relative n -chains with coefficients in G to be the group

$$C_n(X, A; G) = \frac{C_n(X; G)}{C_n(A; G)}$$

In particular, notice that the tensor product here just means that $C_n(X; G)$ is just copies of G , where the number of copies is precisely the number of singular n -simplexes in X .

Definition 6.2.2 (Boundary Operator) Let X be a space. Let G be an abelian group. Define the boundary operator of the singular chain complex with coefficients in G to be

$$\partial_G = \partial_n \otimes 1_G : C_n(X; G) \rightarrow C_{n-1}(X; G)$$

defined by $\sigma \otimes g \mapsto \partial_n(\sigma) \otimes g$ and then extended linearly.

Lemma 6.2.3 Let X be a space. Let G be an abelian group. Then $\partial_G \circ \partial_G = 0$. Moreover, the groups $C_n(X; G)$ together with the boundary operator ∂_G is a chain complex.

Definition 6.2.4 (Singular Homology with Coefficients) Let X be a space. Let G be an abelian group. Define the singular homology groups of X to be the homology groups of the chain complex $(C_\bullet(X; G), \partial_G)$. This means that

$$H_n(X; G) = \frac{\ker(\partial_G : C_n(X; G) \rightarrow C_{n-1}(X; G))}{\operatorname{im}(\partial_G : C_{n+1}(X; G) \rightarrow C_n(X; G))} = H_n(C_\bullet(X; G))$$

for $n \in \mathbb{N}$.

A natural question would be to ask whether there is a relation between homology with integer coefficients $H_n(X)$ and homology with arbitrary coefficients $H_n(X; G)$. Namely, the chain complex for singular homology with coefficients simply replaces the integral coefficients of the original singular chain complex via the tensor product. Is there a natural association between $H_n(X) \otimes G$ and $H_n(X; G)$?

The answer is yes, but it is not an isomorphism. Moreover, there is a way to measure the failure of the isomorphism in a natural and functorial way. This will be delayed to later sections.

6.3 Axioms for Unreduced Homology

Definition 6.3.1 (Generalized Homology Theory for CW Pairs) A Generalized Homology Theory is a collection of functors and natural transformations

$$h_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta_n : h_n \Rightarrow h_n \circ F$$

where $F(X, Y) = (Y, \emptyset)$, for each $n \in \mathbb{N}$, such that the following are true.

- Homotopy Invariance: If $f \simeq g : (X, A) \rightarrow (Y, B)$ then

$$h_n(f) = h_n(g) : h_n(X, A) \rightarrow h_n(Y, B)$$

- Exactness: There exists an exact sequence

$$\cdots \longrightarrow h_{n+1}(X, A) \xrightarrow{\delta_{n+1}} h_n(A, \emptyset) \xrightarrow{h_n(i)} h_n(X, \emptyset) \xrightarrow{h_n(j)} h_n(X, A) \xrightarrow{\delta_n} h_{n-1}(A, \emptyset) \longrightarrow \cdots$$

where $i : (A, \emptyset) \rightarrow (X, \emptyset)$ and $j : (X, \emptyset) \rightarrow (X, A)$ are inclusions.

- Excision: If $\overline{E} \subseteq A^\circ \subseteq X$, then there is an isomorphism

$$h_n(X \setminus E, A \setminus E) \cong h_n(X, A)$$

induced by the inclusion map.

The homotopy invariance axiom means that homology descends to a functor $\mathbf{HoCW}^2 \rightarrow \mathbf{Ab}$. Therefore some authors write the axiom implicit in the definition of the functors h_n , and then saying that a homology theory consists of functors $h_n : \mathbf{HoCW} \rightarrow \mathbf{Ab}$ instead.

Lemma 6.3.2 The excision axiom is equivalent to saying that $X = A^\circ \cup B^\circ$ with inclusion map $\iota : (B, A \cap B) \rightarrow (X, A)$ implies $h_n(\iota) : h_n(B, A \cap B) \rightarrow h_n(X, A)$ is an isomorphism.

Definition 6.3.3 (Additive Homology Theory) Let $h_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory. We say that h_n is additive if the following axiom is satisfied.

- Additivity: If $(X, A) = \coprod_{i \in I} (X_i, A_i)$, then the direct sum of the inclusion maps

$$\bigoplus_{i \in I} h_n(X_i, A_i) \cong h_n(X, A)$$

is an isomorphism

We mention for once and for all that the additivity axiom is required only when the CW complexes are non-finite. In particular, in order for the homology theory to be meaningful, we must restrict the underlying category of spaces to be finite CW complexes if one drops the additivity axiom.

7 Algebra of Cochain Complexes

7.1 Cochain Complexes

Definition 7.1.1 (Cochain Complexes) Let R be a ring. A cochain complex $(C^\bullet, \partial^\bullet)$ over R is a family of R -modules C^n for $n \in \mathbb{Z}$ and R -module homomorphisms $\delta^n : C^{n-1} \rightarrow C^n$ such that $\delta^{n+1} \circ \delta^n = 0$ for all n . In other words, we have the diagram:

$$\cdots \longleftarrow C^{n+1} \xleftarrow{\delta^{n+1}} C^n \xleftarrow{\delta^n} C^{n-1} \longleftarrow \cdots$$

Notice that algebraically, there is no difference between a chain complex and a cochain complex, other than the fact that the boundary maps run in the other direction. For $(C_\bullet, \partial_\bullet)$ a chain complex, we can form a cochain complex by setting $C^n = C_{-n}$ and then using the same boundary maps.

Definition 7.1.2 (Category of Cochain Complexes) Let R be a ring. Define the category of cochain complexes

$$\mathbf{CCh}(R)$$

to consist of the following data.

- The objects are the cochain complexes over R
- The morphisms are the chain maps
- Composition is given by the composition of maps on all levels of the chain map.

Because cochain complexes is the same data as a chain complex, we can also define homology groups for cochain complexes. We call them the cohomology groups of the cochain complex.

Definition 7.1.3 (Cohomology Groups) Let R be a ring. Let $(C^\bullet, \partial^\bullet)$ be a cochain complex over R . Define the n th cohomology group of the cochain complex to be

$$H^n(C^\bullet, \partial^\bullet) = \frac{\ker(\partial^{n+1} : C^n \rightarrow C^{n+1})}{\operatorname{im}(\partial^n : C^{n-1} \rightarrow C^n)} = H_n(C^\bullet, \partial^\bullet)$$

for $n \in \mathbb{N}$.

The construction of cohomology groups are functorial because the homology groups of a chain complex also are.

Definition 7.1.4 (The Cohomology Functor) Let R be a ring. Let $n \in \mathbb{N}$. Define the n th cohomology functor to be the functor

$$H^n : \mathbf{CCh}(R) \rightarrow \mathbf{Ab}$$

to consist of the following data.

- For each cochain complex $(C^\bullet, \delta^\bullet)$ over R , define $H^n(C^\bullet, \delta^\bullet)$ to be the n th cohomology group of the cochain complex.
- For each chain map $f^\bullet : (C^\bullet, \delta^\bullet) \rightarrow (D^\bullet, \delta^\bullet)$, the induced map

$$f^* : H^n(C^\bullet, \delta^\bullet) \rightarrow H^n(D^\bullet, \delta^\bullet)$$

is defined by $[z] \mapsto [f^n(z)]$.

Notice that obtaining cohomology from chain complexes is covariant.

7.2 Producing a Cochain Complex from a Chain Complex

Let R be a ring. Let M, N be R -modules. Recall that $\operatorname{Hom}_R(M, N)$ is only an abelian group, and at most a $Z(R)$ -module in general. However, when R is commutative, $\operatorname{Hom}_R(M, N)$ becomes an R -module. Here we treat the more general case and consider the case when R is a general ring.

Definition 7.2.1 (Dualization of a Chain Complex) Let R be a commutative ring. Let $(C_\bullet, \partial_\bullet)$ be a chain complex. Let M be an R -module. For each R -module C_n , define the associated cochain group to be

$$C^n = \text{Hom}_R(C_n, M)$$

Also define the coboundary map $\delta^n = \partial_n^* : C^{n-1} \rightarrow C^n$ by

$$\delta^n(\phi)(\alpha) = \phi(\partial_{n+1}(\alpha))$$

Lemma 7.2.2 Let $(C_\bullet, \partial_\bullet)$ be a chain complex and let $C^\bullet$ and $\delta^\bullet$ be the associated cochain groups and coboundary maps respectively. Then

$$\delta^n \circ \delta^{n-1} = 0$$

for all n such that $(C^\bullet, \delta^\bullet)$ is a cochain complex.

Definition 7.2.3 (The Hom Functor on Chain Complexes) Let R be a ring. Let M be an R -module. Define the (contravariant) Hom functor on chain complexes to be the functor

$$\text{Hom}_R(-; M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(\mathbb{Z})$$

consisting of the following data.

- For $(C_\bullet, \partial_\bullet)$ a chain complex, define a cochain complex by $C^n = \text{Hom}_R(C_n, M)$ and coboundary map by $\delta_n = \partial_n^*$
- For a chain map $f_\bullet : (C_\bullet, \delta_\bullet) \rightarrow (D_\bullet, \delta_\bullet)$, define a chain map $f^* : D^\bullet \rightarrow C^\bullet$ by $\varphi \mapsto \varphi \circ f$.

If R is a commutative ring and M, N are R -modules, then $\text{Hom}_R(M, N)$ can be given the structure of an R -module. Therefore the above Hom functor upgrades into a functor

$$\text{Hom}_R(-; M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(R)$$

Proposition 7.2.4 Let $C_\bullet, D_\bullet, E_\bullet \in \mathbf{Ch}(R)$ be chain complexes such that the short exact sequence

$$0 \longrightarrow C_\bullet \xrightarrow{f} D_\bullet \xrightarrow{g} E_\bullet \longrightarrow 0$$

of chain complexes is degree wise split (this means that $D_n \cong C_n \oplus E_n$ for all $n \in \mathbb{N}$). Then the dual of the chain complex

$$0 \longrightarrow E^\bullet \xrightarrow{g^*} D^\bullet \xrightarrow{f^*} C^\bullet \longrightarrow 0$$

is also a short exact sequence of chain complexes.

We have now defined a series of functors. We also recap on the material in section 1.

- We have defined the homology functor of chain complexes

$$H_n : \mathbf{Ch}(R) \rightarrow \mathbf{Mod}_R$$

for each $n \in \mathbb{N}$.

- Using the very same functor, but thinking about cochain complexes as chain complexes in the wrong direction, we have also defined the cohomology functor of cochain complexes

$$H^n : \mathbf{CCh}(R) \rightarrow \mathbf{Mod}_R$$

for each $n \in \mathbb{N}$.

- We have a functorial method of dualizing chain complexes into cochain complexes by the functor

$$\text{Hom}_R(-, M) : \mathbf{Ch}(R) \rightarrow \mathbf{CCh}(Z)$$

for any R -module M .

7.3 Free Resolutions

Definition 7.3.1 (Free Resolutions) Let R be a commutative ring. Let M be an R -module. An exact sequence of the form

$$\cdots \longrightarrow F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

is said to be a free resolution of M if F_n is a free R -module for $n \in \mathbb{N}$.

Proposition 7.3.2 Let R be a commutative ring. Let M, N be R -modules. Let $f : M \rightarrow N$ be an R -module homomorphism. Suppose that there are free resolutions

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_1 & \xrightarrow{p_1} & F_0 & \xrightarrow{p_0} & M \longrightarrow 0 \\ & & & & & & \downarrow f \\ \cdots & \longrightarrow & E_1 & \xrightarrow{q_1} & E_0 & \xrightarrow{q_0} & N \longrightarrow 0 \end{array}$$

for M and N respectively. Then there exists a chain map extending f such that the following diagram commutes:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & F_1 & \xrightarrow{p_1} & F_0 & \xrightarrow{p_0} & G \longrightarrow 0 \\ & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f \\ \cdots & \longrightarrow & E_1 & \xrightarrow{q_1} & E_0 & \xrightarrow{q_0} & H \longrightarrow 0 \end{array}$$

Moreover, any two such chain maps are homotopic.

Proof We will define the maps $f_n : F_n \rightarrow E_n$ for $n \in \mathbb{N}$ inductively. Since each F_i is free, we only need to define the map on the basis elements of F_i . Consider the base case $n = 0$. Since $E_0 \rightarrow H$ is surjective, for any $x \in F_0$, there exists $x' \in E_0$ such that $q_0(x') = f(p_0(x))$. Define the map $f_0 : F_0 \rightarrow E_0$ by sending any basis element x of F_0 to such an x' . (The x' here may not be unique but we will show that any choice here leads to homotopic maps).

Now suppose that $f_{k-1} : F_{k-1} \rightarrow E_{k-1}$ is defined such that $q_{k-1} \circ f_{k-1} = f_{k-2} \circ p_{k-1}$. Let $y \in F_k$. Then $f_{k-1}(p_k(y)) \in E_{k-1}$. Now we have

$$q_{k-1}((f_{k-1} \circ p_k)(y)) = (f_{k-2} \circ p_{k-1} \circ p_k)(y) = f_{k-2}(0) = 0$$

which shows that $(f_{k-1} \circ p_k)(y) \in \ker(q_{k-1}) = \text{im}(q_k)$. But this means that there exists $y' \in E_k$ such that $q_k(y') = (f_{k-1} \circ p_k)(y)$. Therefore we define $f_k : F_k \rightarrow E_k$ by sending any basis element y of F_k to the defined y' . (Again the choice does not matter because any two choices of maps are homotopic).

For uniqueness up to homotopy, suppose that $(g_k)_{k \in \mathbb{N}}$ is another such extension. Now consider the map $\alpha_k = f_k - g_k : F_k \rightarrow E_k$. The proof will be complete if the new chain map α is homotopic to the 0 map. Because then $\alpha \simeq 0$ implies $f - g \simeq 0$ which implies $f \simeq g$. To show the null-homotopy, we need to define chain homotopies $\lambda_k : F_k \rightarrow E_{k+1}$ such that $\alpha_k = q_{k+1} \circ \lambda_k + \lambda_{k-1} \circ p_k$. We proceed by induction.

For the map $\lambda_{-1} : G \rightarrow E_0$ we define it to be the 0 map. When $k = 0$, define $\lambda_0 : F_0 \rightarrow E_1$ as follows. Let $x \in F_0$ be a basis element. Then $f(p_0(x)) = 0$ and $g(p_0(x)) = 0$. Now we have that

$$q_0(\alpha_0(x)) = q_0((f_0 - g_0)(x)) = q_0(f_0(x)) - q_0(g_0(x)) = f(p_0(x)) - g(p_0(x)) = (f - g)(p_0(x)) = 0$$

because f and g are the same map. This means that $\alpha_0(x) \in \ker(q_0) = \text{im}(q_1)$. Thus there exists $x' \in E_1$ such that $q_1(x') = \alpha_0(x)$. Therefore we define $\lambda_0 : F_0 \rightarrow E_1$ by sending any basis element $x \in F_0$ to the constructed $x' \in E_1$. But this is exactly what we want because it satisfies the chain condition:

$$q_1(\lambda_0(x)) + \lambda_{-1}(p_0(x)) = q_0(\lambda_0(x)) = q_1(x') = \alpha_0(x)$$

Inductively, suppose that $\lambda_{k-1} : F_{k-1} \rightarrow E_k$ is already defined such that the chain homotopy

condition is satisfied. Let $x \in F_k$ be a basis element. Then notice that

$$\begin{aligned}
 q_k((\alpha_k - \lambda_{k-1} \circ p_k)(x)) &= (q_k \circ \alpha_k)(x) - (q_k \circ \lambda_{k-1} \circ p_k)(x) \\
 &= (q_k \circ f_k - q_k \circ g_k)(x) - (q_k \circ \lambda_{k-1} \circ p_k)(x) \\
 &= (f_{k-1} \circ p_k - g_{k-1} \circ p_k)(x) - (q_k \circ \lambda_{k-1} \circ p_k)(x) \\
 &= (f_{k-1} - g_{k-1})(p_k(x)) - (q_k \circ \lambda_{k-1} \circ p_k)(x) \\
 &= (\alpha_{k-1} \circ p_k)(x) - (q_k \circ \lambda_{k-1} \circ p_k)(x) \\
 &= (\alpha_{k-1} - q_k \circ \lambda_{k-1})(p_k(x)) \\
 &= (\lambda_{k-2} \circ p_{k-1} \circ p_k)(x) \quad (\text{Chain Homotopy Condition}) \\
 &= \lambda_{k-2}(0) \\
 &= 0
 \end{aligned}$$

This shows that $(\alpha_k - \lambda_{k-1} \circ p_k)(x) \in \ker(q_k) = \text{im}(q_{k+1})$, and hence there exists $x' \in E_{k+1}$ such that $q_{k+1}(x') = (\alpha_k - \lambda_{k-1} \circ p_k)(x)$. So we define $\lambda_k : F_k \rightarrow E_{k+1}$ by sending any basis element $x \in F_k$ to the constructed $x' \in E_{k+1}$. But this is exactly what we want because it satisfies the chain condition:

$$\begin{aligned}
 (q_{k+1} \circ \lambda_k)(x) + (\lambda_{k-1} \circ p_k)(x) &= q_{k+1}(x') + (\lambda_{k-1} \circ p_k)(x) \\
 &= (\alpha_k - \lambda_{k-1} \circ p_k)(x) + (\lambda_{k-1} \circ p_k)(x) \\
 &= \alpha_k(x)
 \end{aligned}$$

And so we conclude. ■

Corollary 7.3.3 Let R be a commutative ring. Let M be an abelian group. Then any two free resolutions $F_\bullet \rightarrow M$ and $E_\bullet \rightarrow M$ of M are chain homotopy equivalent.

Proof By the above, the chain map $F_\bullet \rightarrow E_\bullet \rightarrow F_\bullet$ induced by the identity map is chain homotopic to the identity map. Similarly for the chain map $E_\bullet \rightarrow F_\bullet \rightarrow E_\bullet$ induced by the identity map. Hence $F_\bullet$ and $E_\bullet$ are chain homotopy equivalent. ■

It follows from the algebra of chain complexes that any two free resolutions induces isomorphic homology groups.

Lemma 7.3.4 Let A be an abelian group. Then the following is a free resolution of A :

$$0 \longrightarrow \ker(\pi) \longrightarrow \bigoplus_{a \in A} \mathbb{Z} \cdot a \xrightarrow{\pi} A \longrightarrow 0$$

where $\pi : \bigoplus_{a \in A} \mathbb{Z} \cdot a \rightarrow A$ is the projection map.

Proof Clearly the sequence is exact. Moreover, every subgroup of a free abelian group is free abelian. Hence $\ker(\pi)$ is free abelian and the sequence is indeed a free resolution. ■

7.4 Universal Coefficient Theorem for Cohomology

Definition 7.4.1 (Ext Group) Let R be a commutative ring. Let M, N be R -modules. Choose a free resolution $F_\bullet$ of M :

$$0 \longrightarrow F_1 \longrightarrow F_0 \longrightarrow H \longrightarrow 0$$

Define the ext group of M with coefficients in N to be the first cohomology group of the cochain complex with coefficients in N associated to this chain complex:

$$\text{Ext}_R(M, N) = \frac{\text{Hom}(F_1, N)}{\text{im}(\text{Hom}(F_0, N) \rightarrow \text{Hom}(F_1, N))} = H^1(\text{Hom}(F_\bullet; N))$$

The Ext group is well defined because of any choice of free resolution are chain homotopic by a unique chain map. Moreover, chain homotopic maps induce isomorphisms on homology. This isomorphism is moreover canonical because the chain homotopy is given by a unique chain map.

This leads to the second way of extending it. One can consider not using free groups for the resolution, and instead in the setting of R -modules, consider using projective / injective modules (again in Homological Algebra).

For us, we are only considering abelian groups which are $\mathbb{Z}$ -modules. Be wary of this assumption throughout the entire section.

Example 7.4.2 Let $n, m \geq 1$. Then we have

$$\mathrm{Ext}_{\mathbb{Z}}\left(\frac{\mathbb{Z}}{n\mathbb{Z}}, \frac{\mathbb{Z}}{m\mathbb{Z}}\right) \cong \frac{\mathbb{Z}}{\mathrm{gcd}(m, n)\mathbb{Z}}$$

Proof Notice that a free resolution for $\mathbb{Z}/n\mathbb{Z}$ is given by

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0$$

Applying the functor $\mathrm{Hom}(-, \mathbb{Z}/m\mathbb{Z})$ gives a chain complex of the form

$$0 \longrightarrow \mathbb{Z}/\mathrm{gcd}(m, n)\mathbb{Z} \xrightarrow{1 \mapsto \frac{m}{\mathrm{gcd}(m, n)}} \mathbb{Z}/m\mathbb{Z} \xrightarrow{\times n} \mathbb{Z}/m\mathbb{Z} \longrightarrow 0$$

Then the first cohomology group of the chain complex is precisely $\frac{\mathbb{Z}/m\mathbb{Z}}{n\mathbb{Z}/m\mathbb{Z}}$. I claim that this $\mathbb{Z}$ -module is equal to $\frac{\mathbb{Z}/m\mathbb{Z}}{\mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}}$. Let $[nk] \in \mathbb{Z}/m\mathbb{Z}$. Then $[nk] = \left[\mathrm{gcd}(n, m) \frac{nk}{\mathrm{gcd}(n, m)}\right]$ so that $[nk] \in \mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}$. Conversely, let $[\mathrm{gcd}(n, m)k] \in \mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}$. By Bezout's lemma, we know that $\mathrm{gcd}(n, m) = nx + my$ for some $x, y \in \mathbb{Z}$. Then $[\mathrm{gcd}(n, m)k] = [nmx + mky] = [nmx] \in \mathrm{gcd}(n, m)\mathbb{Z}$. Hence $[\mathrm{gcd}(n, m)k] \in n\mathbb{Z}/m\mathbb{Z}$. Thus we have $\frac{\mathbb{Z}/m\mathbb{Z}}{n\mathbb{Z}/m\mathbb{Z}} = \frac{\mathbb{Z}/m\mathbb{Z}}{\mathrm{gcd}(n, m)\mathbb{Z}/m\mathbb{Z}}$. By the third isomorphism theorem, the first cohomology group is isomorphic to $\mathbb{Z}/\mathrm{gcd}(m, n)\mathbb{Z}$. ■

Proposition 7.4.3 Let R be a commutative ring. Let M, N, K be R -modules. Then the following are true.

- $\mathrm{Ext}_R(M \oplus N, K) \cong \mathrm{Ext}_R(M, K) \oplus \mathrm{Ext}_R(N, K)$
- $\mathrm{Ext}_R(M, N) = 0$ if M is a free R -module.
- If A is an abelian group, then $\mathrm{Ext}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A/nA$
- If A is an abelian group, then $\mathrm{Ext}_{\mathbb{Z}}(A, \mathbb{Q}) = 0$

Proof

- Let $A_{\bullet}$ be a free resolution of H . Let $B_{\bullet}$ be a free resolution of K . I claim that

$$0 \longrightarrow A_1 \oplus B_1 \longrightarrow A_0 \oplus B_0 \longrightarrow H \oplus K \longrightarrow 0$$

with the direct sum of the obvious maps is a free resolution of $H \oplus K$. But this is obvious because the direct sum of free groups is again a free group. Now we have that

$$\begin{aligned} \mathrm{Ext}(H \oplus K) &= \frac{\mathrm{Hom}(H \oplus K, G)}{\mathrm{im}(A_0 \oplus B_0 \rightarrow H \oplus K)} \\ &\cong \frac{\mathrm{Hom}(H, G) \oplus \mathrm{Hom}(K, G)}{\mathrm{im}(A_0 \rightarrow H_0) \oplus \mathrm{im}(B_0 \rightarrow K_0)} \\ &\cong \frac{\mathrm{Hom}(H, G)}{\mathrm{im}(A_0 \rightarrow H_0)} \oplus \frac{\mathrm{Hom}(K, G)}{\mathrm{im}(B_0 \rightarrow K_0)} \\ &= \mathrm{Ext}(H, G) \oplus \mathrm{Ext}(K, G) \end{aligned}$$

- If H is already a free group, then $0 \rightarrow H \rightarrow H \rightarrow 0$ with the identity map is a free resolution of H . Then by the definition of Ext groups we conclude that $\mathrm{Ext}(H, G) = 0$

Proposition 7.4.4 Let R be a commutative ring. Let M, N, K be R -modules. Let $f : M \rightarrow N$ be an R -module homomorphism. Define the induced map

$$f^* : \text{Ext}_R(N, K) \rightarrow \text{Ext}_R(M, K)$$

as follows. Then f^* is a group homomorphism. Moreover, the following are true.

- Let T be an R -module and let $g : N \rightarrow T$ be an R -module homomorphism. Then

$$(g \circ f)^* = f^* \circ g^*$$

- We have that

$$\text{id}^* = \text{id}_{\text{Ext}_R(M, K)} : \text{Ext}_R(M, K) \rightarrow \text{Ext}_R(M, K)$$

We conclude that Ext is a functor of the first variable. It is also a functor of the second variable but we will not prove it.

Proposition 7.4.5 Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be a short exact sequence of abelian groups. Then there is a six term exact sequence:

$$0 \longrightarrow \text{Hom}(C, G) \longrightarrow \text{Hom}(B, G) \longrightarrow \text{Hom}(A, G) \longrightarrow \text{Ext}(C, G) \longrightarrow \text{Ext}(B, G) \longrightarrow \text{Ext}(A, G) \longrightarrow 0$$

The point of the Ext group is that it forms an error term to measure how far away is the cohomology of the dualized chain complex from the homology of the chain complex.

Theorem 7.4.6 (Universal Coefficient Theorem for Cohomology) Let R be a PID. Let $(C_\bullet, \partial_\bullet)$ be a chain complex. Let M be an R -module. Then there is a split exact sequence (that does not split naturally) of the form

$$0 \longrightarrow \text{Ext}_R(H_{n-1}(C_\bullet; R), M) \longrightarrow H^n(C_\bullet; M) \xrightarrow{h} \text{Hom}(H_n(C_\bullet; R), M) \longrightarrow 0$$

where h is the map defined by $[\alpha] \mapsto [\overline{\alpha|_{Z_n(C_\bullet)}}]$ for any $n \in \mathbb{N}$.

Proof I claim that $\alpha|_{Z_n}$ descends to a map in homology. This means that there exists a unique map $\overline{\alpha|_{Z_n}} : H_n(C_\bullet) \rightarrow G$ such that the following diagram commutes

$$\begin{array}{ccc} Z_n & \xrightarrow{p} & \frac{Z_n}{B_n} = H_n(C_\bullet) \\ & \searrow \alpha|_{Z_n} & \downarrow \exists! \overline{\alpha|_{Z_n}} \\ & & G \end{array}$$

where p is the projection map. But this is true since for any $x \in C_{n+1}$ such that $\partial x \in B_n$, we have that

$$\alpha(\partial(x)) = \delta(\alpha(x)) = 0$$

where the last equality comes from the fact that $\alpha \in \ker(\delta : C^n \rightarrow C^{n+1})$. By the universal property of quotients, $\overline{\alpha|_{Z_n}}$ exists and is unique. We finally define h by

$$h([\alpha]) = [\overline{\alpha|_{Z_n}}]$$

I claim that this is well defined. This means to show that if $[\alpha] = [\beta]$ then $h([\alpha]) = h([\beta])$. So suppose that $[\alpha] = [\beta]$, then $\alpha + y = \beta$ for some $y \in B^n = \text{im}(\delta : C^{n-1} \rightarrow C^n)$. This means that

there exists $\bar{y} \in C^{n-1}$ such that $\delta(\bar{y}) = y$. Now we have that

$$\begin{aligned}
 h([\beta]) &= [\overline{\beta}|_{Z_n}] \\
 &= [\overline{(\alpha + y)}|_{Z_n}] \\
 &= [\overline{(\alpha + \delta(\bar{y}))}|_{Z_n}] \\
 &= [\overline{(\alpha + \bar{y} \circ \partial)}|_{Z_n}] \\
 &= [\overline{(\alpha)}|_{Z_n}] \quad (\partial \text{ restricted to } Z_n \text{ is just } 0) \\
 &= h([\alpha])
 \end{aligned}$$

Thus h is well defined.

We now show that h is split surjective. Define a map $v : \text{Hom}(H_n(C_\bullet, G)) \rightarrow H^n(C_\bullet, G)$ as follows. Let $\theta \in \text{Hom}(H_n(C_\bullet), G)$. Precompose with the projection map $p : Z_n(C_\bullet) \rightarrow H_n(C_\bullet)$ to obtain a map $\theta \circ p : Z_n \rightarrow G$. Now C_n being free implies that B_{n-1} is free. It follows that there is a split exact sequence

$$0 \longrightarrow Z_n \hookrightarrow C_n \xrightarrow{\partial} B_{n-1} \longrightarrow 0$$

by lmm 1.2.10. By the characterization of split exactness we conclude that there is a map $q : C_n \rightarrow Z_n$ such that the map $Z_n \hookrightarrow C_n \xrightarrow{q} Z_n$ is the identity (so q is the identity on Z_n). Now $\theta \circ p \circ q : C_n \rightarrow G$ vanishes on B_n because $\theta \circ p$ does. This means that for all $x \in C_{n+1}$, we have that $(\theta \circ p \circ q)(\partial x) = 0$ for $\partial x \in B_n$. This means that $\delta(\theta \circ p \circ q) = 0$ and $\theta \circ p \circ q \in \ker(\delta)$. Now define $v : \text{Hom}(H_n(C_\bullet, G)) \rightarrow H^n(C_\bullet, G)$ by

$$v(\theta) = [\theta \circ p \circ q]$$

Now we have that

$$\begin{aligned}
 h(v(\theta)) &= h([\theta \circ p \circ q]) \\
 &= [\overline{\theta \circ p \circ q}|_{Z_n}] \\
 &= [\overline{\theta \circ p}] \quad (q \text{ is the identity on } Z_n)
 \end{aligned}$$

Recall that $\overline{\theta \circ p}$ is the unique map such that $\overline{\theta \circ p} \circ p = \theta \circ p$ (first diagram of this proof). Since θ satisfies the description we have that $\overline{\theta \circ p} = \theta$. We conclude that $h(v(\theta)) = \theta$. From this, we have that for any $\theta \in \text{Hom}(H_n(C_\bullet), G)$ we have $v(\theta) \in H^n(C_\bullet)$ such that $h(v(\theta)) = \theta$. This proves that h is surjective. Moreover, the existence of the map v proves that the sequence

$$0 \longrightarrow \ker(h) \hookrightarrow H^n(C_\bullet) \xrightarrow{h} \text{Hom}(H_n(C_\bullet), G) \longrightarrow 0$$

is split exact.

We finally need to show the involvement of the Ext group.

We begin with a short exact sequence of chain complexes of the form

$$0 \longrightarrow Z_\bullet \hookrightarrow C_\bullet \xrightarrow{\partial} B_{\bullet-1} \longrightarrow 0$$

where its expanded form is given by

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & Z_n & \hookrightarrow & C_n & \xrightarrow{\partial} & B_{n-1} \longrightarrow 0 \\
 & & \downarrow 0 & & \downarrow \partial & & \downarrow 0 \\
 0 & \longrightarrow & Z_{n-1} & \hookrightarrow & C_{n-1} & \xrightarrow{\partial} & B_{n-2} \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \vdots & & \vdots & & \vdots
 \end{array}$$

Dualizing the entire diagram gives

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \uparrow & & \uparrow & & \uparrow & \\
 0 & \longleftarrow & \text{Hom}(Z_n, G) & \longleftrightarrow & \text{Hom}(C_n, G) & \xleftarrow{\partial^*} & \text{Hom}(B_{n-1}, G) \longrightarrow 0 \\
 & \uparrow 0 & & \uparrow \partial^* & & \uparrow 0 & \\
 0 & \longleftarrow & \text{Hom}(Z_{n-1}, G) & \longleftrightarrow & \text{Hom}(C_{n-1}, G) & \xleftarrow{\partial^*} & \text{Hom}(B_{n-2}, G) \longrightarrow 0 \\
 & \uparrow & & \uparrow & & \uparrow & \\
 & \vdots & & \vdots & & \vdots &
 \end{array}$$

Now each row is split exact because it is split exact before dualizing and the Hom functor preserves split exact sequences. We think of the above as a short exact sequence of chain complexes. This induces a long exact sequence in homology given by

$$\cdots \longrightarrow H_{n-1}(\text{Hom}(Z_{n-1}, G)) \longrightarrow H_n(\text{Hom}(B_{n+1}, G)) \longrightarrow H^n(C_\bullet, G) \longrightarrow H_n(\text{Hom}(Z_n, G)) \longrightarrow H_{n-1}(\text{Hom}(B_n, G)) \longrightarrow \cdots$$

Since $\text{Hom}(Z_\bullet, G)$ and $\text{Hom}(B_\bullet, G)$ have 0 as the boundary map, the above long exact sequence reduces to

$$\cdots \longrightarrow \text{Hom}(Z_{n-1}, G) \longrightarrow \text{Hom}(B_{n-1}, G) \longrightarrow H^n(C_\bullet, G) \longrightarrow \text{Hom}(Z_n, G) \longrightarrow \text{Hom}(B_n, G) \longrightarrow \cdots$$

Let us work out the connecting homomorphism $\text{Hom}(Z_n, G) \rightarrow \text{Hom}(B_{n-1}, G)$. For $\alpha \in \text{Hom}(Z_n, G)$, the connecting homomorphism sends $\alpha : Z_n \rightarrow G$ to $\gamma : B_{n-1} \rightarrow G$ where $\partial(\beta) = \alpha$ and $i_n(\gamma) = \partial(\beta)$. Now notice that

$$i_n(\gamma) = \partial(\beta) = \alpha$$

for $i_n : B_{n-1} \rightarrow C_n$ the inclusion. Thus the connecting homomorphism is precisely the dual of i_n , denoted i_n^* .

I claim that there is an isomorphism

$$\ker(i_n^* : \text{Hom}(Z_n, G) \rightarrow \text{Hom}(B_n, G)) \cong \text{Hom}(H_n(C_\bullet), G)$$

Let $\alpha \in \ker(i_n^*)$. Then $\alpha : Z_n \rightarrow G$ is such that $\alpha \circ i_n : B_n \hookrightarrow Z_n \rightarrow G$ is the zero map. Or in other words, $\alpha|_{B_n} = 0$. But then α is clearly an element of $\text{Hom}(H_n(C_\bullet), G)$ by definition. The converse is also true.

Now by definition of $H^n(C_\bullet)$, we get an exact sequence

$$0 \longrightarrow \ker(h) \hookrightarrow H^n(C_\bullet) \xrightarrow{h} \text{Hom}(H_n(C_\bullet), G) \longrightarrow 0$$

Going back to the long exact sequence

$$\cdots \longrightarrow \text{Hom}(Z_{n-1}, G) \xrightarrow{i_{n-1}^*} \text{Hom}(B_{n-1}, G) \longrightarrow H^n(C_\bullet, G) \longrightarrow \text{Hom}(Z_n, G) \xrightarrow{i_n^*} \text{Hom}(B_n, G) \longrightarrow \cdots$$

We can restrict the $\text{Hom}(B_{n-1}, G)$ term to $\text{coker}(i_{n-1}^*)$ and the $\text{Hom}(Z_n, G)$ term to $\ker(i_n^*)$ to obtain a short exact sequence

$$0 \xrightarrow{i_{n-1}^*} \text{coker}(i_{n-1}^*) \longrightarrow H^n(C_\bullet, G) \longrightarrow \ker(i_n^*) \xrightarrow{i_n^*} 0$$

Now using the isomorphism $\ker(i_n^*) \cong \text{Hom}(H_n(C_\bullet), G)$ it is also easy to see that the map $H^n(C_\bullet, G) \rightarrow \ker(i_n^*)$ is actually just the map h . We conclude that the above long exact sequence is precisely the long exact sequence we wanted to analyse, and in particular $\text{coker}(i_{n-1}^*) \cong \ker(h)$.

Now consider the following exact sequence:

$$0 \longrightarrow B_{n-1} \xrightarrow{i_{n-1}} Z_{n-1} \twoheadrightarrow H_{n-1}(C_\bullet) \longrightarrow 0$$

Since C_n is free abelian we conclude that B_{n-1} and Z_{n-1} are free. This means that this exact sequence is actually a free resolution of $H_{n-1}(C_\bullet)$. Using this free resolution, we obtain an isomorphism

$$\text{Ext}(H_{n-1}(C_\bullet), G) \cong \text{coker}(i_{n-1}^*)$$

Thus we conclude the proof. ■

Corollary 7.4.7 Let $(C_\bullet, \partial_\bullet)$ be a chain complex of free abelian groups. Let k be a field. Then there is an isomorphism

$$H^n(C_\bullet, k) \cong \text{Hom}(H_n(C_\bullet), k)$$

given by the universal coefficient theorem.

8 Singular Cohomology

8.1 Singular Cohomology

We now have all the pieces to define singular cohomology. This is a three step functorial process so that the end result is also functorial.

Definition 8.1.1 (The Singular Cochain Complex) Let X be a space. Let G be an abelian group. Define the singular cochain complex $(C^\bullet(X; G), \partial^\bullet)$ with coefficients in G to be the cochain complex given by the following data.

- The n th term is given by

$$C^n(X; G) = \text{Hom}_{\mathbb{Z}}(C_n(X); G)$$

the group of singular n -cochains of X .

- The boundary operator $\partial^n : C^{n-1}(X; G) \rightarrow C^n(X; G)$ is given by the formula

$$\partial^n(\phi) = \phi \circ \partial_n$$

for $\phi \in C^{n-1}(X; G)$.

It is the dual cochain complex of the singular chain complex $(C_\bullet(X), \partial_\bullet)$ of X .

Definition 8.1.2 (Singular Cohomology Groups of a Space) Let X be a topological space. Let G be a group. The n th singular cohomology group of X with coefficients in G is defined to be

$$H^n(X; G) = H_n(C^\bullet(X; G), \partial^\bullet) = \frac{\ker(\partial^{n+1} : C^n(X; G) \rightarrow C^{n+1}(X; G))}{\text{im}(\partial^n : C^{n-1}(X; G) \rightarrow C^n(X; G))}$$

Recall that for a space X , its singular chain complex $C_\bullet^{\text{Sing}}(X)$ is defined to be free abelian group

$$C_n^{\text{Sing}}(X) = \mathbb{Z}[\Delta^n]$$

on the set of singular n -simplices on X . By dualizing it, one obtains the singular cochain complex

$$C_{\text{Sing}}^n(X) = \text{Hom}(\mathbb{Z}[\Delta^n], G)$$

for any abelian group G . Such a map $f \in \text{Hom}(\mathbb{Z}[\Delta^n], G)$ is determined on the generators of the free abelian group by its universal property. And so f is essentially determined by the choice of an element $g \in G$ for every singular n -simplex. In other words, every element of $C_{\text{Sing}}^n(X; G)$ of X is just a labelling of every singular n -simplex with an element of G . Now if $f : \mathbb{Z}[\Delta^n] \rightarrow G$ is a cocycle, we simply require that the sum of the values that f takes on the boundary of Δ^n to be 0. In particular,

- Elements of $C_{\text{Sing}}^0(X; G)$ simply labels every point of X with an element of G .
- Elements of $Z^0(X; G)$ labels every point of X with an element of G such that $f(v_0) = f(v_1)$, thus forming a 1-simplex $v : [0, 1] \rightarrow X$ (Indeed, if $f \in Z^0(X; G)$, then $\delta(f) = f \circ \partial = 0$. Choosing the 1-simplex $[0, 1]$ shows that $0 = f(\partial([0, 1])) = f(v_0 - v_1) = f(v_0) - f(v_1)$ which is exactly what we wanted). But this means that elements of $Z^0(X; G)$ labels every point of X with an element of G such that f is constant on path components of X .

Definition 8.1.3 (Induced Homomorphisms) Let $f : X \rightarrow Y$ be a continuous map. Let G be an abelian group. Define the induced map on cohomology

$$f^* : H^n(Y; G) \rightarrow H^n(X; G)$$

by the formula $[\alpha] \mapsto [\alpha \circ f]$.

For an R -module M , we can describe the construction of singular cohomology groups as a three step functorial process

$$H^n(-; M) : \mathbf{Top} \xrightarrow{C_\bullet(-)} \mathbf{Ch}(R) \xrightarrow{\text{Hom}_R(-, M)} \mathbf{CCh}(\mathbb{Z}) \xrightarrow{H_n(-)} \mathbf{Ab}$$

Therefore the construction of singular cohomology groups are also functorial. In the case that R is a commutative ring, we may upgrade the three step process as

$$H^n(-; M) : \mathbf{Top} \xrightarrow{C_\bullet(-)} \mathbf{Ch}(R) \xrightarrow{\mathrm{Hom}_R(-, M)} \mathbf{CCh}(R) \xrightarrow{H_n(-)} {}_R\mathbf{Mod}$$

Theorem 8.1.4 (Functoriality of Cohomology) Let X, Y be spaces. Let $f : X \rightarrow Y$ be a continuous map. Let G be an abelian group. Then the induced homomorphism

$$f^* : H^n(Y; G) \rightarrow H^n(X; G)$$

is well defined and is a group homomorphism. Moreover, the following are true.

- If Z is a space and $g : Y \rightarrow Z$ is another map, then

$$(g \circ f)^* = f^* \circ g^*$$

- We have that

$$(\mathrm{id}_X)^* = \mathrm{id}_{H^n(X; G)}$$

Proof As seen above cohomology is a three step functorial process, hence the induced map satisfies all the functorial properties. ■

Theorem 8.1.5 (Homotopy Invariance) Let X, Y be spaces. Let $f, g : X \rightarrow Y$ be continuous maps. Let G be an abelian group. If f and g are homotopic, then they induce the same map

$$f^* = g^* : H^n(Y; G) \rightarrow H^n(X; G)$$

on singular cohomology.

Proof We have seen that in the proof of 4.4.3, $f_*, g_* : C_n(X) \rightarrow C_n(Y)$ are chain homotopic by some chain homotopy $\eta : C_n(X) \rightarrow C_{n+1}(Y)$. This means that have

$$g_* - f_* = \partial \circ \eta + \eta \circ \partial : C_n(X) \rightarrow C_n(Y)$$

Both maps are group homomorphisms hence we can dualize and applying the hom functor to get

$$g^* - f^* = \partial^* \circ \eta^* + \eta^* \circ \partial^*$$

This means that the maps $f^*, g^* : C^n(X; G) \rightarrow C^n(Y; G)$ are chain homotopic, hence they induce the same map on cohomology. ■

Theorem 8.1.6 (Mayer-Vietoris Sequence) Let X be a topological space and U_1, U_2 be open sets of X such that $X = U_1 \cup U_2$ and that $U_1 \cap U_2 \neq \emptyset$. Write $i_1 : U_1 \cap U_2 \rightarrow U_1, i_2 : U_1 \cap U_2 \rightarrow U_2, j_1 : U_1 \rightarrow X$ and $j_2 : U_2 \rightarrow X$ the inclusion maps. Let G be an abelian group. Then there is a long exact sequence

$$\cdots \longrightarrow H^{n-1}(X; G) \xrightarrow{\partial} H^n(U_1 \cap U_2; G) \xrightarrow{(i_1)^* - (i_2)^*} H^n(U_1; G) \oplus H^n(U_2; G) \xrightarrow{(j_1)^* + (j_2)^*} H^n(X; G) \xrightarrow{\partial} H^{n+1}(U_1 \cap U_2; G) \longrightarrow \cdots$$

in cohomology.

Proof

Definition 8.1.7 (Reduced Cohomology Groups) Let X be a space. Let G be an abelian group. Define the reduced cohomology groups

$$\tilde{H}^n(X; G)$$

of X with coefficients in G to be the n th cohomology groups of the augmented singular chain complex.

Proposition 8.1.8 Let X be a space. Let R be a ring. Let M be an R -module. Then the following are true regarding the reduced cohomology of X .

- For all $n > 0$,

$$\tilde{H}^n(X; M) = H^n(X; M)$$

- When $n = 0$, the universal coefficient theorem induces an isomorphism

$$\tilde{H}^0(X; M) \cong \text{Hom}(\tilde{H}_0(X), M)$$

Example 8.1.9 Let $n \in \mathbb{N}$. Let A be an abelian group. The singular cohomology of S^n with coefficients in A is given by

$$H^k(S^n; A) = \begin{cases} A & \text{if } k = 0 \text{ or } n \\ 0 & \text{otherwise} \end{cases}$$

Example 8.1.10 Let $n \geq 1$. Let A be an abelian group. The singular cohomology of $\mathbb{RP}^n$ with coefficients in A is given by

$$H^k(\mathbb{RP}^n; A) = \begin{cases} A & \text{if } k = 0 \\ A/2A & \text{if } 0 < k < n \text{ and } k \text{ even} \\ \{a \in A \mid 2 \text{ divides } |a|\} & \text{if } 0 < k < n \text{ and } k \text{ odd} \\ 0 & \text{otherwise} \end{cases}$$

Example 8.1.11 Let $n \geq 1$. Let A be an abelian group. The singular cohomology of $\mathbb{CP}^n$ with coefficients in A is given by

$$H^k(\mathbb{CP}^n; A) = \begin{cases} A & \text{if } 0 \leq k \leq 2n \text{ even} \\ 0 & \text{otherwise} \end{cases}$$

8.2 Relative Cohomology

Definition 8.2.1 (Relative Cochain Complex) Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Define the relative cochain complex $(C^\bullet(X, A; G), \partial^*)$ with coefficients in G to be given by the following data.

- The n th term is given by

$$C^n(X, A; G) = \text{Hom}_{\mathbb{Z}}(C_n(X, A), G)$$

the group of singular n -cochains relative to A .

- The boundary operator $\partial^n : C^{n-1}(X; G) \rightarrow C^n(X; G)$ is given by the formula

$$\partial^n(\phi) = \phi \circ \partial_n$$

for $\phi \in C^{n-1}(X, A; G)$.

It is the dual cochain complex of the relative singular chain complex $(C_\bullet(X, A), \partial_\bullet)$ of X with respect to A .

Lemma 8.2.2 Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Then the n th term of the relative cochain complex is given by

$$C^n(X, A; G) = \{\alpha \in C^n(X; G) \mid \alpha(\sigma) = 0 \text{ for all } \text{im}(\sigma) \subseteq A\}$$

Definition 8.2.3 (Relative Cohomology Groups of a Space) Let X be a topological space. Let G be an abelian group. The n th relative cohomology group of X with coefficients in G is defined to

be

$$H^n(X, A; G) = H_n(C^\bullet(X, A; G), \partial^\bullet) = \frac{\ker(\partial^{n+1} : C^n(X, A; G) \rightarrow C^{n+1}(X, A; G))}{\operatorname{im}(\partial^n : C^{n-1}(X, A; G) \rightarrow C^n(X, A; G))}$$

TBA: Functoriality

Theorem 8.2.4 (Excision) Let (X, A) be a pair of spaces. Let $Z \subseteq A$ be a subspace such that $\bar{Z} \subseteq A^\circ$. Let G be an abelian group. Then the inclusion map $\iota : (X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces an isomorphism

$$\iota^* : H^n(X, A; G) \xrightarrow{\cong} H^n(X \setminus Z, A \setminus Z; G)$$

in cohomology groups for all $n \in \mathbb{N}$.

Theorem 8.2.5 (Long Exact Sequence of Relative Cohomology Groups) Let X be a space and let $A \subseteq X$ be a subspace. Let G be an abelian group. Then there is an exact sequence

$$\cdots \longrightarrow H^{n-1}(A; G) \xrightarrow{\partial} H^n(X, A; G) \xrightarrow{j^*} H^n(X; G) \xrightarrow{i^*} H^n(A; G) \longrightarrow \cdots$$

where $\iota : A \rightarrow X$ is the inclusion map, $j : X \rightarrow X \setminus A$ is the quotient map and ∂ is the connecting homomorphism induced by the snake lemma.

Proof Notice that

$$0 \longrightarrow C^\bullet(X, A; G) \xrightarrow{j^*} C^\bullet(X; G) \xrightarrow{i^*} C^\bullet(A; G) \longrightarrow 0$$

is a short exact sequence by the above lemma. Thus it induces a long exact sequence in cohomology groups.

Consider $[z] = z + C_n(A) \in C_n(X, A)$ a cycle in $C_n(X, A)$. The surjective map j is the projection map so $j(z) = [z]$. Now $d(z) \in \ker(j)$ because

$$j(d(z)) = d(j(z)) = d([z]) = 0$$

by assumption. So $d(z) \in \ker(j) = \operatorname{im}(i)$. Since i is the inclusion, $[d(z)] \in C_{n-1}(A)$ is precisely the element that ∂ maps $[z]$ to. ■

Theorem 8.2.6 (Long Exact Sequence of Triplet of Spaces) Let X be a space. Let $B \subseteq A \subseteq X$ be subspaces. Let $j : B \hookrightarrow A$ and $i : A \hookrightarrow X$ be inclusions. Let M be a R -module for some ring R . Then there is a long exact sequence in cohomology

$$\cdots \longrightarrow H^{n-1}(A, B; M) \xrightarrow{\delta} H^n(X, A; M) \xrightarrow{j^*} H^n(X, B; M) \xrightarrow{i^*} H^n(A, B; M) \xrightarrow{\delta} H^{n+1}(X, A; M) \longrightarrow \cdots$$

induced by the dual of the short exact sequence of the above proposition. The map δ is given by the connecting homomorphism.

The connecting homomorphism ∂ in homology is related to the connecting homomorphism δ in the following way.

Proposition 8.2.7 Let (X, A) be a pair of space. Let M be a R -module for some ring R . Let $\partial : H_{n+1}(X, A) \rightarrow H_n(A)$ denote the connecting homomorphism in relative homology. Let $\delta : H^n(A; M) \rightarrow H^{n+1}(X, A; M)$ denote the connecting homomorphism in relative cohomology. Then the following diagram is commutative:

$$\begin{array}{ccc} H^n(A; M) & \xrightarrow{\delta} & H^{n+1}(X, A; M) \\ h \downarrow & & \downarrow h \\ \operatorname{Hom}_R(H_n(A), M) & \xrightarrow{\partial^*} & \operatorname{Hom}_R(H_{n+1}(X, A), M) \end{array}$$

where h is the canonical homomorphism given by the universal coefficient theorem.

9 Variants of Cohomology

9.1 Cellular Cohomology

Definition 9.1.1 (Cellular Cochain Complex) Let X be a CW complex. Let R be a ring. Let M be an R -module. Define the cellular cochain complex of X with coefficients in M to be

$$C_{CW}^n(X; M) = \text{Hom}_R(C_n^{CW}(X), M)$$

the dual of the cellular chain complex.

TBA: We can splice the dual of the exact sequence to obtain the cellular chain complex similar to that of cellular homology.

Definition 9.1.2 (Cellular Cohomology Groups) Let X be a CW complex. Let R be a ring. Let M be an R -module. Define the n cellular cohomology group of X with coefficients in M

$$H_{CW}^n(X; M) = H^n(\text{Hom}(C_n^{CW}(X), M))$$

to be the n th cohomology group of the dual of the cellular chain complex.

9.2 Cohomology with Compact Support

Definition 9.2.1 (Cochain with Compact Supports) Let X be a space. Let A be an abelian group. Define the n th cochain with compact supports to be the cochain subcomplex given by

$$C_c^n(X; A) = \left\{ c \in C^n(X; A) \mid \begin{array}{l} \text{there exists } K \text{ compact} \\ \text{such that } c(x)=0 \text{ for } x \notin K \end{array} \right\} \subseteq C^n(X; A)$$

and the induced coboundary map.

Definition 9.2.2 (Cohomology with Compact Supports) Let X be a space. Let A be an abelian group. Define the n th cohomology of compact supports of X to be the n th cohomology

$$H_c^n(X; A) = H^n(C_c^n(X; A)) = \frac{\ker(\delta : C_c^n(X; A) \rightarrow C_c^{n+1}(X; A))}{\text{im}(\delta : C_c^{n-1}(X; A) \rightarrow C_c^n(X; A))}$$

of the cochain with compact supports.

We will need an explicit construction of colimits from category theory. Recall that a directed set consists of a set I and a partial order $\leq$ such that for any $i, j \in I$, there exists $k \in I$ such that $i \leq k$ and $j \leq k$. Such a set can be made into a small category whose hom sets are given by $\text{Hom}(i, j) = * = \{f_{ij}\}$ if $i \leq j$ and $\emptyset$ otherwise. A direct system of abelian groups is then a functor $F : I \rightarrow \mathbf{Ab}$.

The colimit of the functor is then a certain cone satisfying certain universal properties. Explicitly, we can define it as

$$\text{colim}_I F = \frac{\bigoplus_{i \in I} F(i)}{\sim}$$

where the equivalence relation is generated by $x_i \sim F(f_{ij})(x_i)$. An element of the colimit is an equivalence class $[x]$ for some $x \in F(i)$ for some i . Moreover, two elements $x \in F(i)$ and $y \in F(j)$ in the direct sum $\bigoplus_{i \in I} F(i)$ are equivalent under the colimit if there exists $k \in I$ such that $i, j \leq k$ and

$$f_{ik}(x) = f_{jk}(y)$$

Proposition 9.2.3 Let X be a space. Let A be an abelian group. Consider the directed system

$I = \{K \subseteq X \mid K \text{ is compact}\}$ of subspaces of X . Then there is an canonical isomorphism

$$\operatorname{colim}_{K \in I} H^n(X, X \setminus K; A) \cong H_c^n(X; A)$$

induced by the universal property for any $n \in \mathbb{N}$.

Proof Firstly, notice that since each $C_c^n(X, X \setminus K; A)$ is a subgroup of $C_c^n(X; A)$, we have that

$$C^n(X; A) = \bigcup_{\substack{K \subseteq X \\ K \text{ compact}}} C_c^n(X, X \setminus K; A) = \operatorname{colim}_{K \in I} C_c^n(X, X \setminus K; A)$$

I claim that if I is a directed set, and $A, B, C : I \rightarrow \mathbf{Ab}$ are functors with natural transformation $f : A \Rightarrow B$ and $g : B \Rightarrow C$ such that

$$A(i) \xrightarrow{f(i)} B(i) \xrightarrow{g(i)} C(i)$$

is exact for all i , then

$$\operatorname{colim}_{i \in I} A(i) \xrightarrow{f} \operatorname{colim}_{i \in I} B(i) \xrightarrow{g} \operatorname{colim}_{i \in I} C(i)$$

Let $[x]$ be an element of $\operatorname{colim}_{i \in I} A(i)$. Suppose that $x \in A(i)$. Then $g(f([x])) = g([f_i(x)]) = [g_i(f_i(x))] = [0]$. Hence $\operatorname{im}(f) \subseteq \ker(g)$. Now assume that $[x] \in \ker(g)$, where $x \in F(i)$. This means that $[g_i(x)] = g([x]) = [0]$. Write $\operatorname{Hom}(i, j) = \{\eta_{ij}\}$ for the unique morphism in the category I when $i \leq j$. Then $[g_i(x)] = [0]$ means that there exists $j \in I$ such that $i \leq j$ and $C(\eta_{ij})(g_i(x)) = 0$. By commutativity of the following diagram:

$$\begin{array}{ccc} B(i) & \xrightarrow{g_i} & C(i) \\ B(\eta_{ij}) \downarrow & & \downarrow C(\eta_{ij}) \\ B(j) & \xrightarrow{g_j} & C(j) \end{array}$$

we have that $0 = C(\eta_{ij})(g_i(x)) = g_j(B(\eta_{ij})(x))$. Since $\operatorname{im}(f_j) = \ker(g_j)$, there exists $y \in A(j)$ such that $f_j(y) = B(\eta_{ij})(x)$. Then we have

$$f([y]) = [f_j(y)] = [B(\eta_{ij})(x)] = [x]$$

so that $\ker(g) \subseteq \operatorname{im}(f)$. Thus we are done. The result then extends to short exact sequences of abelian groups.

For the isomorphism on cohomology, we use the fact that there are exact sequences of the form:

$$0 \longrightarrow \ker(f) \longrightarrow A \xrightarrow{f} B \longrightarrow 0$$

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow \operatorname{coker}(f) \longrightarrow 0$$

$$0 \longrightarrow B \xrightarrow{\operatorname{incl.}} A \xrightarrow{\operatorname{proj.}} A/B \longrightarrow 0$$

to conclude that $\operatorname{colim}_{i \in I}$ commutes with kernels, cokernels and quotients. And then we compute

that

$$\begin{aligned}
 H_c^n(X; A) &= \frac{\ker(\delta_c^n : C_c^n(X; A) \rightarrow C_c^{n+1}(X; A))}{\operatorname{im}(\delta_c^n : C_c^{n-1}(X; A) \rightarrow C_c^n(X; A))} \\
 &\cong \frac{\ker\left(\operatorname{colim}_{K \in I} \delta_K : C_c^n(X, X \setminus K; A) \rightarrow C_c^{n+1}(X, X \setminus K; A)\right)}{\operatorname{im}\left(\operatorname{colim}_{K \in I} \delta_K : C_c^{n-1}(X, X \setminus K; A) \rightarrow C_c^n(X, X \setminus K; A)\right)} \\
 &\cong \frac{\operatorname{colim}_{K \in I} \ker(\delta_K)}{\operatorname{colim}_{K \in I} \operatorname{im}(\delta_K)} \\
 &\cong \operatorname{colim}_{K \in I} \frac{\ker(\delta_K)}{\operatorname{im}(\delta_K)} \\
 &= \operatorname{colim}_{K \in I} H^n(X, X \setminus K; A)
 \end{aligned}$$

■

9.3 Axioms for Unreduced Cohomology

Definition 9.3.1 (Generalized Cohomology Theory for CW Pairs) A Generalized cohomology theory is a collection of contravariant functors and natural transformations

$$h^n : \mathbf{CW}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta^n : h^n \circ F \Rightarrow h^{n+1}(X, A)$$

where $F(X, A) = (A, \emptyset)$, for each $n \in \mathbb{N}$ such that the following are true.

- Homotopy Invariance: If $f \simeq g : (X, A) \rightarrow (Y, B)$ then

$$h^n(f) = h^n(g) : h^n(X, A) \rightarrow h^n(Y, B)$$

- Exactness: If X is a CW-complex and $A \subseteq X$, then there is a short exact sequence

$$\cdots \longrightarrow h^n(X, A) \longrightarrow h^n(X, \emptyset) \longrightarrow h^n(A, \emptyset) \xrightarrow{\partial_n} h^{n+1}(X, A) \longrightarrow h^{n+1}(X, \emptyset) \longrightarrow \cdots$$

- Excision: If $\overline{E} \subseteq A^\circ \subseteq X$, then

$$h^n(X \setminus E, A \setminus E) \cong h^n(X, A)$$

induced by the inclusion map

Definition 9.3.2 (Additive Cohomology Theories) Let $h_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$ be a generalized cohomology theory. We say that h_n is additive if it satisfies the following axiom.

- Additivity: Denote the inclusion map as $\iota_k : (X_k, A_k) \rightarrow \coprod_{k \in I} (X_k, A_k)$. Then the product of the induced maps of inclusions induce an isomorphism

$$\prod_{k \in I} (\iota_k)^* : h^n\left(\prod_{k \in I} (X_k, A_k)\right) \xrightarrow{\cong} \prod_{k \in I} h^n(X_k, A_k)$$

10 The Ring Structure in Singular Cohomology

10.1 The Cup Product

What separates singular cohomology from singular homology is that the abelian group arising from the cohomology groups has a natural structure of a ring. This in turns makes singular cohomology an invariant that associates to every space a graded ring, instead of just a sequence of abelian groups.

For this, we define the cup product which serves as multiplication between elements of the graded abelian groups.

Definition 10.1.1 (Cup Product) Let X be a space. Let R be a ring. Let $\phi \in C^k(X; R)$ and $\psi \in C^l(X; R)$. Define the cup product to be $\phi \smile \psi : C^{k+l}(X; R) \rightarrow R$ where for $\sigma = [v_0, \dots, v_{k+l}]$, we have that

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0, \dots, v_k]}) \cdot \psi(\sigma|_{[v_k, \dots, v_{k+l}]})$$

and then extended R -linearly.

Notice that the cup product is defined in terms of the singular complexes, therefore this definition only works for chain complexes that arise from a space. Moreover, the cup product is only defined when we take singular cohomology in a ring instead of an abelian group.

Lemma 10.1.2 Let X be a space. Let R be a ring. Let $\phi \in C^k(X; R)$ and $\psi \in C^l(X; R)$. Then we have that

$$\delta(\phi \smile \psi) = \delta\phi \smile \psi + (-1)^k \phi \smile \delta\psi$$

Proof Let $\sigma : \Delta^{k+1+l} \rightarrow X$ be a singular $(k+1+l)$ -simplex. We have that

$$\begin{aligned} (\delta\phi \smile \psi)(\sigma) &= \delta(\phi)(\sigma|_{v_0, \dots, v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) \\ &= \phi(\partial(\sigma|_{v_0, \dots, v_{k+1}})) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) \\ &= \phi\left(\sum_{i=0}^{k+1} (-1)^i \sigma|_{v_0, \dots, \hat{v}_i, \dots, v_{k+1}}\right) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) \\ &= \sum_{i=0}^{k+1} (-1)^i \phi(\sigma|_{v_0, \dots, \hat{v}_i, \dots, v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) \end{aligned}$$

and

$$\begin{aligned} (-1)^k (\phi \smile \delta\psi)(\sigma) &= \phi(\sigma|_{v_0, \dots, v_k}) \cdot (\psi(\partial\sigma|_{v_k, \dots, v_{k+1+l}})) \\ &= \phi(\sigma|_{v_0, \dots, v_k}) \cdot ((-1)^k \psi(\partial\sigma|_{v_k, \dots, v_{k+1+l}})) \\ &= \phi(\sigma|_{v_0, \dots, v_k}) \cdot \psi\left(\sum_{j=k}^{k+1+l} (-1)^j \sigma|_{v_k, \dots, \hat{v}_j, \dots, v_{k+1+l}}\right) \\ &= \phi(\sigma|_{v_0, \dots, v_k}) \cdot \sum_{j=k}^{k+1+l} (-1)^j \psi(\sigma|_{v_k, \dots, \hat{v}_j, \dots, v_{k+1+l}}) \end{aligned}$$

Notice that the last term of the first sum cancels out the first term of the second sum. Thus we have

$$\begin{aligned} &(\delta\phi \smile \psi)(\sigma) + (-1)^k (\phi \smile \delta\psi)(\sigma) \\ &= \sum_{i=0}^{k+1} (-1)^i \phi(\sigma|_{v_0, \dots, \hat{v}_i, \dots, v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) + \phi(\sigma|_{v_0, \dots, v_k}) \cdot \sum_{j=k}^{k+1+l} (-1)^j \psi(\sigma|_{v_k, \dots, \hat{v}_j, \dots, v_{k+1+l}}) \\ &= \sum_{i=0}^k (-1)^i \phi(\sigma|_{v_0, \dots, \hat{v}_i, \dots, v_{k+1}}) \cdot \psi(\sigma|_{v_{k+1}, \dots, v_{k+1+l}}) + \phi(\sigma|_{v_0, \dots, v_k}) \cdot \sum_{j=k+1}^{k+1+l} (-1)^j \psi(\sigma|_{v_k, \dots, \hat{v}_j, \dots, v_{k+1+l}}) \\ &= (\phi \smile \psi)(\partial\sigma) \\ &= \delta(\phi \smile \psi)(\sigma) \end{aligned}$$

and so we conclude. ■

Proposition 10.1.3 Let X be a space. Let R be a ring. Then the cup product

$$\smile: H^m(X; R) \times H^n(X; R) \rightarrow H^{m+n}(X; R)$$

descends to homology and is well defined.

Proof By the universal property of quotients, we just have to show the following:

- If ϕ and ψ are cochains, then $\phi \smile \psi$ is a cochain
- If either ϕ or ψ are coboundaries, then $\phi \smile \psi$ is a coboundary.

So let ϕ and ψ are cochains. Then $\delta(\phi) = 0$ and $\delta(\psi) = 0$. By the above lemma, we have that

$$\delta(\phi \smile \psi) = 0 \smile \psi + (-1)^n \phi \smile 0 = 0$$

Hence we are done.

If ϕ is a coboundary, then there exists $\alpha \in C^{n-1}(X; R)$ such that $\delta(\alpha) = \phi$. Then we have that

$$\delta(\alpha \smile \psi) = \phi \smile \psi + (-1)^{n-1} \alpha \smile 0 = \phi \smile \psi$$

so that $\phi \smile \psi$ is a coboundary. If ψ is a coboundary, then there exists $\beta \in C^{m-1}(X; R)$ such that $\delta(\beta) = \psi$. Then we have that

$$\delta(\phi \smile \beta) = 0 \smile \beta + (-1)^n \phi \smile \psi$$

so that $\phi \smile \psi$ is a coboundary. ■

Proposition 10.1.4 Let X be a space and let R be a commutative ring. If $[\phi] \in H^m(X; R)$ and $[\psi] \in H^n(X; R)$, then

$$[\phi] \smile [\psi] = (-1)^{mn} [\psi] \smile [\phi]$$

Proof Let $\varepsilon_n = (-1)^{\frac{n(n+1)}{2}}$. Let L be the linear homeomorphism reversing the order of the vertices of $[v_0, \dots, v_n]$. It sends the vertex v_i to v_{n-i} . Define $\rho: C_n(X) \rightarrow C_n(X)$ by

$$\rho(\sigma) = \varepsilon_n(\sigma \circ L)$$

We first show that ρ is a chain map. On one hand, we have

$$\partial(\rho(\sigma)) = \varepsilon_n \sum_{k=0}^n (-1)^k (\sigma \circ L)|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]}$$

and

$$\begin{aligned}
\rho(\partial\sigma) &= \rho\left(\sum_{k=0}^n (-1)^k \sigma|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]}\right) \\
&= \sum_{k=0}^n (-1)^k \varepsilon_{n-1}(\sigma|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]} \circ L) \\
&= \varepsilon_{n-1} \sum_{k=0}^n (-1)^k (\sigma \circ L)|_{[v_0, \dots, \widehat{v}_{n-k}, \dots, v_n]} \\
&= \varepsilon_{n-1} \sum_{k=0}^n (-1)^{n-k} (\sigma \circ L)|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]} \\
&= (-1)^n \varepsilon_{n-1} \sum_{k=0}^n (-1)^k (\sigma \circ L)|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]} \\
&= \varepsilon_n \sum_{k=0}^n (-1)^k (\sigma \circ L)|_{[v_0, \dots, \widehat{v}_k, \dots, v_n]} \\
&= \partial(\rho(\sigma))
\end{aligned}$$

so that ρ is a chain map. We now want to show that ρ is chain homotopic to $\text{id} : C_n(X) \rightarrow C_n(X)$. To show this, define $P : C_n(X) \rightarrow C_{n+1}(X)$ by the formula

$$P(\sigma|_{[v_0, \dots, v_n]}) = \sum_{k=0}^n (-1)^k \varepsilon_{n-k}(\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]}$$

where $\pi : \Delta^n \times I \rightarrow \Delta^n$ is the projection map. We compute that

$$\begin{aligned}
\partial(P(\sigma|_{[v_0, \dots, v_n]})) &= \partial\left(\sum_{k=0}^n (-1)^k \varepsilon_{n-k}(\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]}\right) \\
&= \sum_{k=0}^n (-1)^k \varepsilon_{n-k} \partial((\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]}) \\
&= \sum_{k=0}^n (-1)^k \varepsilon_{n-k} \left(\sum_{j=0}^k (-1)^j (\sigma \circ \pi)|_{[v_{00}, \dots, \widehat{v}_{j0}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]} \right. \\
&\quad \left. + \sum_{j=k+1}^{n+1} (-1)^j (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, \widehat{v}_{n-j+1+k, 1}, \dots, v_{k1}]} \right) \\
&= \sum_{k=0}^n (-1)^k \varepsilon_{n-k} \left(\sum_{j=0}^k (-1)^j (\sigma \circ \pi)|_{[v_{00}, \dots, \widehat{v}_{j0}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]} \right. \\
&\quad \left. + \sum_{j=k}^n (-1)^{n-j+1+k} (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, \widehat{v}_{j1}, \dots, v_{k1}]} \right)
\end{aligned}$$

The term for which $k = j$ is given by

$$\begin{aligned}
& \varepsilon_n(\sigma \circ \pi)|_{[v_{n1}, \dots, v_{k1}]} + \sum_{k=1}^n (-1)^k \varepsilon_{n-k} (-1)^k (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k-1,0}, v_{n1}, \dots, v_k]} \\
& + \sum_{k=0}^{n-1} (-1)^k (-1)^{n-1} \varepsilon_{n-k} (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k+1,1}]} - [v_{00}, \dots, v_{n0}] \\
& = \varepsilon_n[v_{n1}, \dots, v_{k1}] + \sum_{k=1}^n \varepsilon_{n-k} (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k-1,0}, v_{n1}, \dots, v_k]} \\
& + \sum_{k=0}^{n-1} (-1)^{n+k-1} \varepsilon_{n-k} (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k+1,1}]} - [v_{00}, \dots, v_{n0}]
\end{aligned}$$

Notice that in both sums, the coefficients of $(\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, v_{k+1,1}]}$ are ε_{n-k-1} and $(-1)^{n+k-1} \varepsilon_{n-k}$. Their sum is 0 so these terms in the double sum cancel out. We are left with $\partial(P(\sigma|_{[v_0, \dots, v_n]}))$ equal to

$$\begin{aligned}
& \sum_{k=0}^n \varepsilon_{n-k} \left(\sum_{j=0}^{k-1} (-1)^{k+j} (\sigma \circ \pi)|_{[v_{00}, \dots, \widehat{v}_{j0}, \dots, v_{k0}, v_{n1}, \dots, v_{k1}]} + \sum_{j=k+1}^n (-1)^{n-j+1} (\sigma \circ \pi)|_{[v_{00}, \dots, v_{k0}, v_{n1}, \dots, \widehat{v}_{j1}, \dots, v_{k1}]} \right) \\
& + \varepsilon_n (\sigma \circ \pi)|_{[v_{n1}, \dots, v_{01}]} - (\sigma \circ \pi)|_{[v_0, \dots, v_n]}
\end{aligned}$$

One can compute $P(\partial(\sigma|_{[v_0, \dots, v_n]}))$ to get

$$\partial(P(\sigma|_{[v_0, \dots, v_n]} + P(\partial(\sigma|_{[v_0, \dots, v_n]}))) = \varepsilon_n \sigma|_{[v_n, \dots, v_0]} - \sigma|_{[v_0, \dots, v_n]} = \rho(\sigma|_{[v_0, \dots, v_n]}) - \text{id}(\sigma|_{[v_0, \dots, v_n]})$$

Let $\phi \in C^k(X; R)$ and $\psi \in C^l(X; R)$. Then we have that

$$\begin{aligned}
(\rho^*(\phi) \smile \rho^*(\psi))(\sigma) &= \phi(\varepsilon_k \sigma|_{[v_k, \dots, v_0]}) \psi(\varepsilon_l \sigma|_{[v_{k+l}, \dots, v_k]}) \\
&= \varepsilon_k \varepsilon_l \phi(\sigma|_{[v_k, \dots, v_0]}) \psi(\sigma|_{[v_{k+l}, \dots, v_k]})
\end{aligned}$$

Since $\varepsilon_k \varepsilon_l = (-1)^{kl} \varepsilon_{k+l}$ and R is assumed commutative, we have that

$$\begin{aligned}
\varepsilon_k \varepsilon_l \phi(\sigma|_{[v_k, \dots, v_0]}) \psi(\sigma|_{[v_{k+l}, \dots, v_k]}) &= (-1)^{kl} \varepsilon_{k+l} \psi(\sigma|_{[v_{k+l}, \dots, v_k]}) \phi(\sigma|_{[v_k, \dots, v_0]}) \\
&= (-1)^{kl} \rho^*(\psi \smile \phi)(\sigma)
\end{aligned}$$

Since ρ^* and id are chain homotopic, they descend to the identity

$$[\phi] \smile [\psi] = (-1)^{kl} [\psi] \smile [\phi]$$

■

It is important to note that the above proposition (graded commutativity) is only true when R is a commutative ring.

Theorem 10.1.5 (The Cohomology Ring) Let X be a topological space and R a commutative ring with identity. Then the direct sum of abelian groups

$$H^*(X; R) = \bigoplus_{i=0}^{\infty} H^i(X; R)$$

is a graded commutative ring with identity $1_R \in H^0(X; R)$. The product is given by the cup product

$$\smile: H^m(X; R) \times H^n(X; R) \rightarrow H^{m+n}(X; R)$$

defined by

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0, \dots, v_n]}) \cdot \psi(\sigma|_{[v_n, \dots, v_{n+m}]})$$

Moreover, $H^*(X; R)$ is an R -algebra.

Proof ■

Lemma 10.1.6 Let X, Y be spaces. Let R be a commutative ring. Let $f : X \rightarrow Y$ be a map. Then the induced map

$$f^* : H^*(Y; R) \rightarrow H^*(X; R)$$

is a graded ring homomorphism.

Example 10.1.7 Let $n \in \mathbb{N}$. Then there is an isomorphism of graded rings

$$H^*(S^n; \mathbb{Z}) \cong \frac{\mathbb{Z}[t]}{(t^2)}$$

where t is of degree n in the graded structure.

Example 10.1.8 Let $n \geq 1$. Then the following are true.

- There is an isomorphism of graded rings

$$H^*(\mathbb{R}P^n; \mathbb{Z}/2\mathbb{Z}) \cong \frac{\mathbb{Z}/2\mathbb{Z}[t]}{(t^{n+1})}$$

where t is of degree 1 in the graded structure.

- There is an isomorphism of graded rings

$$H^*(\mathbb{C}P^n; \mathbb{Z}) \cong \frac{\mathbb{Z}[t]}{(t^{n+1})}$$

where t is of degree 2 in the graded structure.

Proposition 10.1.9 Let $\mathbb{R}^n$ be equipped with the structure of a division algebra over $\mathbb{R}$. Then n is a power of 2.

10.2 The Relative Cup Product

Definition 10.2.1 (The Relative Cup Product) Let X be a space. Let $A, B \subseteq X$ be open. Let R be a ring. Define the relative cup product of A and B to be the map

$$\smile : H^k(X, A; R) \times H^l(X, B; R) \rightarrow H^{k+l}(X, A \cup B; R)$$

by the formula

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0, \dots, v_k]}) \cdot \psi(\sigma|_{[v_k, \dots, v_{k+l}]})$$

for $\phi \in C_{\text{Sing}}^k(X, A; R)$ and $\psi \in C_{\text{Sing}}^l(X, B; R)$.

In particular, notice that this is the same formula for the (absolute) cup product, other than the fact that the domain of the two inputs are restricted. But we also want to show that this map is well defined in the sense that the codomain is given correctly.

Lemma 10.2.2 Let X be a space. Let $A, B \subseteq X$ be open. Let R be a ring. Then the relative cup product

$$\smile : H^k(X, A; R) \times H^l(X, B; R) \rightarrow H^{k+l}(X, A \cup B; R)$$

is well defined.

Proof Let $\phi \in C_{\text{Sing}}^k(X, A; R)$ and $\psi \in C_{\text{Sing}}^l(X, B; R)$. Then ϕ vanishes on any simplexes in A . Likewise ψ vanishes on any simplexes in B . Recall in the proof of excision that $C^n(X, A + B; R)$ is the subgroup of $C^n(X; R)$ consisting of cochains that vanish on sums of chains in A and sums of chains in B . We conclude that $\phi \smile \psi \in C^{k+l}(X, A + B; R)$. So now our cup product is well-defined in the following way:

$$\smile: C^k(X, A; R) \times C^l(X, B; R) \rightarrow C^{k+l}(X, A + B; R)$$

Moreover, they descend to their respective cohomology groups in the same way as 11.4.3.

Recall from singular homology that the following two horizontal sequences are exact:

$$0 \longrightarrow C_n(A \cup B) \hookrightarrow C_n(X) \twoheadrightarrow C_n(X, A \cup B) \longrightarrow 0$$

$$0 \longrightarrow C_n(A + B) \hookrightarrow C_n(X) \twoheadrightarrow C_n(X, A + B) \longrightarrow 0$$

Dualizing also gives two short exact sequence of chain complexes:

$$0 \longrightarrow C^n(X, A \cup B; R) \hookrightarrow C^n(X; R) \twoheadrightarrow C^n(A \cup B; R) \longrightarrow 0$$

$$0 \longrightarrow C^n(X, A + B; R) \hookrightarrow C^n(X; R) \twoheadrightarrow C^n(A + B; R) \longrightarrow 0$$

Now recall from 4.3.7 that the inclusion $C_\bullet(A + B) \rightarrow C_\bullet(A \cup B)$ is chain homotopic to the identity map. Dualizing the inclusion gives a restriction map $C^\bullet(A \cup B) \rightarrow C^\bullet(A + B)$ that is chain homotopic to the identity. By naturality of the connecting homomorphism, we now obtain the following diagram:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & H^n(X, A \cup B; R) & \longrightarrow & H^n(X; R) & \longrightarrow & H^n(A \cup B; R) & \xrightarrow{\delta} & H^{n+1}(X, A \cup B; R) & \longrightarrow & H^{n+1}(X; R) & \longrightarrow & \cdots \\ & & \cong \downarrow & & \cong \downarrow & & \downarrow & & \cong \downarrow & & \cong \downarrow & & \\ \cdots & \longrightarrow & H^n(X, A + B; R) & \longrightarrow & H^n(X; R) & \longrightarrow & H^n(A + B; R) & \xrightarrow{\delta} & H^{n+1}(X, A + B; R) & \longrightarrow & H^{n+1}(X; R) & \longrightarrow & \cdots \end{array}$$

By the 5-lemma, we conclude that the middle map is an isomorphism and so we conclude. $\blacksquare$

Theorem 10.2.3 (The Relative Cohomology Ring) Let X be a topological space and R a commutative ring with identity. Then the direct sum of abelian groups

$$H^*(X; R) = \bigoplus_{i=0}^{\infty} H^i(X; R)$$

is a graded commutative ring with identity $1_R \in H^0(X; R)$. The product is given by the cup product

$$\smile: H^m(X; R) \times H^n(X; R) \rightarrow H^{m+n}(X; R)$$

defined by

$$(\phi \smile \psi)(\sigma) = \phi(\sigma|_{[v_0, \dots, v_n]}) \cdot \psi(\sigma|_{[v_n, \dots, v_{n+m}]})$$

Moreover, $H^*(X; R)$ is an R -algebra.

Proof $\blacksquare$

10.3 The Kunneth Formula

Definition 10.3.1 (The Cross Product) Let X, Y be topological spaces. Let R be a commutative ring. Denote $p_1: X \times Y \rightarrow X$ and $p_2: X \times Y \rightarrow Y$ the projection maps. Define the cross product of $x \in H^m(X; R)$ and $y \in H^n(Y; R)$ for a ring R to be

$$x \times y = p_1^*(x) \smile p_2^*(y)$$

where $x \times y \in H^{m+n}(X \times Y; R)$.

Proposition 10.3.2 Let X, Y be topological spaces. Let R be a commutative ring. Let $a, c \in H^*(X; R)$ and $b, d \in H^*(Y; R)$ with $c \in H^m(X; R)$ and $d \in H^n(Y; R)$ we have

$$(a \times b) \smile (c \times d) = (-1)^{mn}(a \smile c) \times (b \smile d)$$

Let R be a commutative ring. Let M and N be graded R -modules. Recall that their tensor product can also be given a grading where

$$(M \otimes N)_n = \bigoplus_{i+j=n} M_i \otimes_R N_j$$

Theorem 10.3.3 (The Kunneth Formula) Let X and Y be CW-complexes. Let R be a commutative ring. Then the cross product

$$\times : H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$$

defined by $a \otimes b \mapsto a \times b$ is graded ring homomorphism. Moreover, it is an isomorphism of graded rings if $H^k(Y; R)$ is a finitely generated free R -module for all k .

Example 10.3.4 There is an isomorphism of graded rings

$$H^*(T; \mathbb{Z}) \cong \frac{\mathbb{Z}\langle s, t \rangle}{(s^2, t^2, st + ts)}$$

where s, t are elements of degree 1.

Theorem 10.3.5 (The Relative Kunneth Formula) Let (X, A) and (Y, B) be CW complex pairs. Let R be a commutative ring. Then the cross product

$$\times : H^*(X, A; R) \otimes_R H^*(Y, B; R) \rightarrow H^*(X \times Y, X \times B \cup A \times Y; R)$$

is graded ring homomorphism. Moreover, it is an isomorphism of graded rings if $H^k(Y, B; R)$ is a finitely generated free R -module for all k .

10.4 Lusternik-Schnirelman Category

Definition 10.4.1 (Lusternik-Schnirelman Category) Let X be a space. Define the Lusternik-Schnirelman category of X to be the smallest $n \in \mathbb{N}$ such that there exists an open cover $X = \bigcup_{i=1}^n U_i$ so that $U_i \hookrightarrow X$ is null-homotopic.

Proposition 10.4.2 Let X be a space. Suppose that X has Lusternik-Schnirelman category $n \in \mathbb{N}$. Let $x_i \in H^{k_i}(X; \mathbb{Z})$ for $k_i > 0$. Then we have

$$x_1 \smile \cdots \smile x_n = 0$$

Proof For each U_k in the open cover, notice that $U_k \hookrightarrow X$ is null-homotopic implies that the long exact sequence in relative cohomology becomes

$$\cdots \longrightarrow H^{i_j}(X, U_k) \longrightarrow H^{i_j}(X) \xrightarrow{0} H^{i_j}(U_k) \longrightarrow \cdots$$

which also implies that $H^{i_j}(X, U_k) \rightarrow H^{i_j}(X)$ is surjective. Thus by taking the product, we have that the map

$$H^{i_1}(X, U_1) \times \cdots \times H^{i_n}(X, U_n) \rightarrow H^{i_1}(X) \times \cdots \times H^{i_n}(X)$$

is surjective. Now by surjectivity, there exists a lift $(\tilde{x}_1, \dots, \tilde{x}_n) \in H^{i_1}(X, U_1) \times \cdots \times H^{i_n}(X, U_n)$ of $(x_1, \dots, x_n)$. Now using the relative version of the cup product, we obtain a commutative diagram:

$$\begin{array}{ccc}
 \prod_{k=1}^n H^{i_k}(X; R) & \xrightarrow{\quad \smile \quad} & H^{i_1+\dots+i_n}(X; R) \\
 \uparrow & & \uparrow \\
 \prod_{k=1}^n H^{i_k}(X, U_k; R) & \xrightarrow{\quad \smile \quad} & H^{i_1+\dots+i_n}(X, U_1 \cup \dots \cup U_n; R)
 \end{array}$$

(why is it natural / commutative) But $X = \bigcup_{i=1}^n U_i$ implies that $H^{i_1+\dots+i_n}(X, U_1 \cup \dots \cup U_n; R) = 0$.
 Thus $(x_1, \dots, x_n)$ is mapped to the 0 element. ■

11 Suspensions and Loopspaces

11.1 Suspensions of a Space

Definition 11.1.1 (Reduced Suspension) Let (X, x_0) be a pointed space. Define the reduced suspension of X to be the space

$$\Sigma X = \frac{X \times I}{(X \times \{0\} \cup X \times \{1\} \cup \{x_0\} \times I)}$$

together with the base point as the equivalence class in the quotient.

By definition of quotients, the reduced suspension defines a canonical continuous map sending a space X to its reduced suspension ΣX .

Proposition 11.1.2 The reduced suspension Σ define functors on $\mathbf{Top}_*$ and $\mathbf{hTop}_*$ as follows.

- Σ sends a pointed space (X, x_0) to

$$(\Sigma X, (x_0, 0))$$

- For the non homotopy category, Σ sends a map $f : X \rightarrow Y$ to

$$\Sigma f : \Sigma X \rightarrow \Sigma Y$$

defined by $\Sigma f([x, t]) = [f(x), t]$

- For the homotopy category, Σ sends a homotopy class of maps $[X, Y]$ to

$$[\Sigma X, \Sigma Y]$$

respectively given by the same formula as above

Example 11.1.3 Let $n \in \mathbb{N}$. Then there is a homeomorphism $\Sigma S^n \cong S^{n+1}$.

Proposition 11.1.4 Let X be a space. Then the suspension map induces natural isomorphisms

$$\tilde{H}_{k+1}(\Sigma X) \cong \tilde{H}_k(X)$$

for $k \geq -1$.

Definition 11.1.5 ((Reduced) Cone) Let (X, x_0) be a pointed space. Define the reduced cone of the space to be

$$CX = \frac{X \times I}{X \times \{1\} \cup \{x_0\} \times I}$$

together with the base point $[(x_0, 1)] = X \times \{1\} \cup \{x_0\} \times I$.

Proposition 11.1.6 The reduced cone defines a functor on

$$C : \mathbf{Top}_* \rightarrow \mathbf{Top}_*$$

as follows.

- C sends a pointed space (X, x_0) to $(CX, [(x_0, 1)])$
- C sends a map $f : X \rightarrow Y$ to

$$Cf : CX \rightarrow CY$$

given by the induced map $f \times \text{id}_I$ by taking quotients.

11.2 Unpointed Suspensions

Definition 11.2.1 (Unreduced Cone) Let X be a space. Define the unreduced cone of X to be

$$CX = \frac{X \times I}{X \times \{1\}}$$

Definition 11.2.2 (Unreduced Suspensions) Let X be a space. Define the unreduced suspension of X to be the space

$$SX = \frac{X \times I}{X \times \{0\}, X \times \{1\}}$$

Proposition 11.2.3 Let X be a space. Then the following diagram is a pushout:

$$\begin{array}{ccc} X \times \{0\} & \hookrightarrow & CX \\ \downarrow & & \downarrow \\ CX & \longrightarrow & SX \end{array}$$

Proposition 11.2.4 Let X be a space. The unreduced suspension map induces natural isomorphisms

$$H_{k+1}(SX) \cong H_k(SX)$$

for $k \geq 1$. Moreover, the following are true.

- There is an isomorphism $H_0(X) \cong H_1(SX) \oplus \mathbb{Z}$
- $H_0(SX) \cong \mathbb{Z}$.

12 Homotopy Groups and (Co)Homology Groups

12.1 Weak Equivalences and Homological Isomorphisms

Proposition 12.1.1 Let X, Y be spaces and let $f : X \rightarrow Y$ be a weak homotopy equivalence. Then f induces isomorphisms

$$f_* : H_n(X; G) \xrightarrow{\cong} H_n(Y; G) \quad \text{and} \quad f^* : H^n(Y; G) \xrightarrow{\cong} H^n(X; G)$$

for any group G and all $n \in \mathbb{N}$.

This theorem shows that the higher homotopy groups is not a weaker invariant than homology and cohomology. Indeed, the theorem states that if the all homotopy groups are isomorphic, then all their (co)homology groups will be isomorphic.

12.2 The Hurewicz Homomorphism

Theorem 12.2.1 (Hurewicz's Homomorphism) Let X be a path connected space. Then for any $n \in \mathbb{N}$, there is a group homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

called the Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 12.2.2 (Hurewicz's Theorem) Let X be a space. Then the following are true regarding Hurewicz's homomorphism.

- Let $n \geq 2$. If X is $(n-1)$ -connected, then $\tilde{H}_k(X) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X) \rightarrow H_n(X)$$

is an isomorphism. Moreover, h_{n+1} is a surjection.

- Let $n = 1$, then Hurewicz's homomorphism induces an isomorphism

$$\bar{h}_1 : \pi_1(X)^{\text{ab}} \rightarrow H_1(X)$$

Theorem 12.2.3 (Relative Hurewicz's Homomorphism) Let (X, A) be a pair of spaces. Then for any $n \geq 1$, there is a group homomorphism

$$h_n : \pi_n(X, A) \rightarrow H_n(X, A)$$

called the relative Hurewicz homomorphism, defined as follows. Let $[u_n] \in H_n(S^n, \partial S^n)$ be a canonical generator. Then $h_n([f]) = f_*(u_n)$.

Theorem 12.2.4 (Relative Hurewicz's Theorem) Let (X, A) be a pair of spaces. Let $n \geq 2$. If X and A are path connected and (X, A) is $(n-1)$ -connected, then $H_k(X, A) = 0$ for all $0 \leq k < n$. Moreover, the Hurewicz homomorphism

$$h_n : \pi_n(X, A, x_0) \rightarrow H_n(X, A)$$

is an isomorphism.

Theorem 12.2.5 (Naturality of Hurewicz's Homomorphism) Let (X, x_0) and (Y, y_0) be pointed spaces and let $f : (X, x_0) \rightarrow (Y, y_0)$ be a map. Then the following diagram is commutative:

$$\begin{array}{ccc}
\pi_k(X, x_0) & \xrightarrow{\pi_k(f)} & \pi_k(Y, y_0) \\
h_k \downarrow & & \downarrow h_k \\
H_k(X) & \xrightarrow{f_*} & H_k(Y)
\end{array}$$

where h is the Hurewicz homomorphism. Moreover, a similar diagram is also commutative for the relative Hurewicz homomorphism.

The connection between the homotopy groups and the homology groups begs the question of whether there is a relationship between the homotopy groups and cohomology groups that is not implicit by the relation between homology and cohomology. This is answered in Stable Homotopy Theory, when we introduced Brown's representability theorem.

12.3 Reading Cell Structures from Homology Groups

Proposition 12.3.1

Let X be a simply connected CW complex. Suppose that $H_n(X)$ is a finitely generated abelian group for all $n \in \mathbb{N}$. Then there exists a CW complex Z and a cellular homotopy equivalence $f : Z \rightarrow X$ such that the cells of Z are of one of the following forms.

- A generator n -cell e_α^n corresponding to a generator α of $H_n(X)$.
- A relator $(n+1)$ -cell whose cellular boundary is equal to a multiple of the generator n -cell e_α^n if α has finite order.

13 The Eilenberg-Steenrod Axioms

We have seen that the homotopy groups, the homology groups and the cohomology groups all satisfy a functorial property. This means that they can be considered as functors from the category of spaces to the category of some algebraic structures. It is meaningful to study all of them at once, and to compare different versions of homology and cohomology.

In general, there are different parameters for (co)homology theories.

- Including “for CW Pairs” means that the theory is tailored for CW pairs. If we want a general theory for any topological spaces, we must add a new axiom so that weak equivalences (isomorphisms in homotopy groups) gives isomorphisms in homology groups. This is true for CW complexes but not for arbitrary topological spaces by Whitehead’s theorem.
- “Generalized” means that the additivity and dimension axioms are dropped. This enables wilder (co)homology theories such as (co)bordism to appear. “Ordinary” means that the dimension axiom is included. In the case that we restrict to CW complexes, this axiom ensures that all (co)homology theories of this type are isomorphic to each other.
- “Reduced” theories typically throw away the relative context in order to gain more concrete computations. There is a theorem that says that determining a generalized (co)homology theory is the “same” determining a reduced theory and vice versa.

13.1 Generalized (Co)Homology Theories for Spaces

Definition 13.1.1 (Generalized Homology Theory for Spaces)

A generalized homology theory is a collection of functors and natural transformations

$$h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta_n : h_n \Rightarrow h_n \circ F$$

for each $n \in \mathbb{N}$, where F is the functor defined by $F(X, A) = (A, \emptyset)$, such that the following are true.

- Homotopy Invariance: Let (X, A) and (Y, B) be pairs of spaces. If $f \simeq g : (X, A) \rightarrow (Y, B)$ are homotopy equivalent maps, then

$$h_n(f) = h_n(g) : h_n(X, A) \rightarrow h_n(Y, B)$$

- Exactness: Let (X, A) be a pair of space. Then there exists an exact sequence

$$\cdots \longrightarrow h_{n+1}(X, A) \xrightarrow{\delta_{n+1}} h_n(A, \emptyset) \xrightarrow{h_n(i)} h_n(X, \emptyset) \xrightarrow{h_n(j)} h_n(X, A) \xrightarrow{\delta_n} h_{n-1}(A, \emptyset) \longrightarrow \cdots$$

where $i : (A, \emptyset) \rightarrow (X, \emptyset)$ and $j : (X, \emptyset) \rightarrow (X, A)$ are the inclusions maps.

- Excision: Let (X, A) be a pair of space. Let $E \subseteq X$ be a subspace of X such that $\overline{E} \subseteq A^\circ \subseteq X$, then the inclusion map induces an isomorphism

$$h_n(i) : h_n(X \setminus E, A \setminus E) \xrightarrow{\cong} h_n(X, A)$$

for all $n \in \mathbb{N}$.

- Weak Equivalence: Let (X, A) and (Y, B) be pairs of spaces. If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then there are isomorphisms

$$h_n(f) : h_n(X, A) \xrightarrow{\cong} h_n(Y, B)$$

for all $n \in \mathbb{N}$.

Note that the axiom for weak equivalence is not present in the definition of generalized homology theories for CW complexes. Indeed, we have just shown that weak equivalences between CW complexes are homotopy equivalences. This also shows that this definition really is redundant:

Proposition 13.1.2

Any generalized homology theory on $\mathbf{Top}^2$ determines and is determined by a generalized homology theory on $\mathbf{CW}^2$.

Definition 13.1.3 (Additive Homology Theory)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory. We say that h_n is additive if the following axiom is satisfied.

- Additivity: If $(X, A) = \coprod_{i \in I} (X_i, A_i)$, then the direct sum of the inclusion maps

$$\bigoplus_{i \in I} h_n(X_i, A_i) \cong h_n(X, A)$$

is an isomorphism

Definition 13.1.4 (Ordinary Homology Theory)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory. We say that h_n is ordinary if there exists an abelian group A such that the following axiom is satisfied.

- Dimension:

$$h_n(*) = \begin{cases} A & \text{if } n = 0 \\ 0 & \text{otherwise} \end{cases}$$

Theorem 13.1.5 (Eilenberg-Steenrod Uniqueness Theorem) Let $T : (h_n, \delta_n) \rightarrow (h'_n, \delta'_n)$ be a natural transformation of generalized homology theories defined on $\mathbf{CW}^2$ such that $h_n(*) \cong h'_n(*)$, then T is a natural isomorphism

$$(h_n, \delta_n) \cong (h'_n, \delta'_n)$$

Definition 13.1.6 (Generalized Cohomology Theory for Spaces)

A generalized cohomology theory is a collection of contravariant functors and natural transformations

$$h^n : \mathbf{Top}^2 \rightarrow \mathbf{Ab} \quad \text{and} \quad \delta^n : h^n \circ F \Rightarrow h^{n+1} \circ F$$

for each $n \in \mathbb{N}$, where F is the functor defined by $F(X, A) = (A, \emptyset)$, such that the following are true.

- Homotopy Invariance: Let (X, A) and (Y, B) be pairs of spaces. If $f \simeq g : (X, A) \rightarrow (Y, B)$ are homotopy equivalent, then

$$h^n(f) = h^n(g) : h^n(X, A) \rightarrow h^n(Y, B)$$

- Exactness: Let (X, A) be a pair of spaces. Then there exists an exact sequence

$$\cdots \longrightarrow h^{n-1}(A, \emptyset) \xrightarrow{\delta^{n-1}} h^n(X, A) \xrightarrow{h^n(j)} h^n(X, \emptyset) \xrightarrow{h^n(i)} h^n(A, \emptyset) \xrightarrow{\delta^n} h^{n+1}(A, \emptyset) \longrightarrow \cdots$$

where $i : (A, \emptyset) \rightarrow (X, \emptyset)$ and $j : (X, \emptyset) \rightarrow (X, A)$ are the inclusion maps.

- Excision: Let (X, A) be a pair of space. Let $E \subseteq X$ be a subspace of X such that $\overline{E} \subseteq A^\circ \subseteq X$, then the inclusion map induces an isomorphism

$$h^n(i) : h^n(X, A) \xrightarrow{\cong} h^n(X \setminus E, A \setminus E)$$

for all $n \in \mathbb{N}$.

- Weak Equivalence: Let (X, A) and (Y, B) be pairs of spaces. If $f : (X, A) \rightarrow (Y, B)$ is a weak equivalence, then there are isomorphisms

$$h^n(f) : h^n(Y, B) \xrightarrow{\cong} h^n(X, A)$$

for all $n \in \mathbb{N}$.

Definition 13.1.7 (Additive Cohomology Theories)

Let $h_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$ be a generalized cohomology theory. We say that h_n is additive if it satisfies the following axiom.

- Additivity: Denote the inclusion map as $\iota_k : (X_k, A_k) \rightarrow \coprod_{k \in I} (X_k, A_k)$. Then the product of the induced maps of inclusions induce an isomorphism

$$\prod_{k \in I} (\iota_k)^* : h^n \left(\prod_{k \in I} (X_k, A_k) \right) \xrightarrow{\cong} \prod_{k \in I} h^n(X_k, A_k)$$

Proposition 13.1.8 Let F and G be two generalized cohomology theories for CW pairs. Let $\lambda : F \Rightarrow G$ be a natural transformation such that

$$\lambda : F(*, \emptyset) \xrightarrow{\cong} G(*, \emptyset)$$

is an isomorphism for $*$ the one-point space. Then λ is a natural isomorphism.

13.2 Reduced (Co)Homology Theory for CW Complexes**Definition 13.2.1** (Reduced Homology Theory for CW Complexes)

A reduced Homology Theory is a collection of functors

$$\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

that satisfies the following.

- Homotopy Invariance: Let (X, x_0) and (Y, y_0) be pointed CW complexes. If $f \simeq g : (X, x_0) \rightarrow (Y, y_0)$ are homotopy equivalent, then

$$\tilde{h}_n(f) = \tilde{h}_n(g) : \tilde{h}_n(X, x_0) \rightarrow \tilde{h}_n(Y, y_0)$$

- Exactness: If (X, x_0) is a pointed CW-complex and (A, x_0) is a subcomplex, then there is an exact sequence

$$\tilde{h}_n(A, x_0) \xrightarrow{\tilde{\iota}} \tilde{h}_n(X, x_0) \xrightarrow{\tilde{p}} \tilde{h}_n(X/A, *)$$

for each $n \in \mathbb{N}$, where $\iota : A \rightarrow X$ is the inclusion and $p : X \rightarrow X/A$ is the projection.

- Suspension: Let (X, x_0) be a pointed CW complex. Then the suspension functor induces a natural isomorphism

$$\Sigma : \tilde{h}_n(X, x_0) \xrightarrow{\cong} \tilde{h}_{n+1}(\Sigma X, *)$$

Definition 13.2.2 (Additive Reduced Homology Theories)

Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory. We say that $\tilde{h}_n$ is additive if it satisfies the following axiom.

- Additivity: Let (X_k, x_k) be a collection of CW complexes for $k \in I$. Denote the inclusion map as $\iota_k : (X_k, x_k) \hookrightarrow \bigvee_{i \in I} (X_i, x_i)$. Then there is an isomorphism

$$\bigoplus_{i \in I} \tilde{h}_n(\iota_i) : \bigoplus_{i \in I} \tilde{h}_n(X_i, x_i) \xrightarrow{\cong} \tilde{h}_n \left(\bigvee_{i \in I} (X_i, x_i) \right)$$

Lemma 13.2.3 Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory. Then

$$\tilde{h}_n(*) = 0$$

Proposition 13.2.4 Let $h_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$ be a generalized homology theory for CW complexes. Define a collection of functors

$$\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

by $(X, x_0) \mapsto \tilde{h}_n(X, x_0) = h_n(X, \{x_0\})$. Then $\tilde{h}_n$ defines a reduced homology theory for CW complexes.

Proposition 13.2.5 Let $\tilde{h}_n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$ be a reduced homology theory for CW complexes. Define a collection of functors

$$\tilde{h}_n : \mathbf{CW}^2 \rightarrow \mathbf{Ab}$$

by $(X, A) \mapsto \tilde{h}_n(X/A, * = A)$ and a collection of natural transformations

$$\delta_n : h_n(X, Y) \rightarrow (Y, \emptyset)$$

by ??? Then h_n and δ_n defines a generalized homology theory for CW complexes.

Proposition 13.2.6 Any generalized homology theory for CW complexes determines and is determined by a reduced homology theory for CW complexes.

Thus we have showed that under the conditions of all the given axioms of all such homology theories, we really just have the same thing:

$$\begin{array}{ccccc} \text{Reduced Homology Theory} & \rightleftharpoons & \text{Generalized Homology Theory} & \rightleftharpoons & \text{Generalized Homology Theory} \\ \text{for CW Complexes} & & \text{for CW Complexes} & & \text{for Spaces} \end{array}$$

Note that such a bijection is only possible when we introduce the weak equivalence axiom for homology theories on spaces. Such an axiom in fact restricts the amount of valid homology theories, but it is only through this axiom that we can declare all such homology theories are essentially determined by one another.

Some authors indeed does not require the weak equivalence axiom to exist. In such cases the bijection between generalized homology theories and reduced homology theories can be constructed, but there is no passage between homology of CW complexes and homology of spaces in general.

Definition 13.2.7 (Reduced Cohomology Theory for CW Complexes)

A reduced cohomology theory is a collection of contravariant functors

$$h^n : \mathbf{CW}_* \rightarrow \mathbf{Ab}$$

,for each $n \in \mathbb{N}$ such that the following are true.

- Homotopy Invariance: If $f \simeq g : X \rightarrow Y$ then

$$h^n(f) = h^n(g) : h^n(X) \rightarrow h^n(Y)$$

- Exactness: If $f : A \rightarrow X$ is an inclusion of CW complexes, then there is an exact sequence

$$h^n(C_f) \xrightarrow{h^n(i)} h^n(X) \xrightarrow{h^n(f)} h^n(A)$$

where $i : X \rightarrow C_f$ is the inclusion map.

- Suspension: There is a natural isomorphism of functors:

$$\sigma : h^{n+1} \circ \Sigma \Rightarrow h^n$$

- Additivity: Let $i_k : X_k \rightarrow \bigvee_{i \in I} X_i$ denote the inclusion map. Their products induce an isomorphism

$$\prod_{k \in I} h^n(i_k) : h^n \left(\bigvee_{i \in I} X_i \right) \xrightarrow{\cong} \prod_{i \in I} h^n(X_i)$$

Lemma 13.2.8 Let $\tilde{h}_n : \mathbf{CW} \rightarrow \mathbf{Ab}$ be a reduced homology theory. Then

$$\tilde{h}_n(*) = 0$$

13.3 Brown Representability Theorem

Theorem 13.3.1 (Brown's Representability Theorem for Singular Cohomology)

Let G be an abelian group. Then there is a natural isomorphism

$$H^n(-; G) \cong [-, K(G, n)]$$

for each $n \in \mathbb{N}$.

14 The Euler Characteristic

The Euler characteristic is similar to a notion of size. In set theory we have cardinality, in linear algebra we have dimensions, in abelian groups we have rank. The Euler characteristic acts similar to these quantities.

14.1 The Characteristic as an Invariant

Definition 14.1.1 (The Euler Characteristic) Let X be a space with only finitely many non-zero homology groups, all of which are finitely generated abelian groups. Then the Euler characteristic is defined as

$$\chi(X) = \sum_{n \geq 0} (-1)^n \text{rank}(H_n(X))$$

Lemma 14.1.2 Let $C_\bullet$ be a chain complex with only finitely many non-zero terms, all of which are finitely generated abelian groups. Then

$$\sum_{n \in \mathbb{Z}} (-1)^n \text{rank}(C_n) = \sum_{n \in \mathbb{Z}} (-1)^n \text{rank}(H_n(C_\bullet))$$

Proof We have two short exact sequences for all n given by

$$0 \longrightarrow Z_n \longrightarrow C_n \longrightarrow B_{n-1} \longrightarrow 0$$

$$0 \longrightarrow B_n \longrightarrow Z_n \longrightarrow H_n \longrightarrow 0$$

We therefore have that

$$\begin{aligned} \sum_{n \in \mathbb{Z}} (-1)^n \text{rank}(C_n) &= \sum_{n \in \mathbb{Z}} (-1)^n (\text{rank}(Z_n) + \text{rank}(B_{n-1})) \\ &= \sum_{n \in \mathbb{Z}} (-1)^n (\text{rank}(Z_n) - \text{rank}(B_n)) \\ &= \sum_{n \in \mathbb{Z}} (-1)^n \text{rank}(H_n) \end{aligned}$$

and so we conclude. ■

Corollary 14.1.3 Let X be a CW complex. Then the alternating sum

$$\chi(X) = \sum_{n \geq 0} (-1)^n |\{n\text{-cells}\}|$$

In particular, $\chi(X)$ is independent of the choice of CW structure.

Proof By the above corollary, we have that the sum is equal to $\sum_{n \in \mathbb{Z}} (-1)^n \text{rank}(H_n(X))$ and the rank of $H_n(X)$ is independent of the cell structure. ■

Lemma 14.1.4 Let X be a space with only finitely many non-zero homology groups, all of which are finitely generated abelian groups. Let $\mathbb{F}$ be a field. Then we have

$$\chi(X) = \sum_{n \geq 0} (-1)^n \dim_{\mathbb{F}}(H^n(X; \mathbb{F}))$$

Definition 14.1.5 (Plane Graph) A plane graph is a finite 1-dimensional CW-complex embedded in the real plane $\mathbb{R}^2$. Equivalently, it is a finite graph in the plane in which the edges do not cross. A planar graph is a finite 1-dimensional CW-complex that exhibits such an embedding.

A face of a graph X is a connected component of $\mathbb{R}^2 \setminus X$.

The Euler characteristic is in fact a generalization of Euler's formula.

Proposition 14.1.6 (Euler's Formula) Let X be a planar graph with v vertices, e edges and f faces. Then we have

$$v - e + f = 2$$

14.2 First Properties of the Euler Characteristic

Proposition 14.2.1 Let $X = U \cup V$ be a space and assume that one of the following holds.

- X is a CW-complex and U, V are subcomplexes
- $U, V \subseteq X$ are open

Then if $\chi(U), \chi(V), \chi(U \cap V)$ are well defined, $\chi(X)$ is also well defined and we have

$$\chi(X) = \chi(U) + \chi(V) - \chi(U \cap V)$$

Proof If U and V are finite CW complexes then so is X . Looking at the number c_n of cells in each dimension n we have that

$$c_n(X) = c_n(U) + c_n(V) - c_n(U \cap V)$$

which completes the proof.

If U and V are open, then using Mayer-Vietoris sequence, we obtain an alternating sum of cells in X, U and V . By corollary 11.1.3, it follows that since the sequence is exact, we have that

$$\sum_{n \in \mathbb{Z}} (-1)^n (\text{rank}(H_n(U \cap V)) - \text{rank}(H_n(U)) - \text{rank}(H_n(V)) + \text{rank}(H_n(X))) = 0$$

and so we conclude. ■

Proposition 14.2.2 Let X and Y be finite CW-complexes. Then so is $X \times Y$ and we have

$$\chi(X \times Y) = \chi(X) \times \chi(Y)$$

Proof We first prove that $c_n(X \times Y) = \sum_{a+b=n} c_a(X)c_b(Y)$. The result then follows by proposition 11.1.3. ■

We can relate the Euler characteristic with covering spaces by the following formula.

Proposition 14.2.3 Let $p : \tilde{X} \rightarrow X$ be a d -sheeted cover and X is a finite CW complex. Then we have that

$$\chi(\tilde{X}) = d \cdot \chi(X)$$

Proof In the proof that $\tilde{X}$ is also a CW complex, we have seen that if X has k amount of n -cells, then $\tilde{X}$ has $d \cdot k$ amount. This applies to all n and so we obtain the formula. ■